



CHAPTER 5

HYDROCARBONS

The slide features abstract geometric decorations. In the top-left corner, there is a cluster of overlapping triangles in shades of blue, green, and red. In the bottom-right corner, there is a cluster of overlapping triangles in various shades of gray.

5.1 ALKANES

INTRODUCTION

HYDROCARBONS

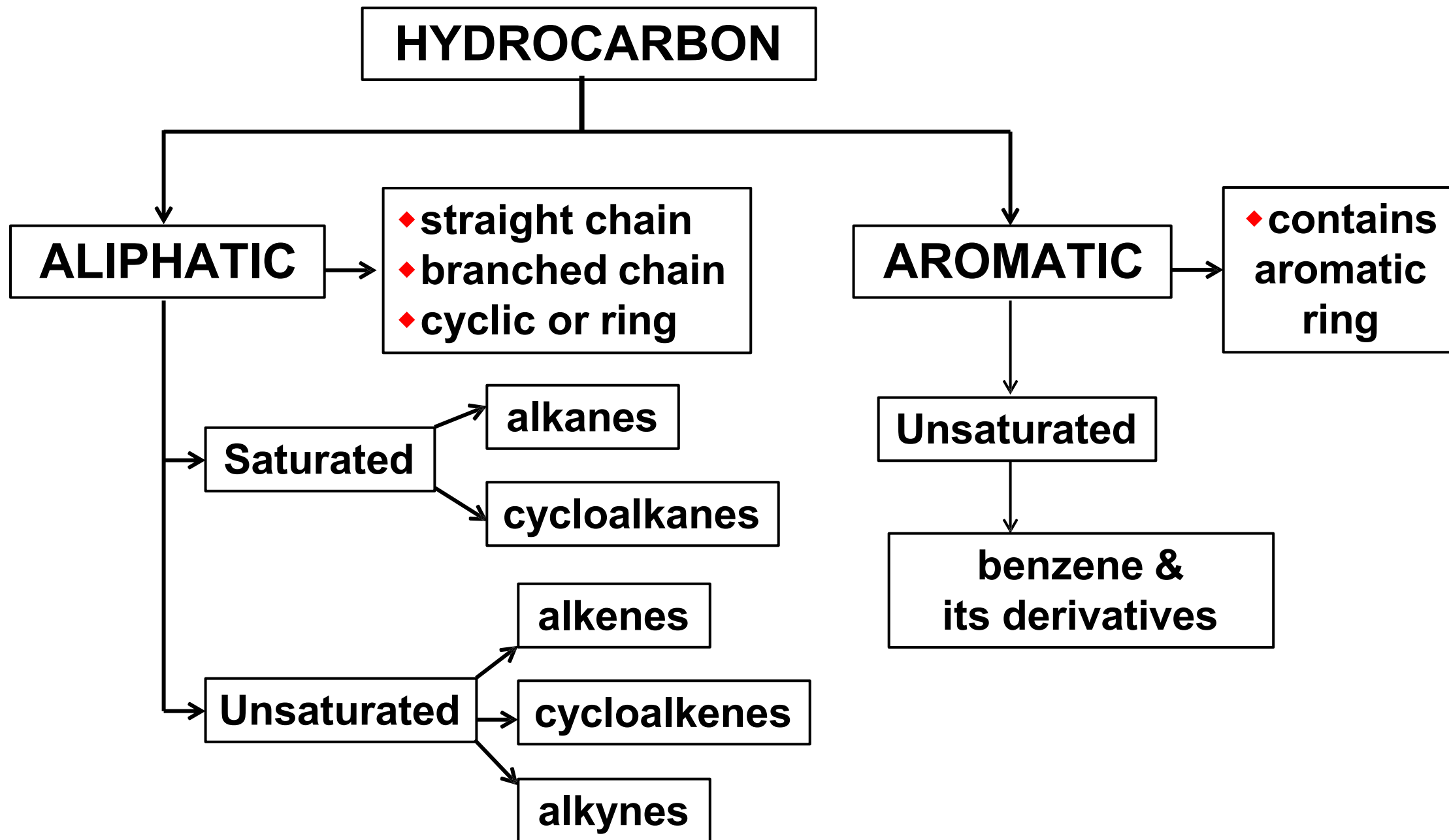
(compounds made up of only C and H elements)

❖ Saturated

- Contains only single bonds between C–C.
- Possess the **maximum number of H atoms** that the compound can have.

❖ Unsaturated

- Contains multiple bonds: double bond, C=C or triple bonds, C≡C.
- Posses **fewer than the maximum** number of H atoms.



ALKANES

- Saturated hydrocarbons, all bonds are single covalent bonds;
- Non-polar molecules, C-C and C-H bonds are non-polar;
- Do not have functional group;
- All C atoms are sp^3 -hybridised and assume the shape of tetrahedral;
- The first 3 members: CH_4 , C_2H_6 and C_3H_8 do not have isomers;
- From C_4H_{10} onwards, structural isomers (chain isomers) exist;
- The number of isomers increases with the increase in the number of C atoms.

Open chain alkane

General formula: C_nH_{2n+2} ($n \geq 1$)

Cyclic alkane
(cycloalkane)

General formula : C_nH_{2n} ($n \geq 3$)

(n = number of C atoms)

IUPAC Nomenclature

- Names end with the suffix –ane.
- Prefixes are based on the number of C atoms.
- Names of first 10 straight chain alkanes:

No. of C	Prefix	Name
1	Meth-	Methane
2	Eth-	Ethane
3	Prop-	Propane
4	But-	Butane
5	Pent-	Pentane

No. of C	Prefix	Name
6	Hex-	Hexane
7	Hept-	Heptane
8	Oct-	Octane
9	Non-	Nonane
10	Dec-	Decane

Some Common Substituent Groups

Alkyl Groups, R ($-\text{C}_n\text{H}_{2n+1}$)		
Molecular formula	Structure	Name
$-\text{CH}_3$	$-\text{CH}_3$	methyl
$-\text{C}_2\text{H}_5$	$-\text{CH}_2\text{CH}_3$	ethyl
$-\text{C}_3\text{H}_7$	$-\text{CH}_2\text{CH}_2\text{CH}_3$	propyl
	$ \begin{array}{c} -\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array} $	isopropyl (1-methylethyl)

Alkyl Groups, R ($-\text{C}_n\text{H}_{2n+1}$)		
Molecular formula	Structure	Name
$-\text{C}_4\text{H}_9$	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	butyl
	$ \begin{array}{c} -\text{CH}_2\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array} $	isobutyl
	$ \begin{array}{c} -\text{CHCH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $	sec-butyl
	$ \begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $	tert-butyl

*2° carbon bond
to parent chain*

*3° carbon bond
to parent chain*

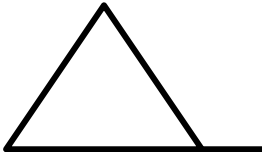
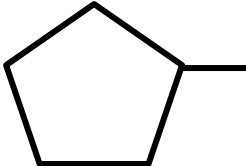
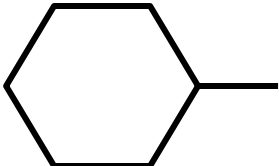
Alkyl Groups, R ($-\text{C}_n\text{H}_{2n+1}$)		
Molecular formula	Structure	Name
$-\text{C}_5\text{H}_{11}$	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	pentyl
	$\begin{array}{c} \text{---CH}_2\text{CH}_2\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array}$	isopentyl
	$\begin{array}{c} \text{---CHCH}_2\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	sec-pentyl

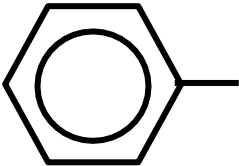
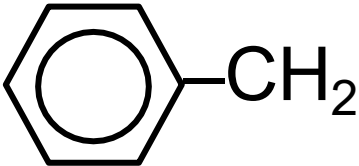
*2° carbon bond
to parent chain*

Alkyl Groups, R ($-\text{C}_n\text{H}_{2n+1}$)		
Molecular formula	Structure	Name
$-\text{C}_5\text{H}_{11}$	$ \begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $	tert-pentyl
	$ \begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $	neopentyl

*3° carbon bond
to parent chain*



Cyclic Alkyls	
Structure	Name
	cyclopropyl
	cyclobutyl
	cyclopentyl
	cyclohexyl

Aromatic Alkyls	
Structure	Name
$-\text{C}_6\text{H}_5$ or 	phenyl
$-\text{CH}_2\text{C}_6\text{H}_5$ or 	benzyl

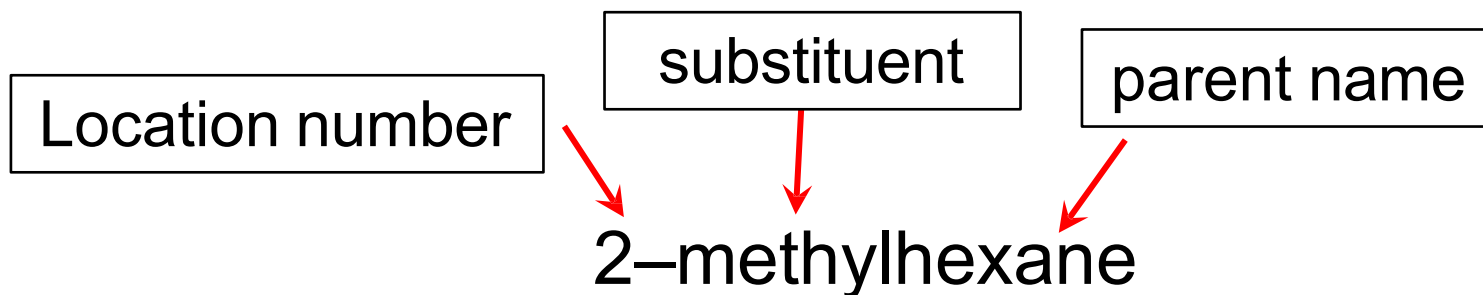
Halogen	
Structure	Name
-Cl	chloro-
-F	fluoro-
-Br	bromo-
-I	iodo-

Other Substituents	
Structure	Name
-OR	alkoxy
-OH	hydroxy
-NO ₂	nitro
-C≡N	cyano
-NH ₂	amino

Rules in Naming Branched Alkanes

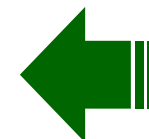
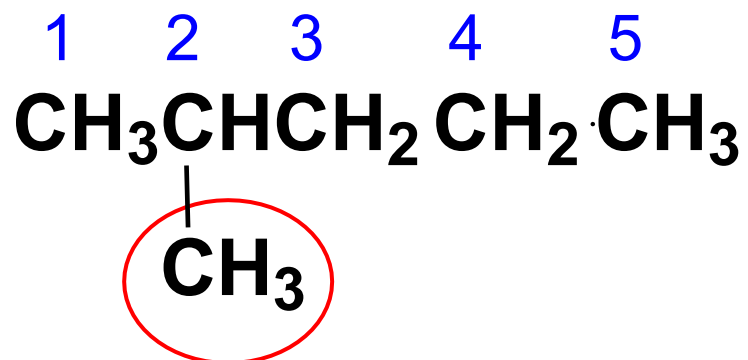
14

1. Find the longest continuous carbon chain (parent chain) which may or may not be shown in straight line and name it.
2. Number the parent chain, starting from the end closer to the branch or substituent(s).
3. Identify each substituent and its location number.
 - ☐ Substituent name precedes parent name and must be joined.
 - ☐ Location number of substituent precedes its name.
 - ☐ Numbers are separated from words by a hyphen.



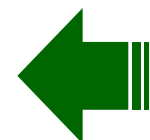
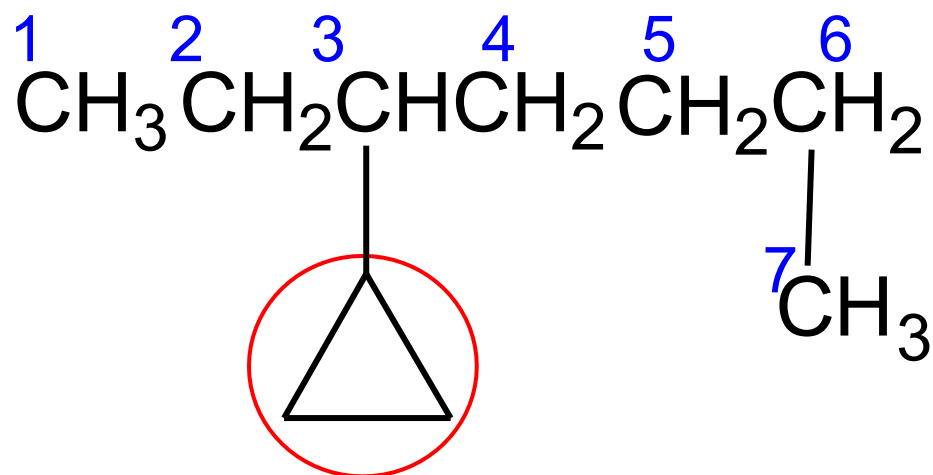
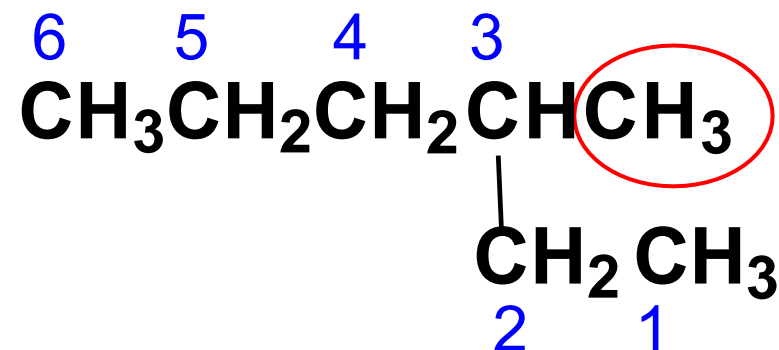
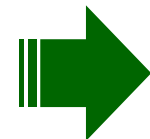
Examples:

15



2-methylpentane

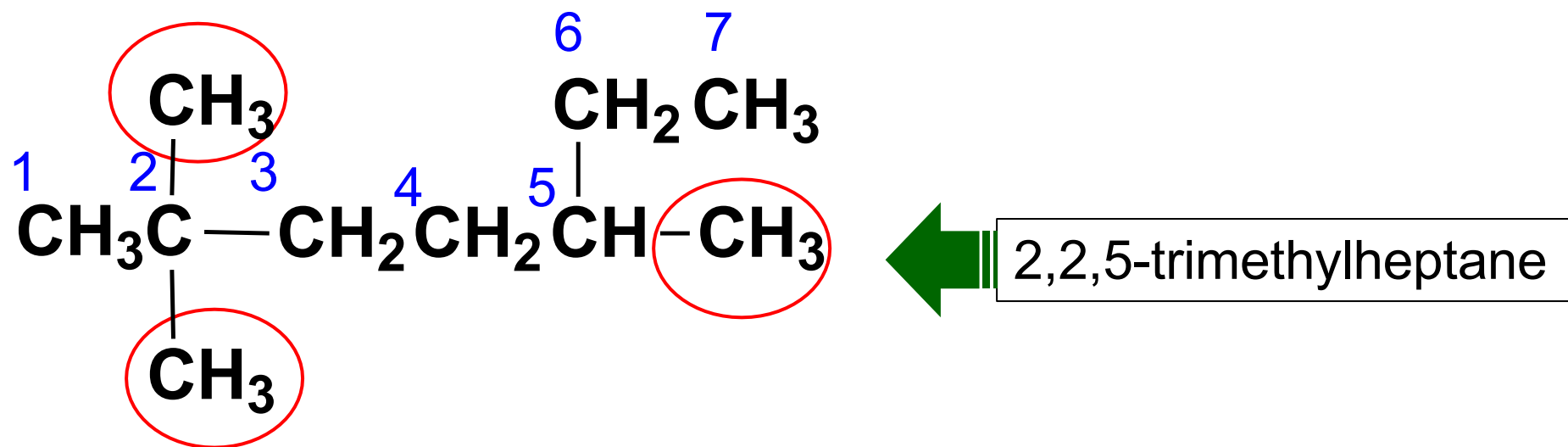
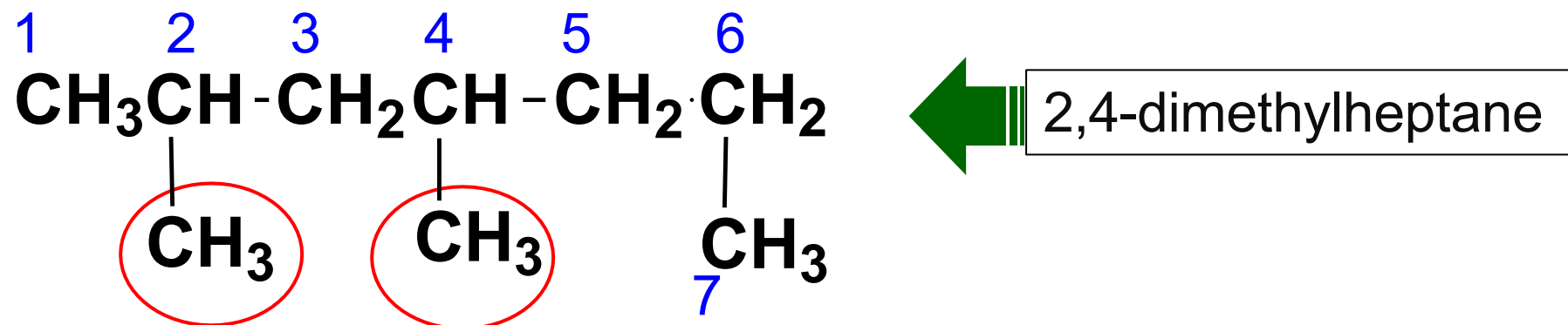
3-methylhexane



3-cyclopropylheptane

4. A repeating substituent is indicated by the use of prefixes di-, tri-, tetra- and so on. Subsequent numbers are separated by comma(s).

Examples:

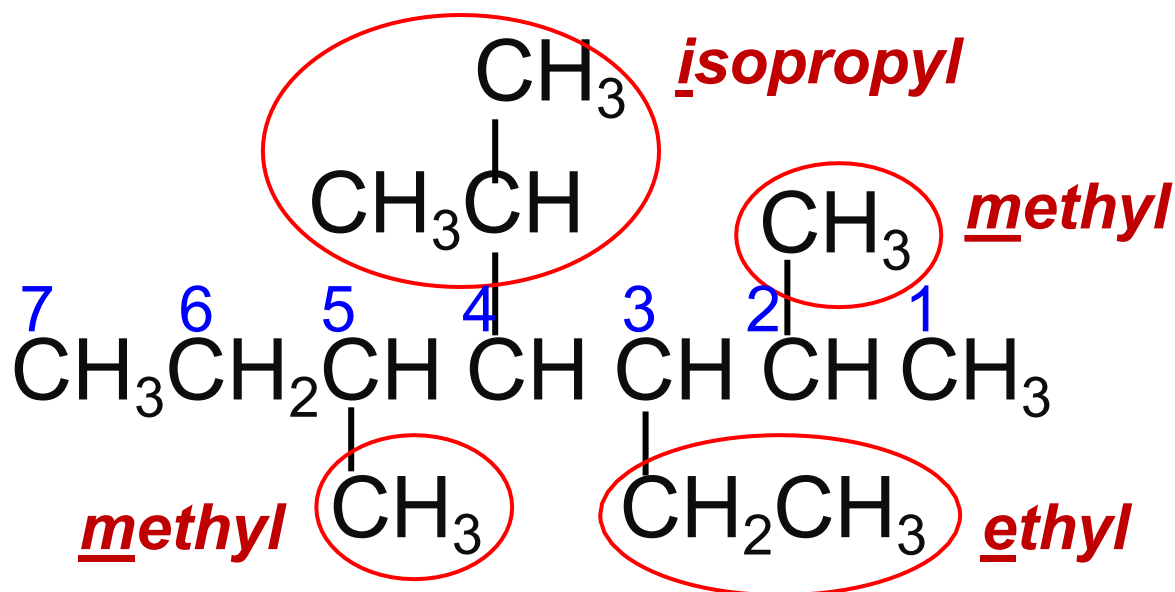


5. When two or more substituents are present, location of each must be numbered and the names are listed alphabetically.

Note

Alphabet for the prefixes di-, tri-, tetra-, sec- and tert- are ignored but 'i' for iso and 'n' for neo are considered.

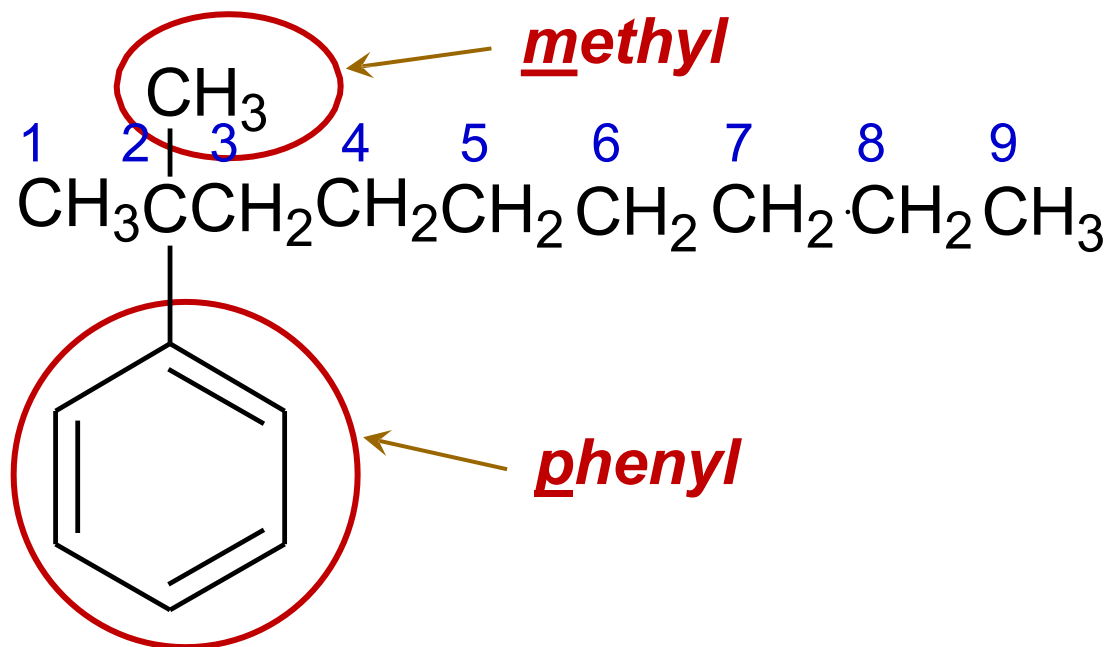
Example:



3-ethyl-4-isopropyl-2,5-dimethylheptane

6. When two substituents are present on the same carbon atom, **use that number twice**.

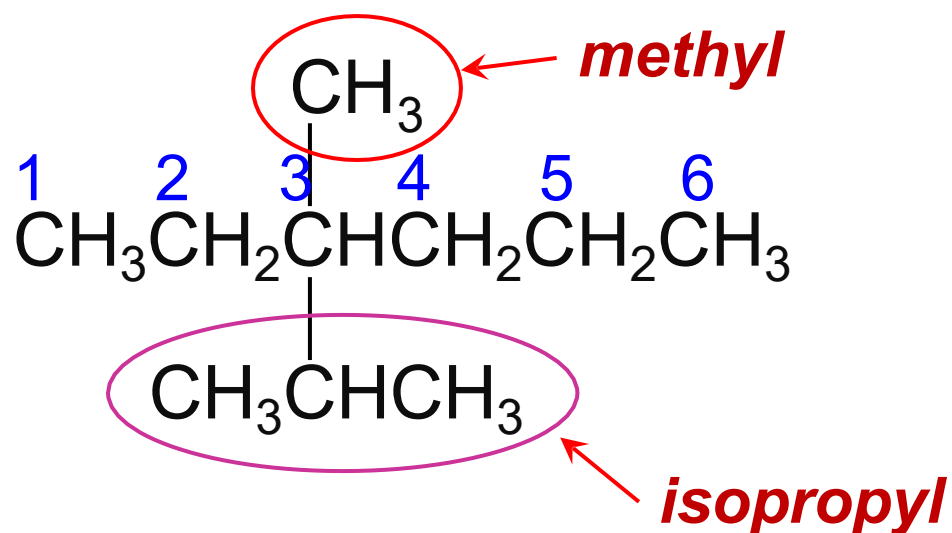
Example:



2-methyl-2-phenylnonane

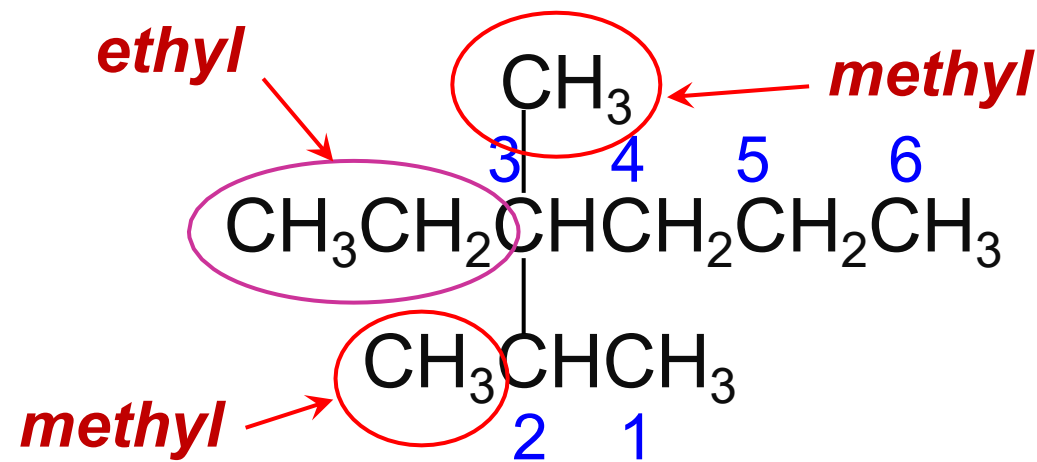
7. When two or more **carbon chains** are of equal length, choose the chain with **more substituents** as the parent

Example:



2 substituents

✗



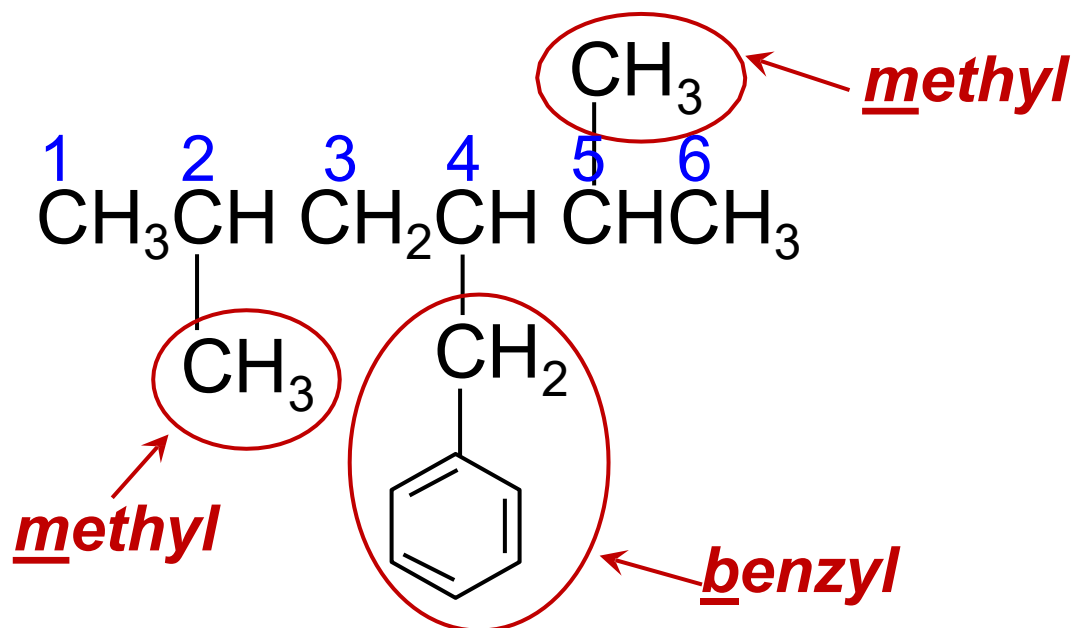
3 substituents

✓

3-ethyl-2,3-dimethylhexane

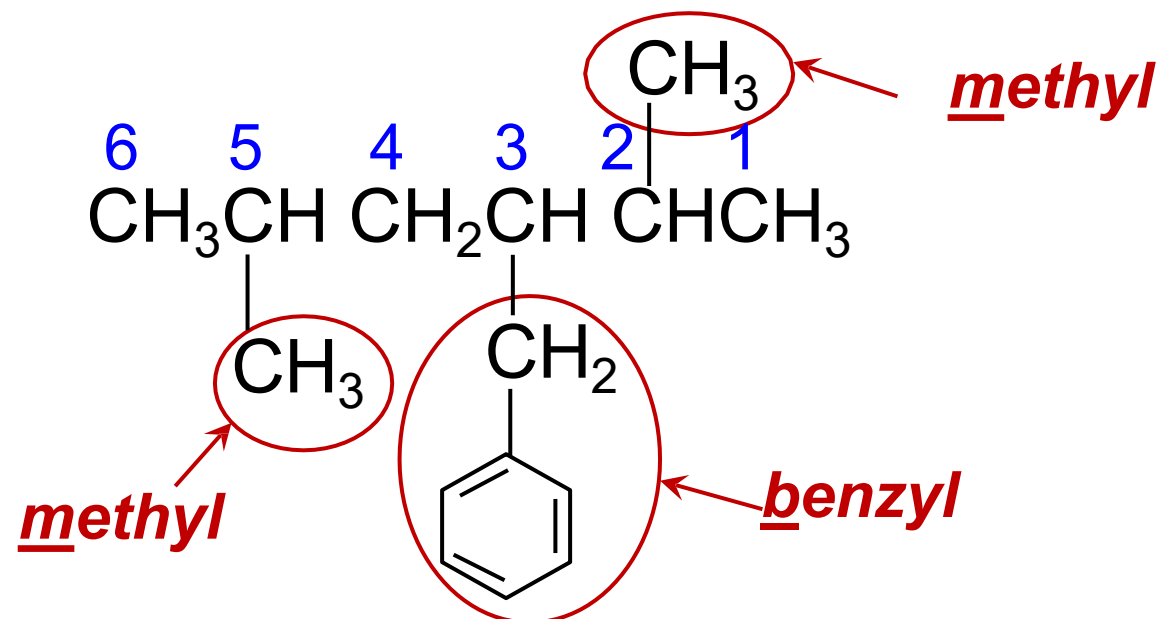
8. When branching first occurs at an equal distance from either end of the longest chain, choose the name that gives the lower number at the first point of difference.

Example:



4-benzyl-2,5-dimethylhexane

✗



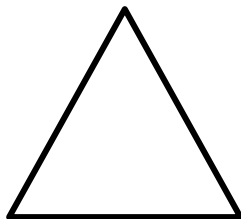
3-benzyl-2,5-dimethylhexane

✓

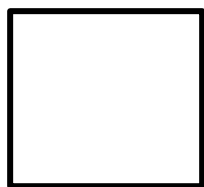
IUPAC NOMENCLATURE FOR CYCLOALKANES

21

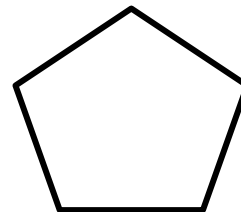
□ A prefix “**cyclo**” is added to the names of alkanes.



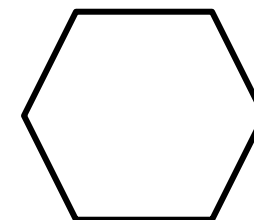
cyclopropane



cyclobutane

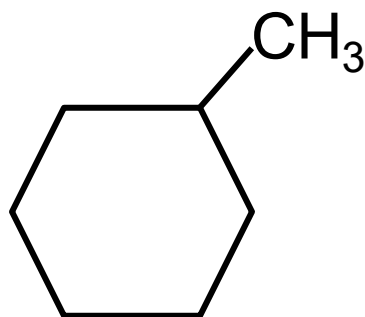


cyclopentane

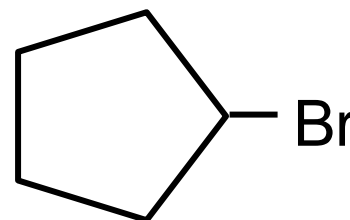


cyclohexane

□ If only one substituent present numbering is not needed.

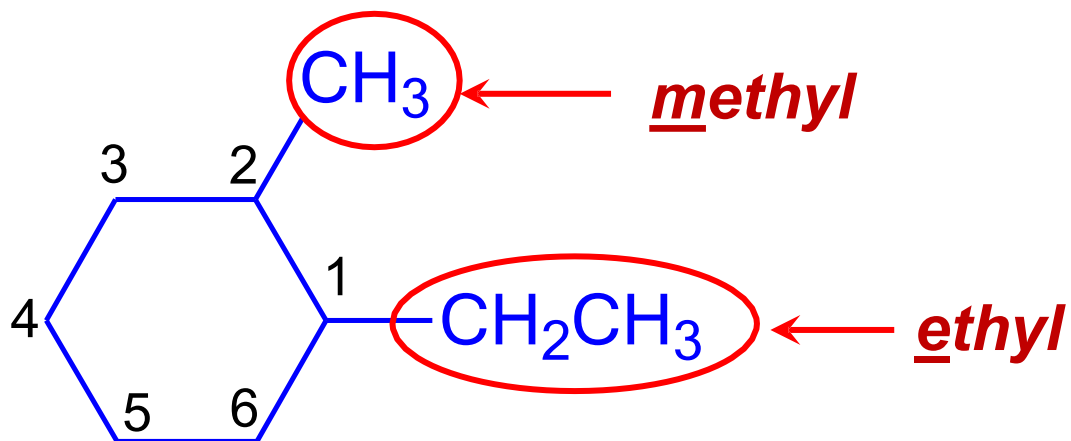


methylcyclohexane

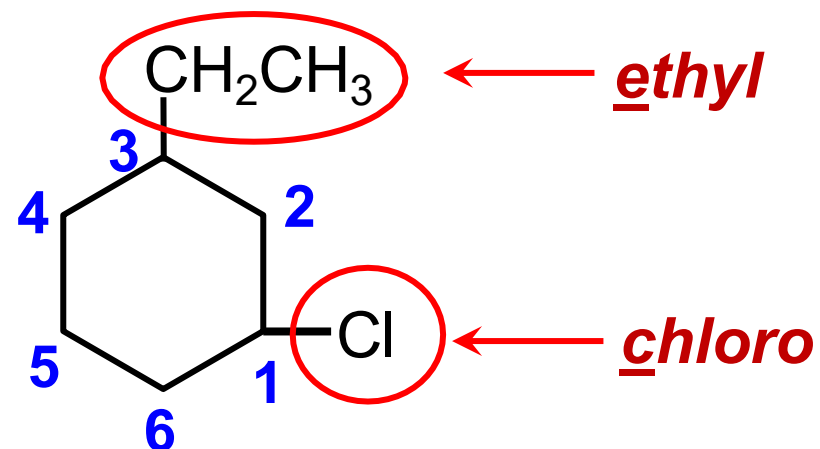


bromocyclopentane

- If 2 substituents present, number from the lower alphabet substituent to give lowest possible number to the next substituent (in clockwise or anticlockwise direction).

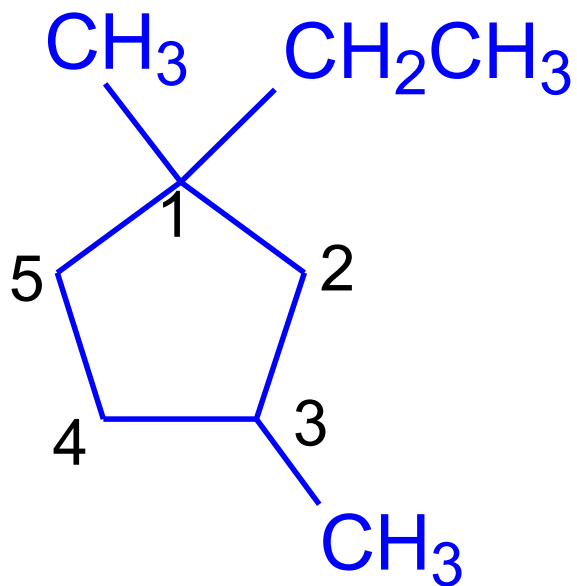


1-ethyl-2-methylcyclohexane

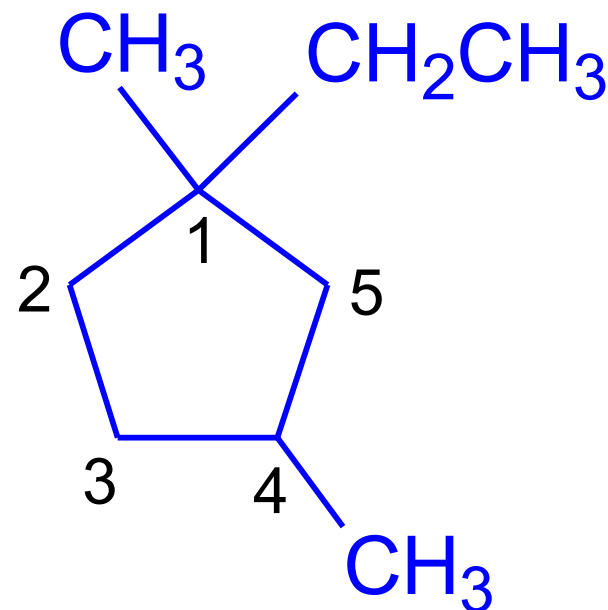


1-chloro-3-ethylcyclohexane

- When 3 or more substituents present, numbering begins from the C with substituent that leads to the lowest set of locants.



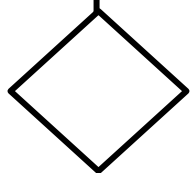
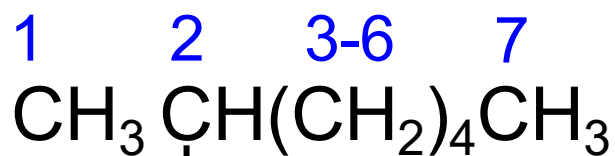
1-ethyl-1,3-dimethylcyclopentane



1-ethyl-1,4-dimethylcyclopentane



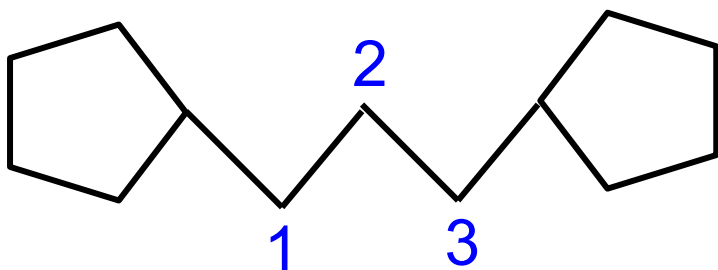
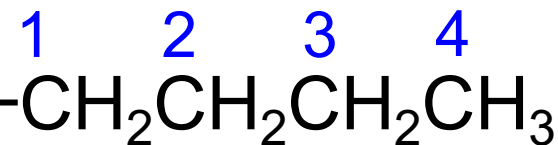
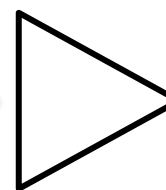
- If the number of C atoms in the open chain is greater than in the ring, or when more than one ring are attached to a single chain, the compound is named as cycloalkylalkane.



← *4 carbon ring*

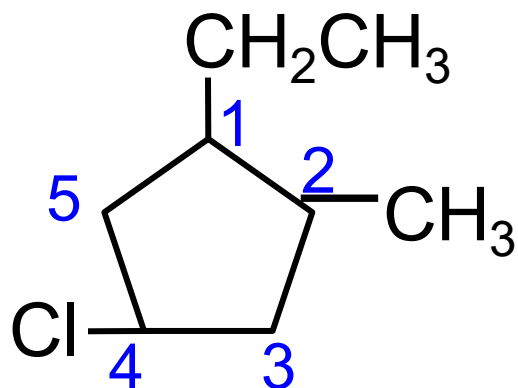
2-cyclobutylheptane

1-cyclopropylbutane



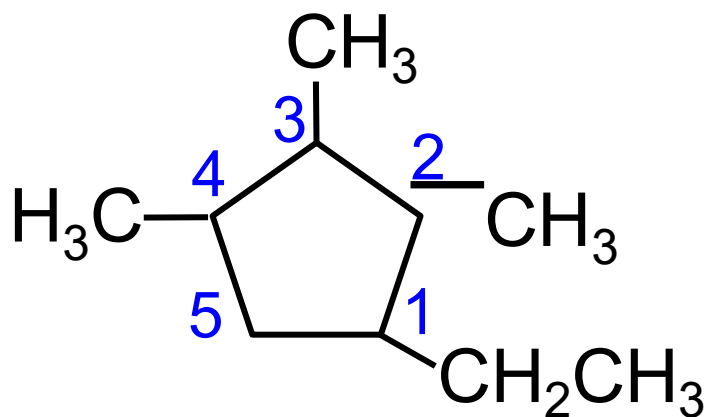
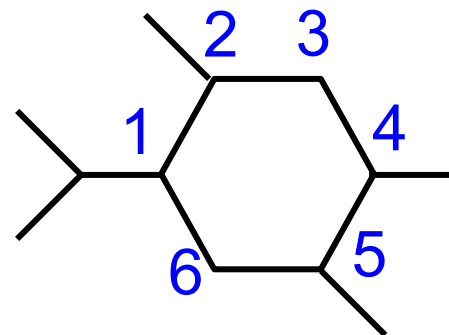
1,3-dicyclopentylpropane

Examples: Name the following compounds:

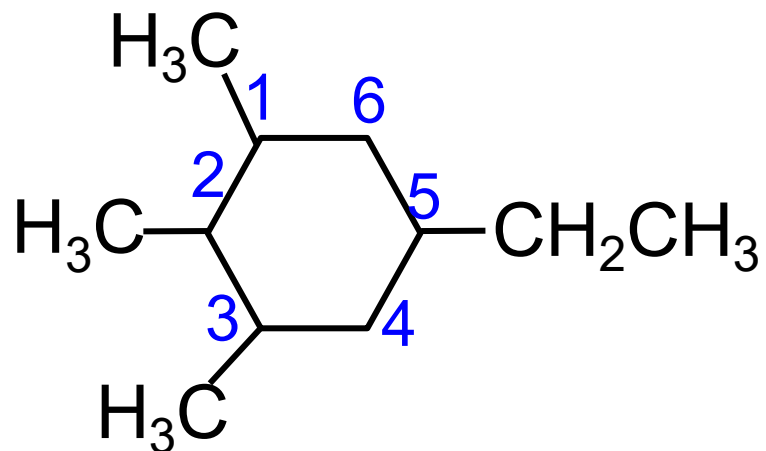


4-chloro-1-ethyl-2-methylcyclopentane

1-isopropyl-2,4,5-trimethylcyclohexane

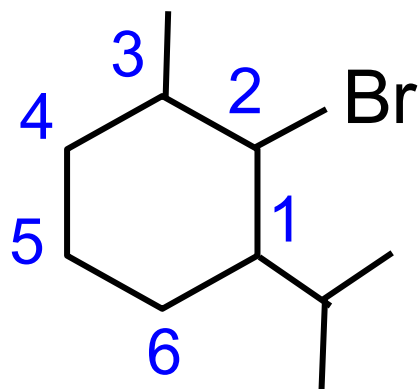
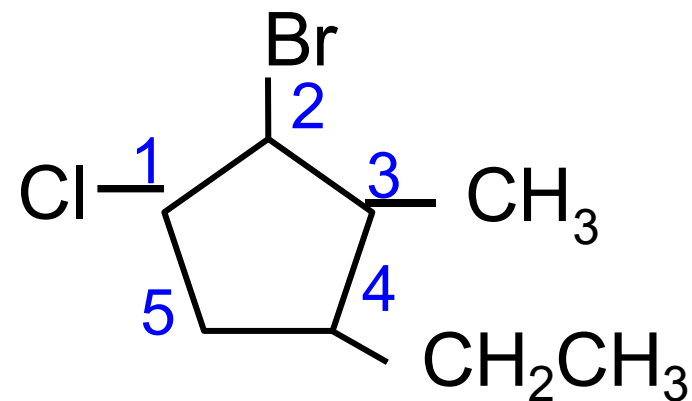


1-ethyl-2,3,4-trimethylcyclopentane



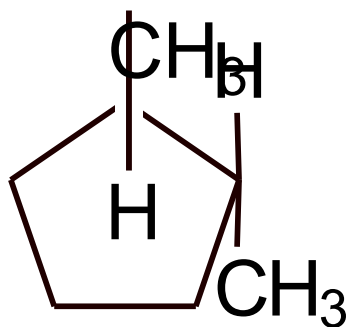
5-ethyl-1,2,3-trimethylcyclohexane

2-bromo-1-chloro-4-ethyl-3-methylcyclopentane

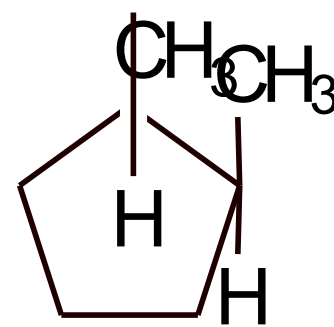


2-bromo-1-isopropyl-3-methylcyclohexane

- Di-substituted cycloalkanes are one class of molecules that exhibit stereoisomerism.
- Stereoisomers require an additional nomenclature prefix be added to the IUPAC name in order to indicate their spatial orientation.
- Di-substituted cycloalkane stereoisomers are designated by the nomenclature prefixes **cis** (Latin, meaning on this side) and **trans** (Latin, meaning across).



trans-1,2-dimethylcyclopentane



cis-1,2-dimethylcyclopentane

Physical Properties of Alkanes

Physical State

- At room temperature (25°C) and atmospheric pressure (1 atm), for unbranched alkanes,

★ $C_1 - C_4$: **gases**

★ $C_5 - C_{17}$: **liquids**

★ $> C_{17}$: **solids**

(i) Boiling Point

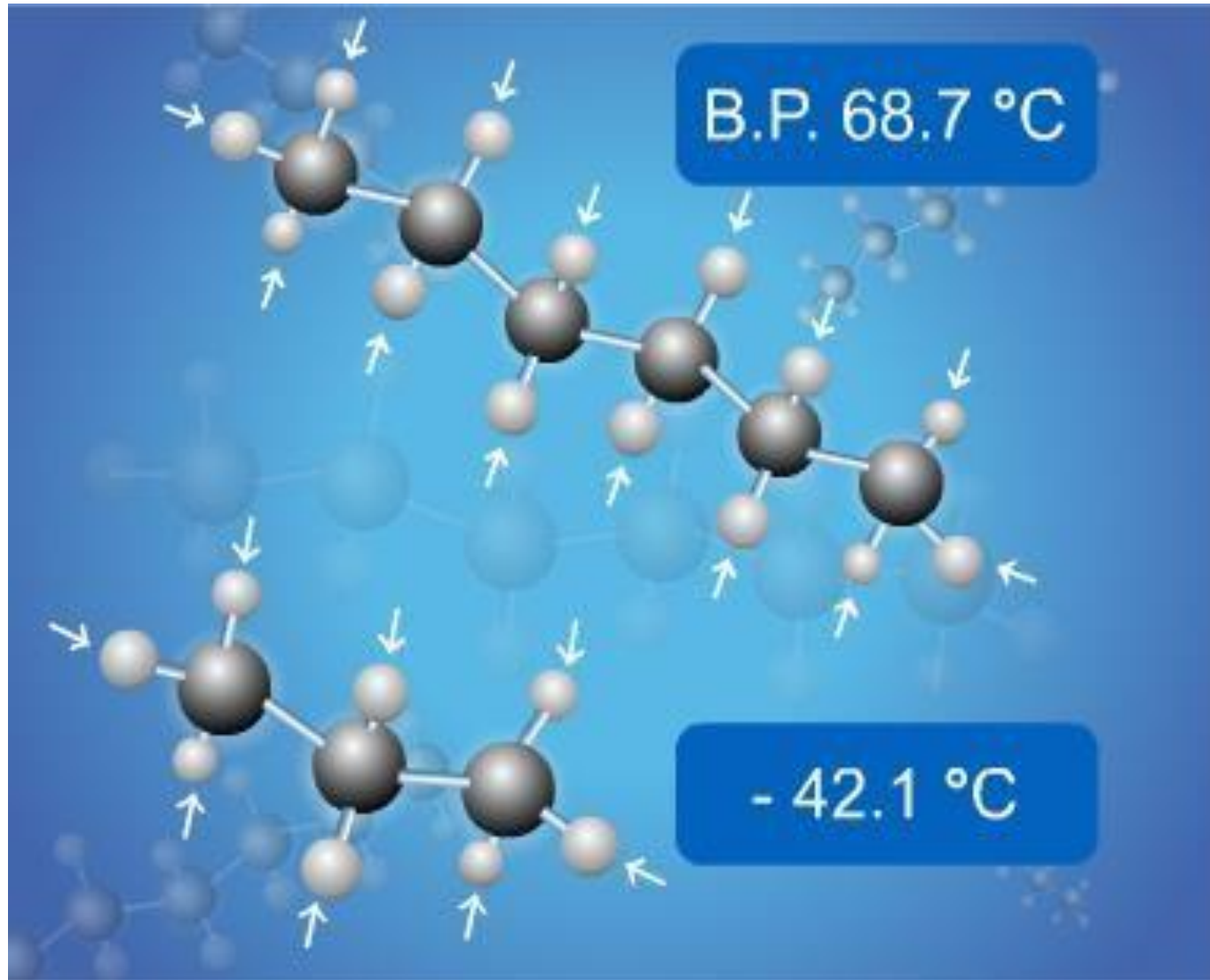
(a) Molecular Weight (size)

- Alkanes are **nonpolar** and thus exhibit only **weak van der Waals forces** (dispersion forces) between molecules.
- The boiling points of the straight chain alkanes show a **regular increase with increasing molecular weight (size)**.

Reason :

As the molecular size increases, the total **surface area increases**, the **strength of van der Waals forces** also increases.

Example



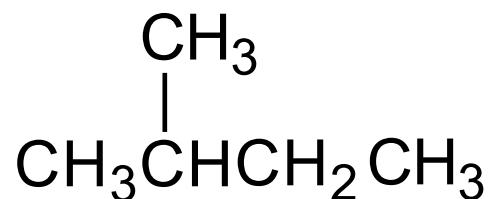
hexane

propane

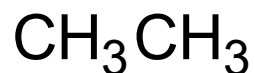
Example

Arrange the following alkanes in order of increasing boiling point.

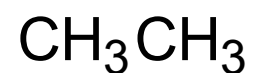
5C



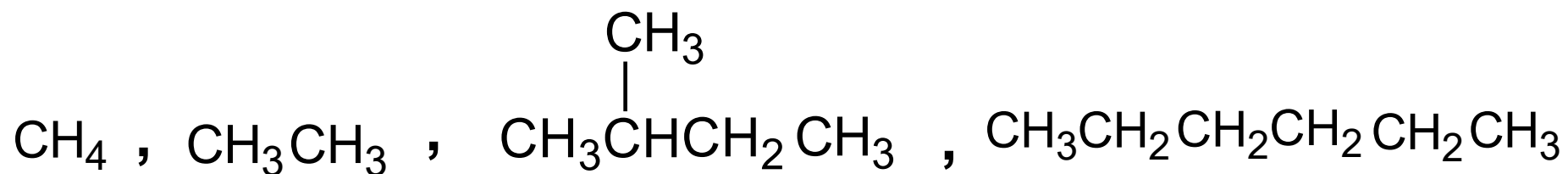
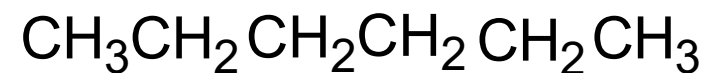
2C



1C



6C



increasing boiling point



(b) Isomeric alkanes

- The boiling point of isomers decreases with branching .

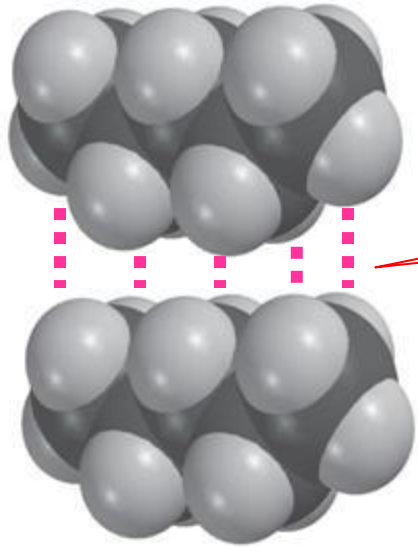
Branching ↑ **Boiling point** ↓

Reason :

For molecules with the same molecular weight (size), branching decreases surface area and strength of van der Waals forces, therefore molecules with more branches will have lower boiling points.

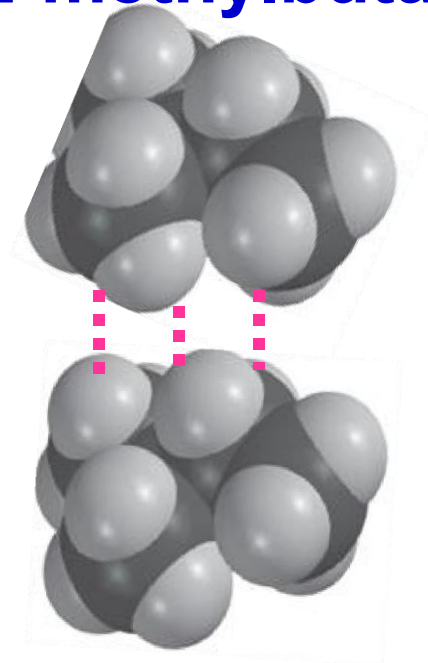
pentane

bp = 36.3 °C



More points for van der Waals forces to act.

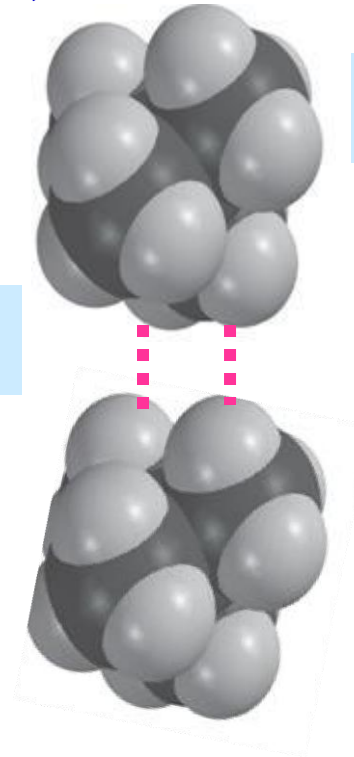
2-methylbutane



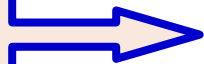
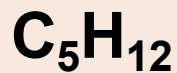
bp = 27.9 °C

2,2-dimethylpropane

bp = 9.5 °C



Example

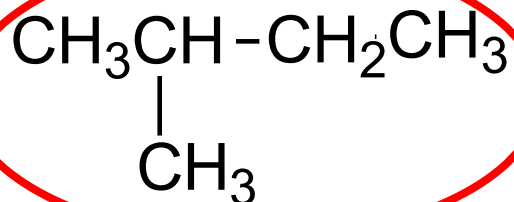


Three isomeric alkanes



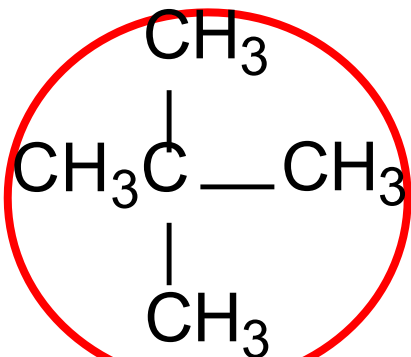
pentane

Boiling point = 36.3 °C



2-methylbutane

Boiling point = 27.9 °C



2,2-dimethylpropane

Boiling point = 9.5 °C

(iii) Cycloalkanes

- The boiling points of cycloalkanes are 10°C to 15°C higher than the corresponding straight chain alkanes.
- The boiling points of cycloalkanes are higher than the corresponding straight chain alkanes due to its larger surface area.

Cycloalkane	Boiling point (°C)	Alkane	Boiling point (°C)
Cyclobutane	13	Butane	-0.5
Cyclopentane	49	Pentane	36. 3

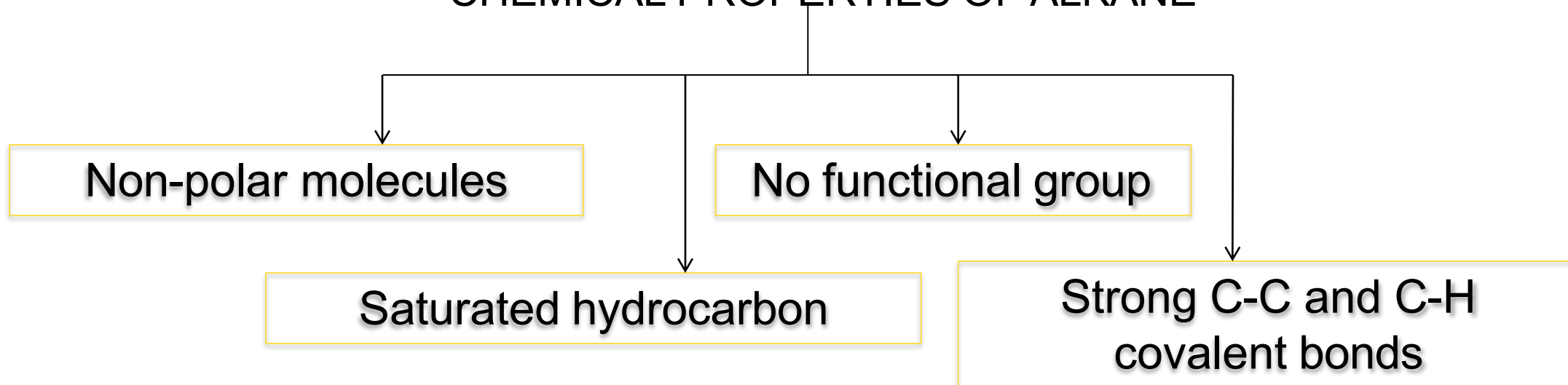
B. Solubility

- Alkanes are **insoluble in water** because they are **nonpolar molecules** that cannot form **hydrogen bonds** with water molecule.
- Liquid alkanes and cycloalkanes are soluble in one another, and they generally **dissolve in non-polar solvents** (e.g. benzene and other hydrocarbon) because they can form **intermolecular van der Waals forces** with the solvent molecules.

ALKANES – CHEMICAL PROPERTIES

37

CHEMICAL PROPERTIES OF ALKANE



Chemical properties of alkanes give reasons why alkanes are non-reactive organic compound

➤ Alkanes DO NOT undergo addition reaction

➤ DO NOT REACT with acids, bases, oxidising and reducing agent, under normal T and P.

However, alkane may undergo FREE RADICAL SUBSTITUTION REACTION UNDER THE RIGHT CONDITION.

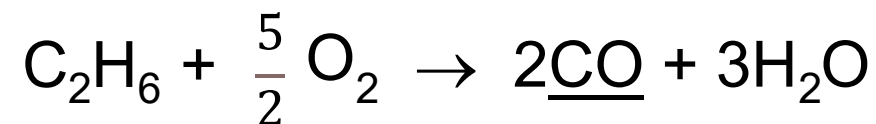
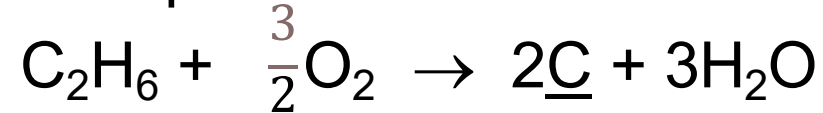
Chemical reactions of alkanes

Combustion

In limited oxygen

- Forms **carbon monoxide @ carbon** and **water**.

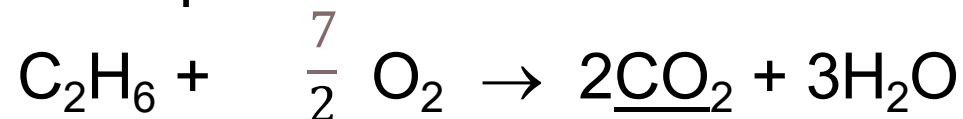
- Example:



In excess oxygen

- Forms **carbon dioxide** and **water**.

- Example:



Halogenation

- Reaction of alkanes with halogens.
- Alkanes react with halogen (Cl_2 , Br_2) to produce haloalkanes in the presence of light or temperature greater than 100°C.
- Example: $\text{CH}_4 + \text{Br}_2 \xrightarrow{\text{uv}} \text{CH}_3\text{Br} + \text{HBr}$
- Type of reaction : **free radical substitution reaction**.

B. Halogenation of Alkanes

- Alkanes react with **halogen** (chlorine and bromine) to produce **haloalkanes** in the presence of **light** or **uv**.



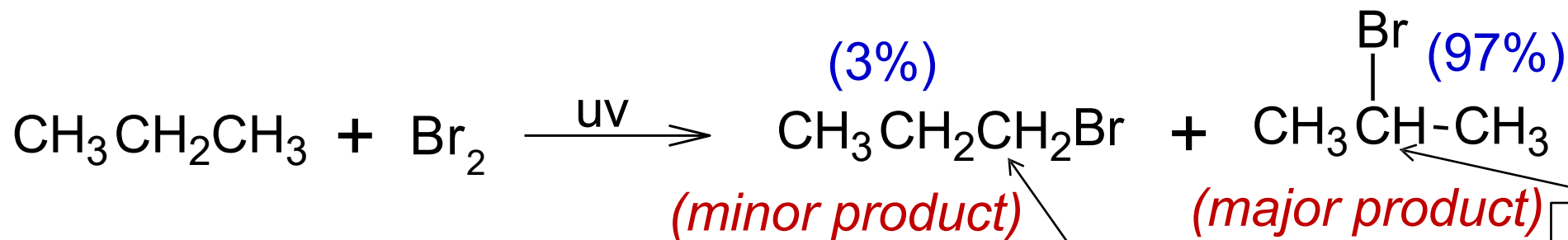
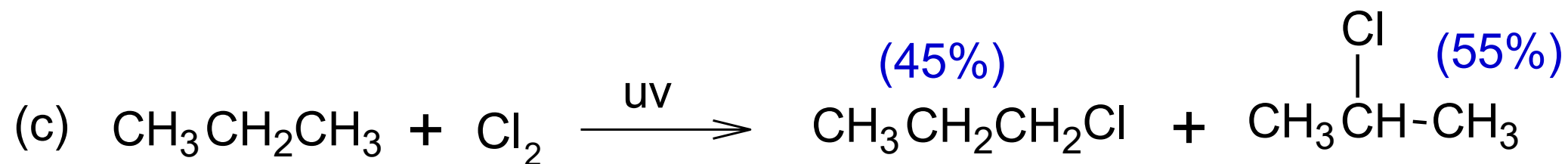
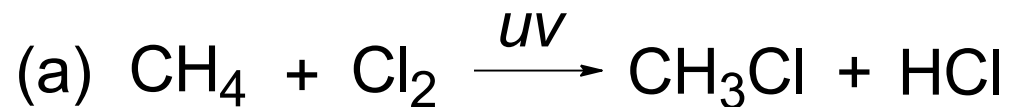
- Function of uv or sunlight : to **provide energy** for the **homolytic cleavage** of the **bond between halogen atoms** (X-X bond).
- Type of reaction : **Free Radical Substitution**
- In chlorination of alkane, **HCl** formed can be observed as **white fume**.
- Reactivity of alkanes towards halogenation : $\text{F} > \text{Cl} > \text{Br} > \text{I}$


too explosive


too slow

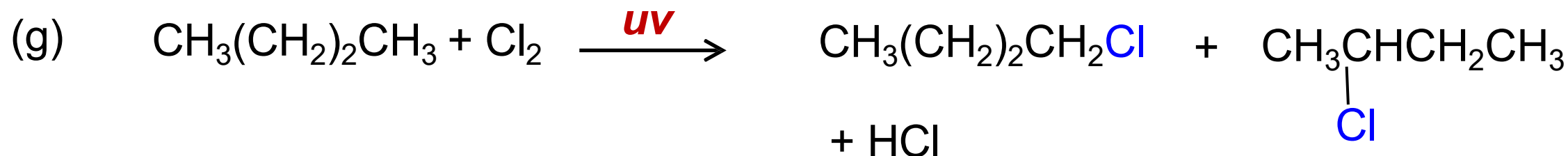
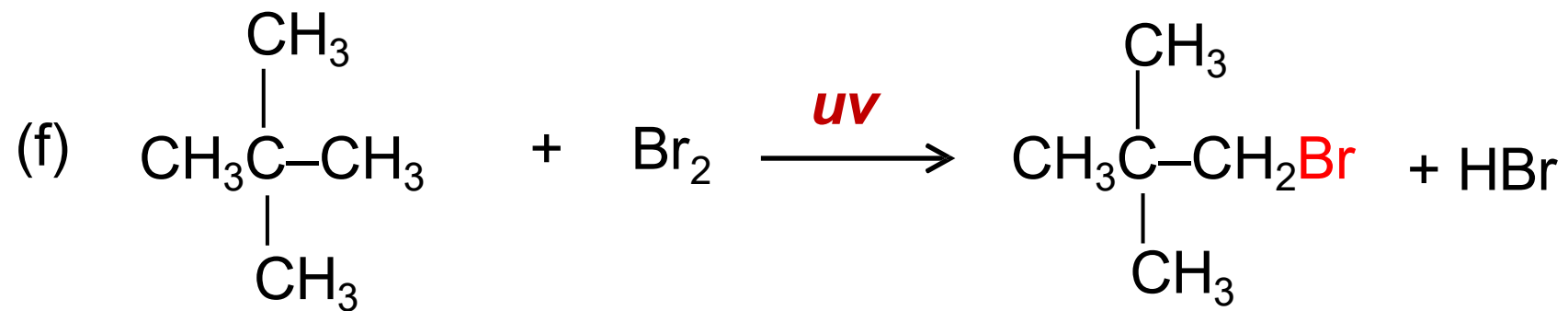
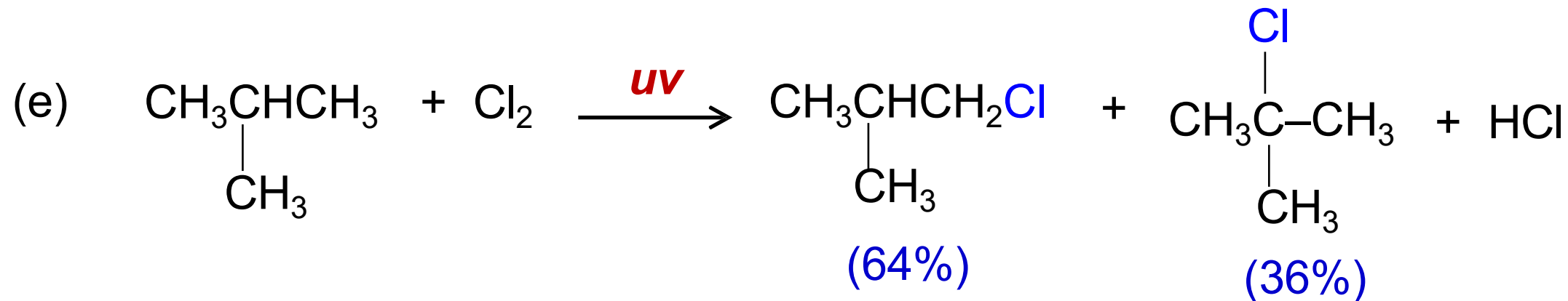
EXAMPLES

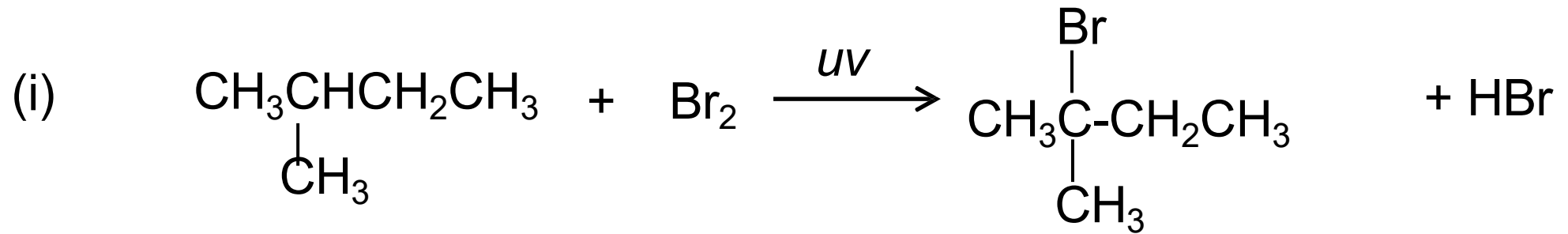
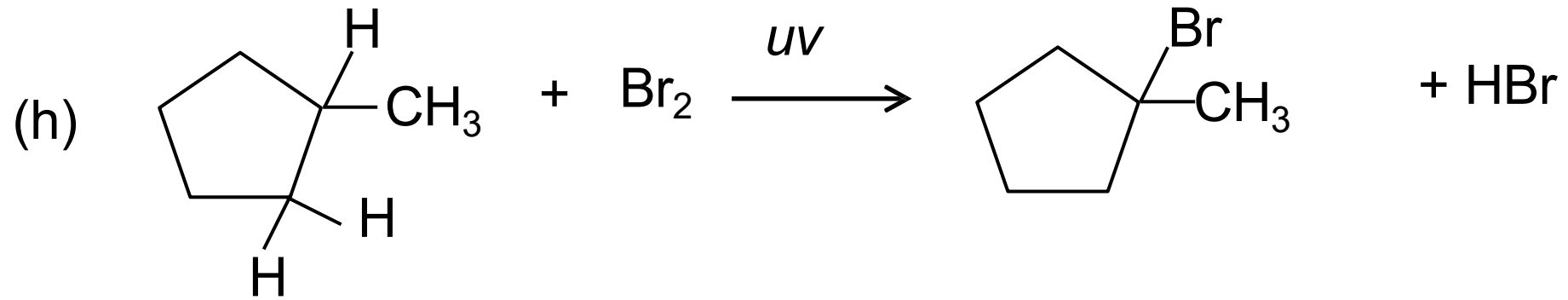
42



Comes from substitution of
*1° H forms 1° carbon free
radical which is unstable*

Comes from
substitution of *2°*
*H forms 2° carbon
free radical which
is stable*





- Bromination of alkanes occurs by a similar mechanism, but is **slower** and more selective.
- Bromination is more selective than chlorination because the major reaction is favored by a larger amount.
- Chlorination (unselective) is faster than bromination (selective)

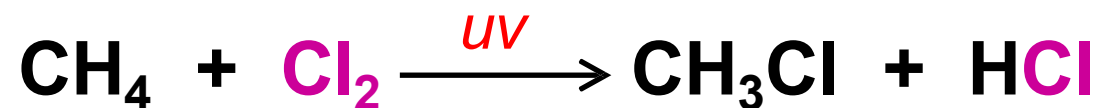


Ability to discriminate among the different types of hydrogen atoms

- Halogenation of alkanes occurs via free-radical substitution.
- With methane, the reaction produces a mixture of **halomethane** and a hydrogen halide.

For monosubstitution of methane

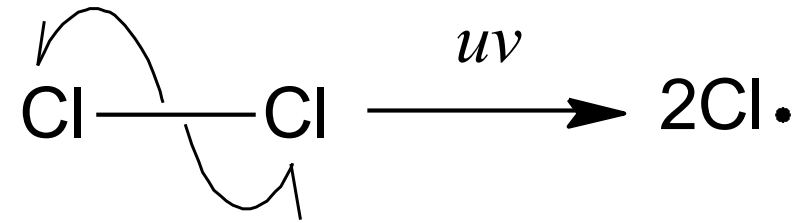
(1) Chlorination of propane



Note :

When asked to draw the products of halogenation of an alkane, draw the products of monosubstitution only, unless specifically directed to do otherwise.

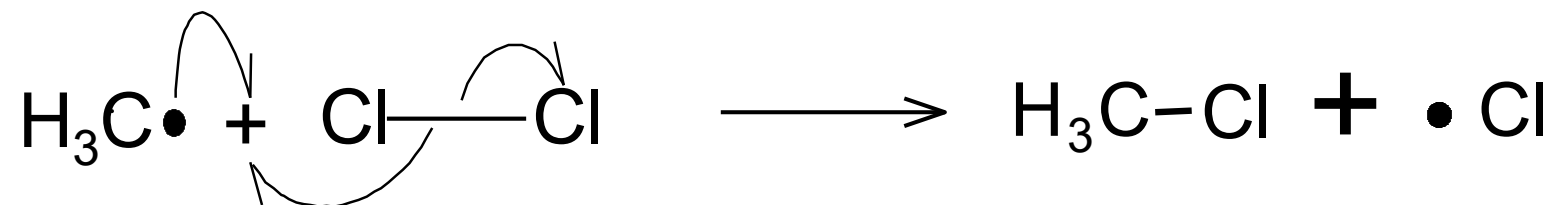
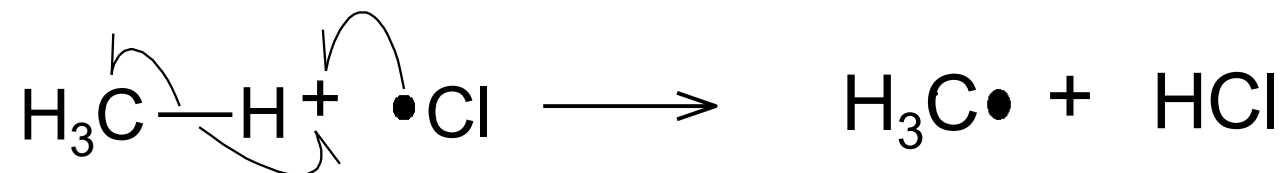
1. Chain Initiation Step



- ▶▶ Fragmentation of halogen molecule by light @ heat, creating new free radicals.
- ▶▶ uv light provide energy for the homolytic cleavage of halogen molecules.

2. Propagation Step

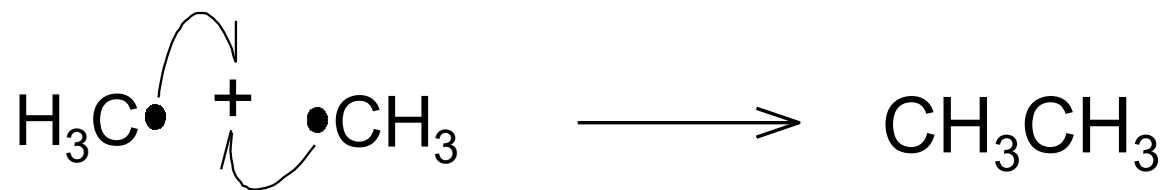
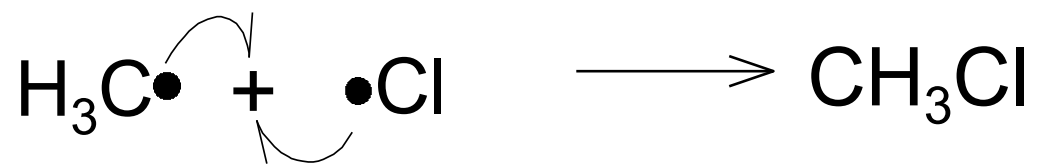
- ▶▶ Each propagation step **consumes** one free radical but **generates** another one.
- ▶▶ Usually combine a free radical and a reactant to form a product and another free radical.

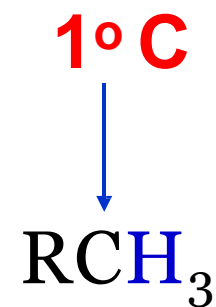
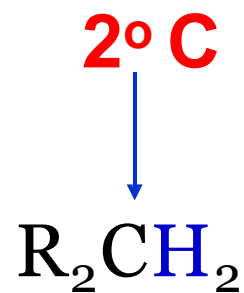
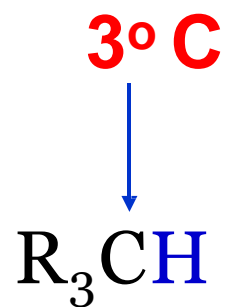


Note : Almost all of the product is formed by repetitive cycles of the two propagation steps.

3. Termination Step

►► Free radicals are **combined** and thus the chain reaction is terminated.





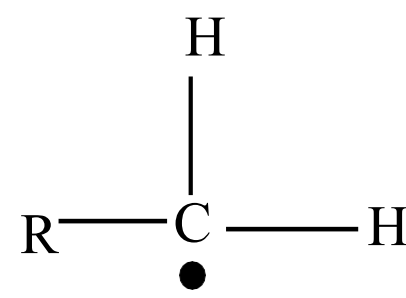
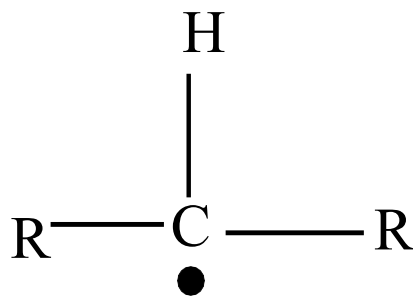
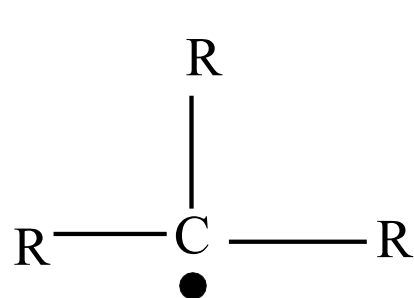
Type of H atom

3°

2°

1°

Free
radical
Formed



→
decreasing stability

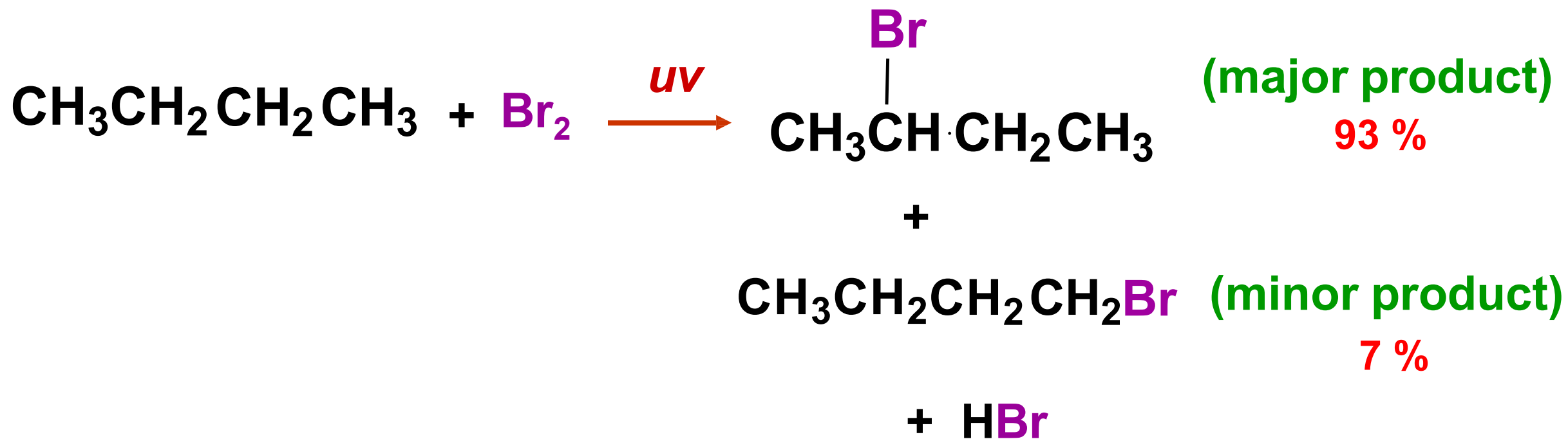
3° free radical

2° free radical

1° free radical

(2) Bromination of butane

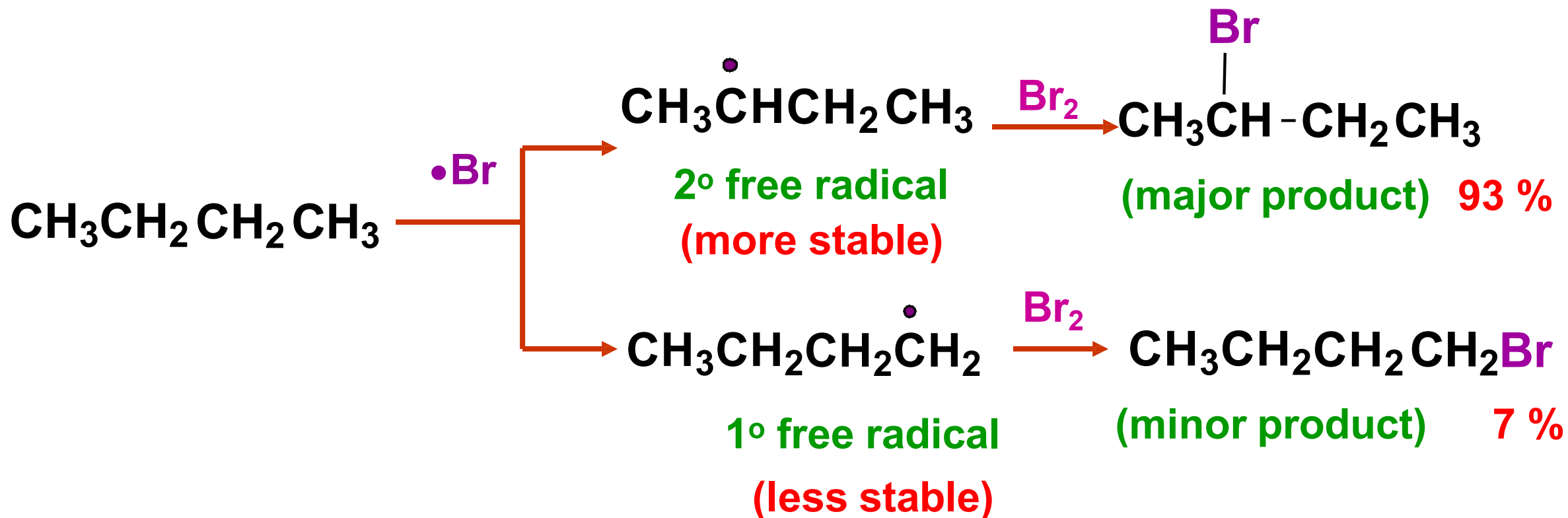
- Halogenation of propane forms a mixture of 2 constitutional isomers of monosubstituted alkane.



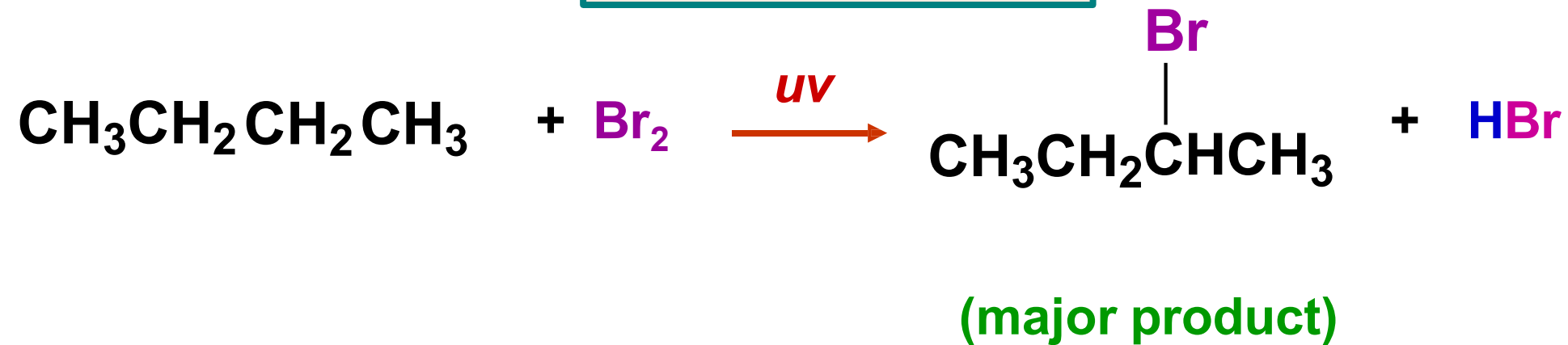
- The major product is 2-bromopropane because it is formed from 2° free radical whereas 1-bromopropane is formed from 1° free radical.

Propagation step

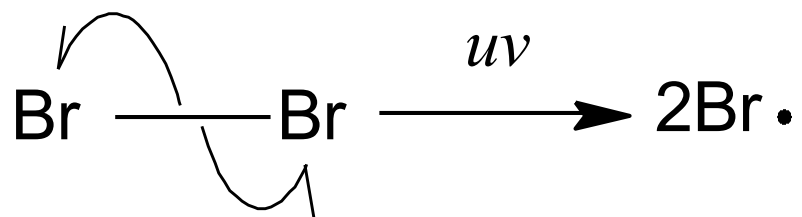
2 possible types
of free radical



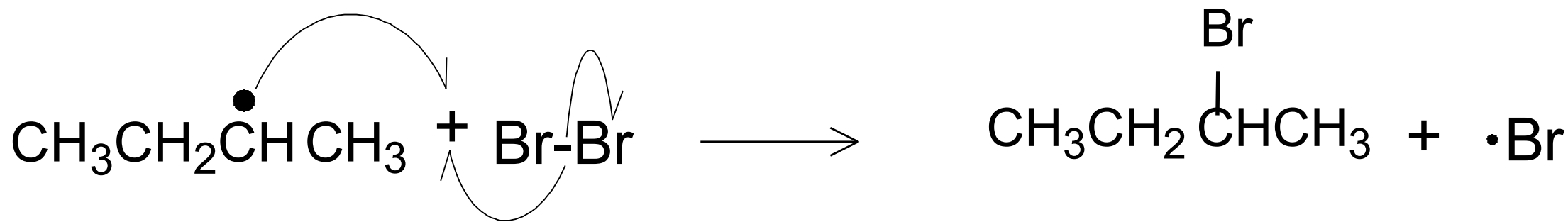
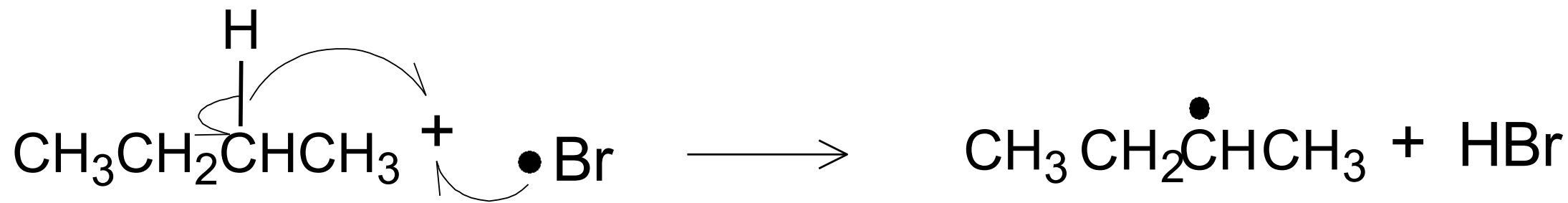
MECHANISM



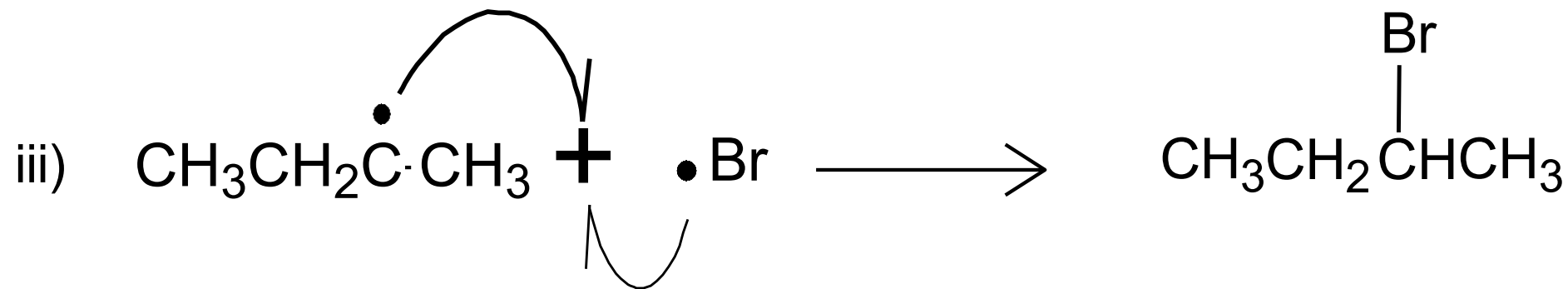
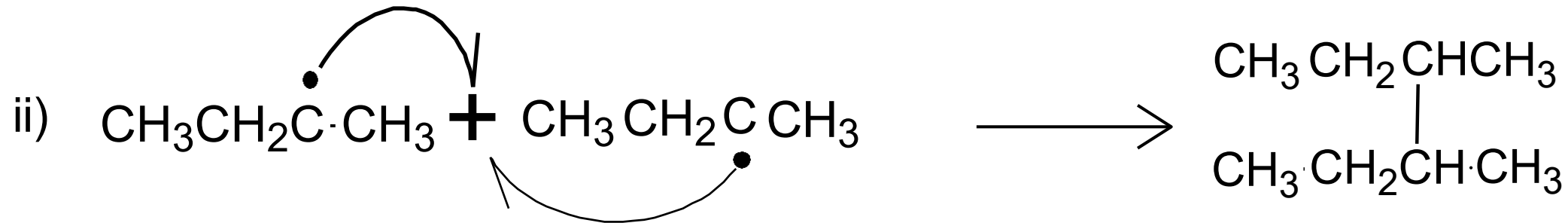
1. Chain Initiation step



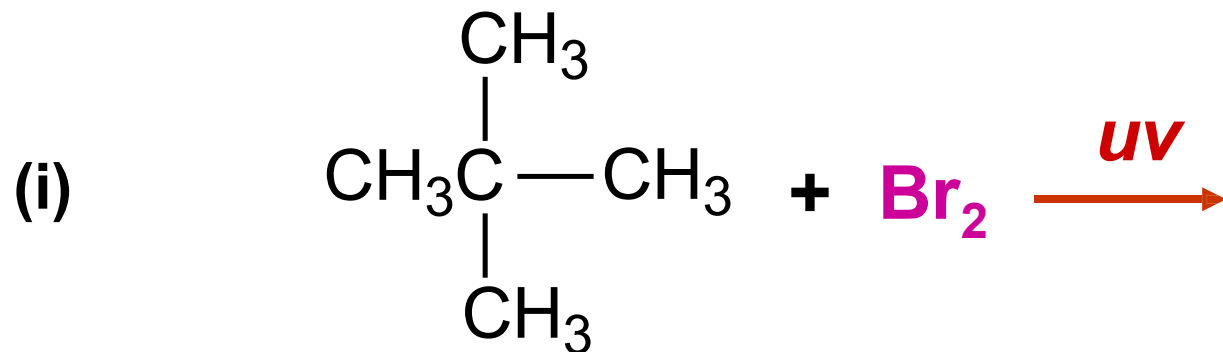
2. Propagation step



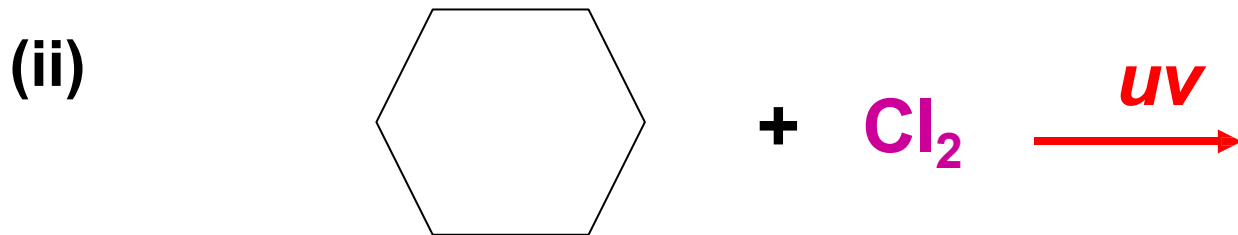
3. Termination step



Monosubstitution of alkanes containing equivalent type of H atoms

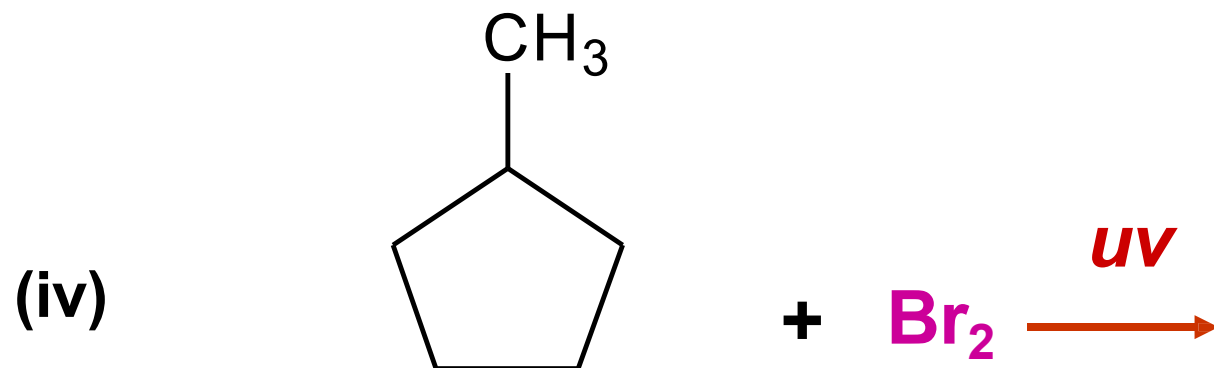
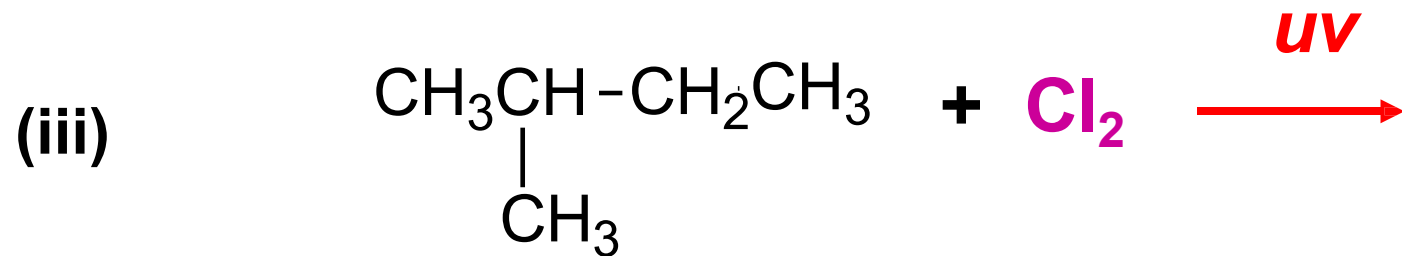


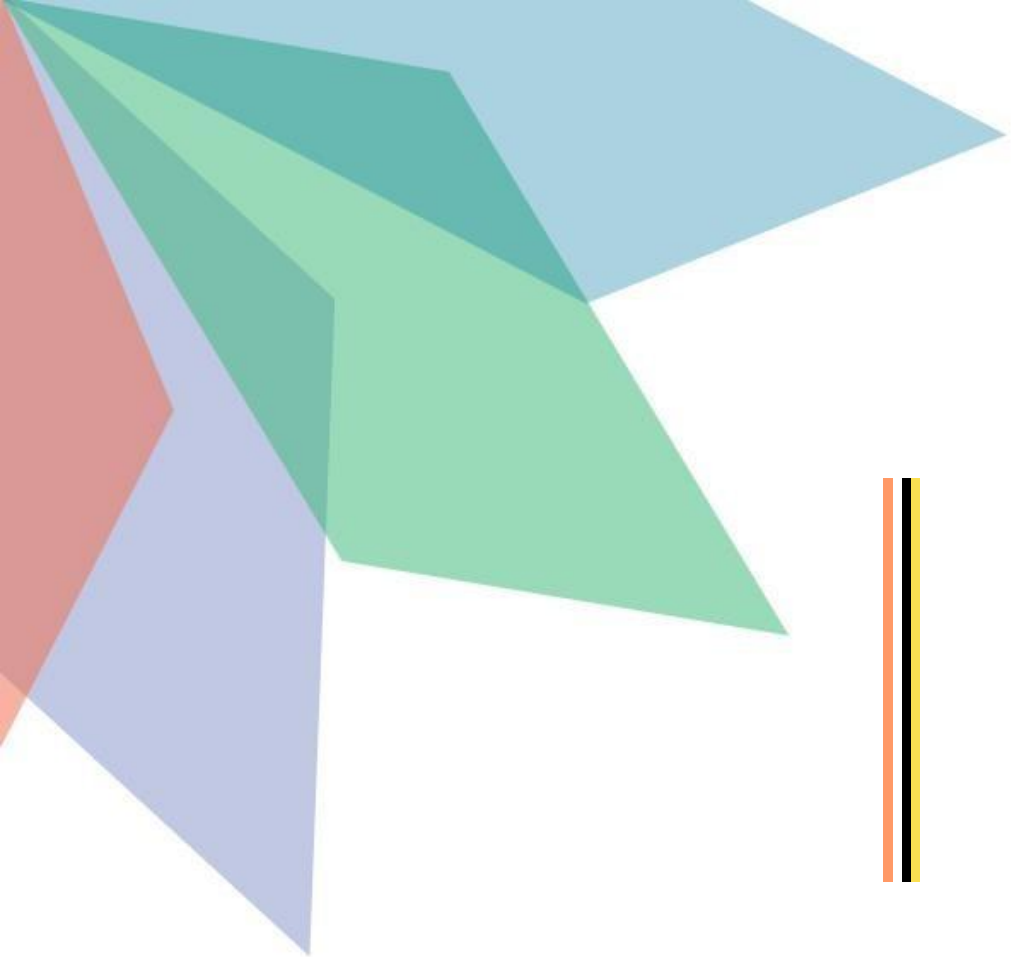
2,2 -dimethylpropane



cyclohexane

Example Monosubstitution of alkanes containing non-equivalent type of H atoms





5.2 ALKENES

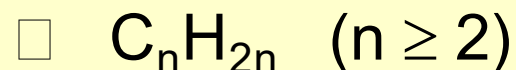
INTRODUCTION

ALKENES

① unsaturated hydrocarbons

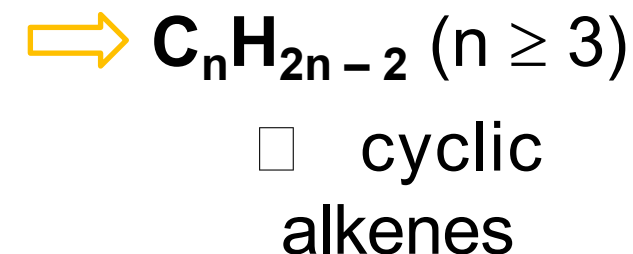
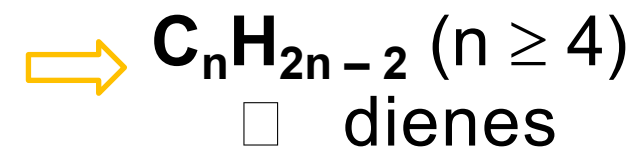
② functional group, C=C.

③ General formula for open chain alkene with one double bond:



④ may exhibit cis-trans isomerism if each of the carbon is attached to two different groups.

⑤ The number of H atoms are reduced by two for each additional C=C double bond in dienes or cyclic alkenes with one C=C.



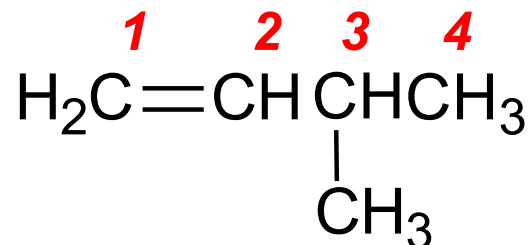
IUPAC Nomenclature

Examples:

① Choose the **longest continuous carbon chain** that contains the **C=C** as the **parent** and the name ends with the suffix **–ene**.

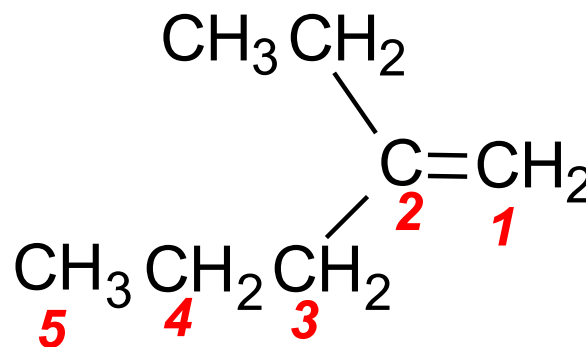
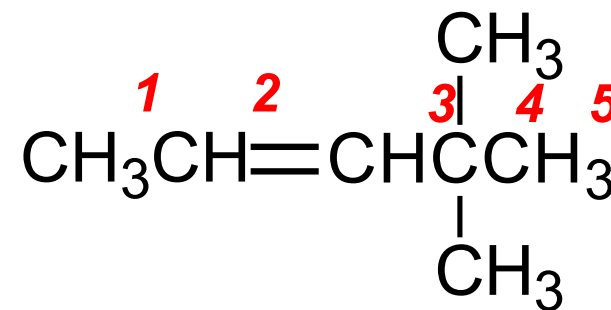
② For alkenes with **4 carbon onwards**, the location of the double bond must be shown.

③ Numbering starts from the end that gives the **lowest number to the double bond**.



3-methyl-1-butene

4,4-dimethyl-2-pentene

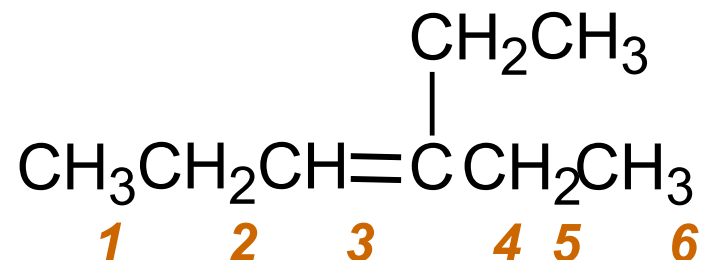


2-ethyl-1-pentene

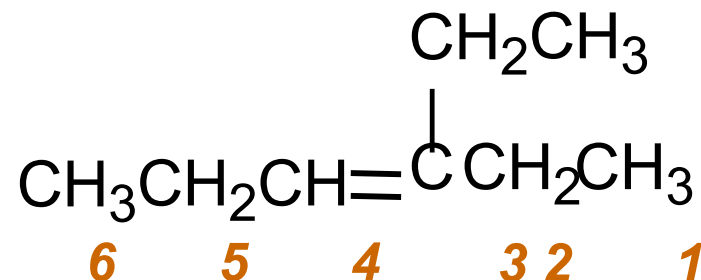
IUPAC Nomenclature

- ④ If the double bond gets the same number from either end, numbering starts from the end that gives lower number to the first encountered substituent.
- ⑤ If two double bonds are present, diene is used in replace of –ene.

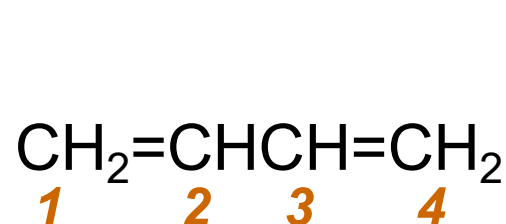
Examples:



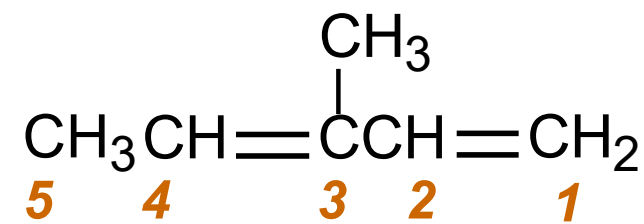
4-ethyl-3-hexene ✗



3-ethyl-3-hexene ✓



1,3-butadiene

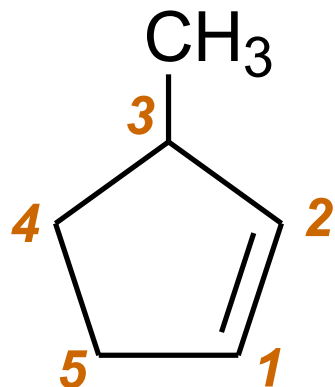


3-methyl-1,3-pentadiene

IUPAC Nomenclature

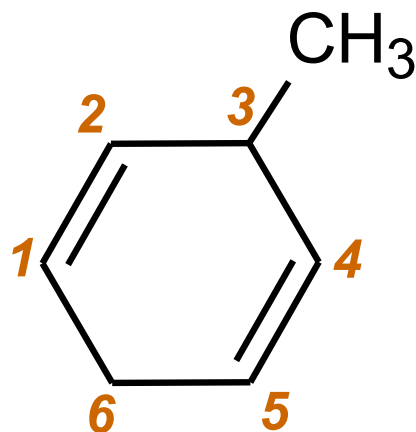
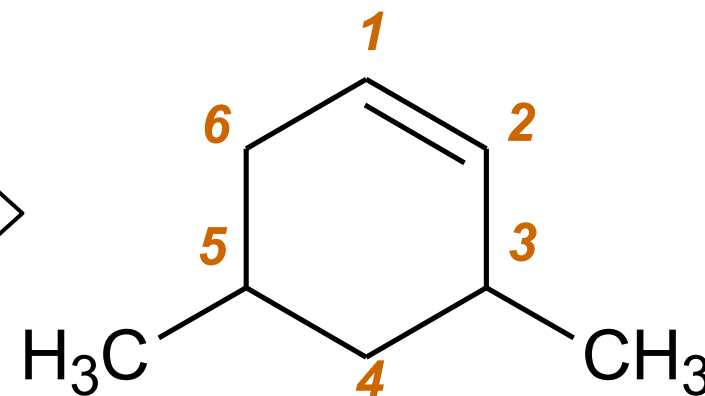
- ⑥ For cycloalkenes, number the doubly bonded carbon atoms with 1 and 2 in the direction that gives a small number for the first encountered substituent.

Examples:



3-methylcyclopentene

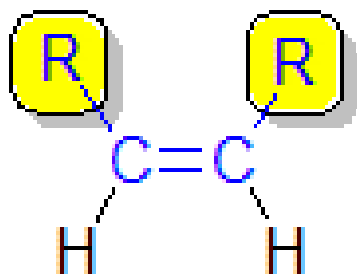
3,5-dimethylcyclohexene



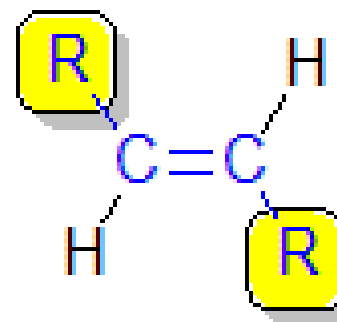
3-methyl-1,4-cyclohexadiene

1,2-disubstituted alkenes are described as:

- *cis*- if the two alkyl groups, R-, are on the same side of the C=C
- *trans*- if the two alkyl groups, R-, are on opposite sides of the C=C.
- these terms are inserted into the name as prefixes.

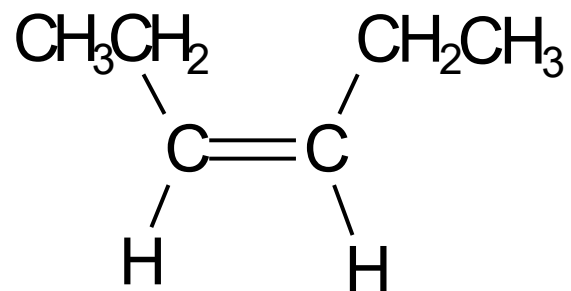


cis-isomer

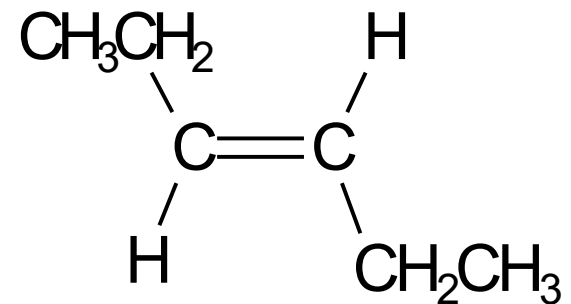


trans-isomer

Examples:



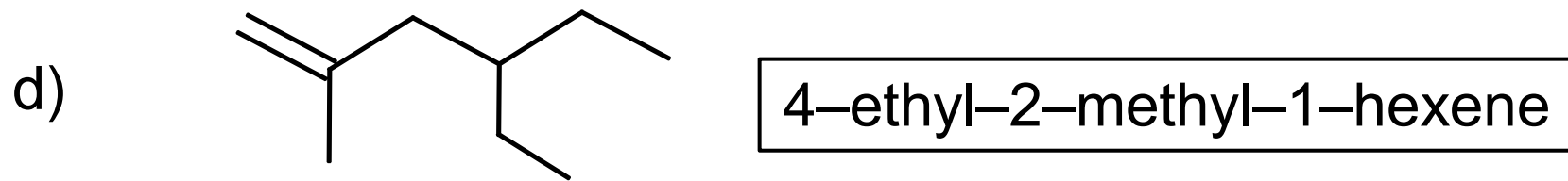
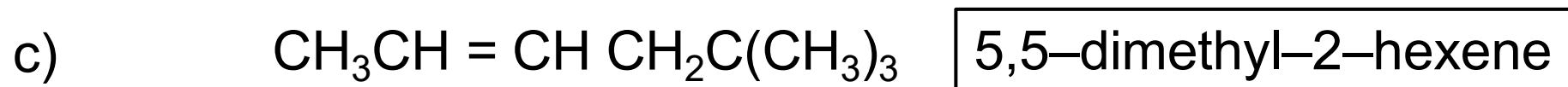
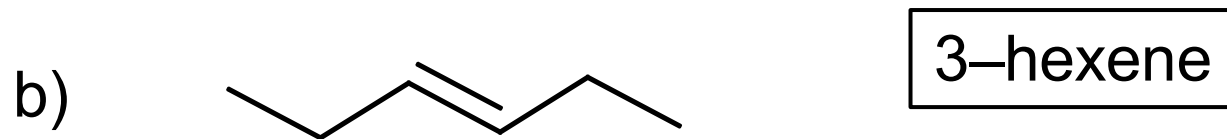
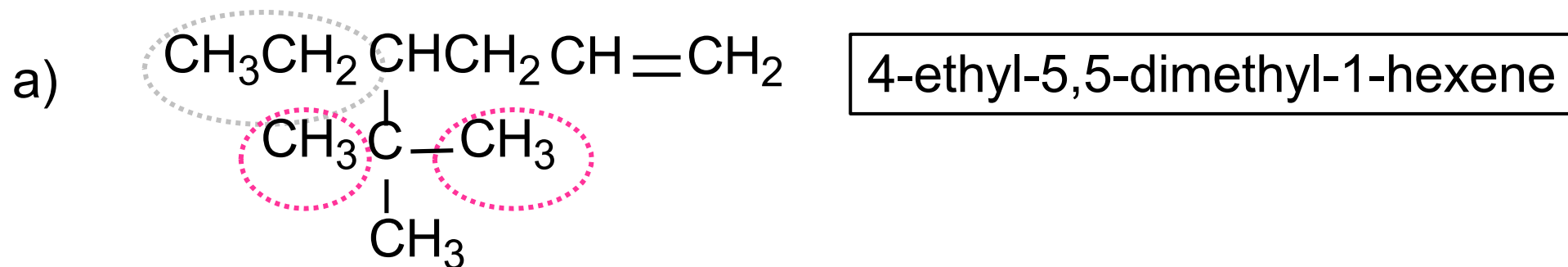
cis-3-hexene

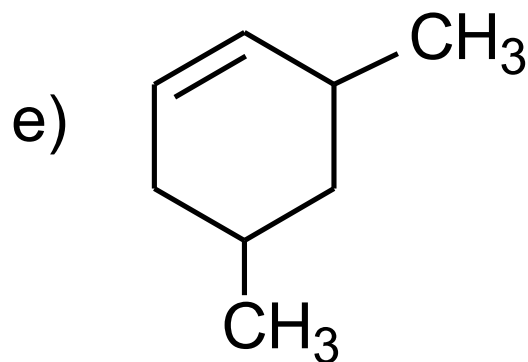


trans-3-hexene

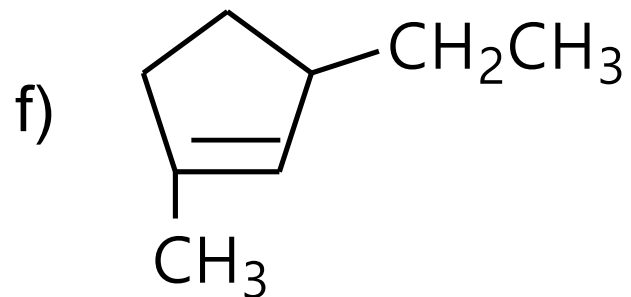
Exercise:

Name the following alkenes:

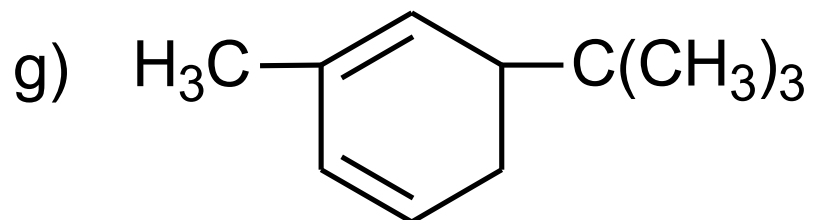




3,5-dimethylcyclohexene



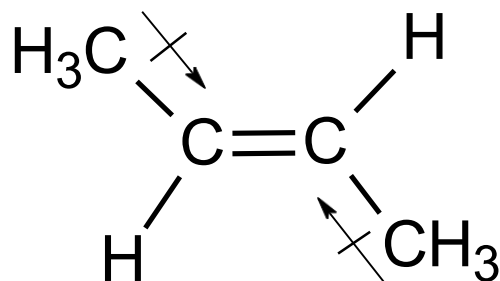
3-ethyl-1-methylcyclopentene



6-tert-butyl-2-methyl-1,3-cyclohexadiene

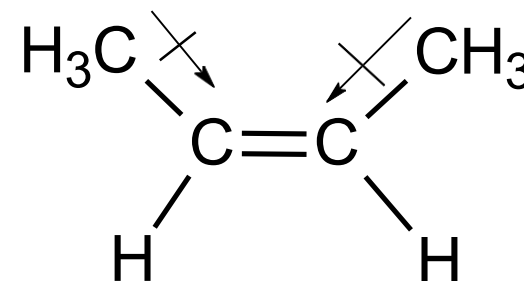
Boiling Points of Isomeric Alkenes

- For **cis-trans** isomers of the same alkene, boiling point of cis-isomer is higher than trans-isomer because cis-isomer is polar whereas trans-isomer not polar.



no net dipole moment, $\mu = 0$

trans-2-butene
(boiling point = 0.9°C)



has net dipole moment, $\mu \neq 0$

cis-2-butene
(boiling point = 3.7°C)

SYNTHESIS (PREPARATION) OF ALKENES

① Dehydration
of alcohols

Reagent:

conc. H_2SO_4 , Δ
or conc. H_3PO_4 , Δ

Alcohol $\rightarrow$ Alkene

② Dehydrohalogenation
of haloalkanes

Reagent:

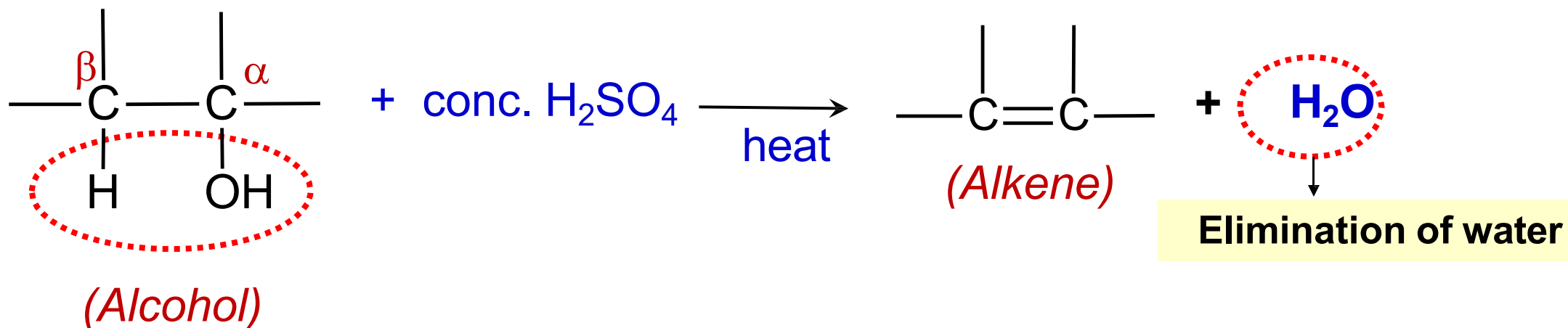
KOH , alcohol, Δ

Haloalkane $\rightarrow$ Alkene

Type of reaction: **Elimination**

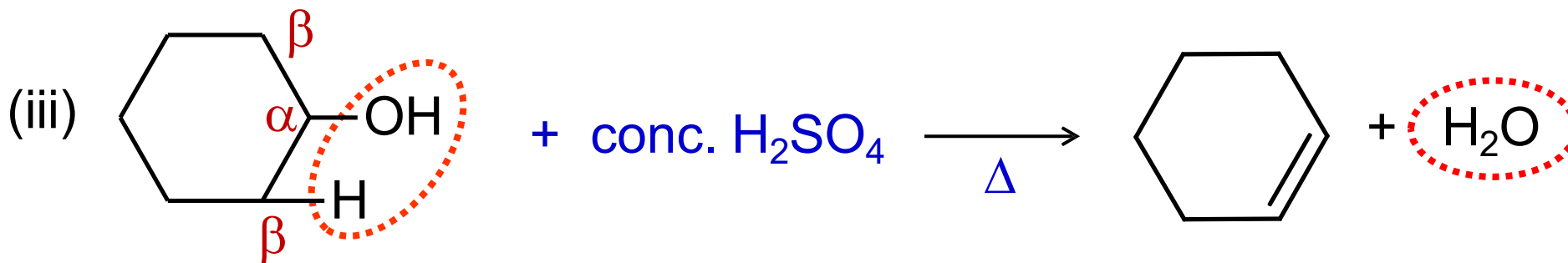
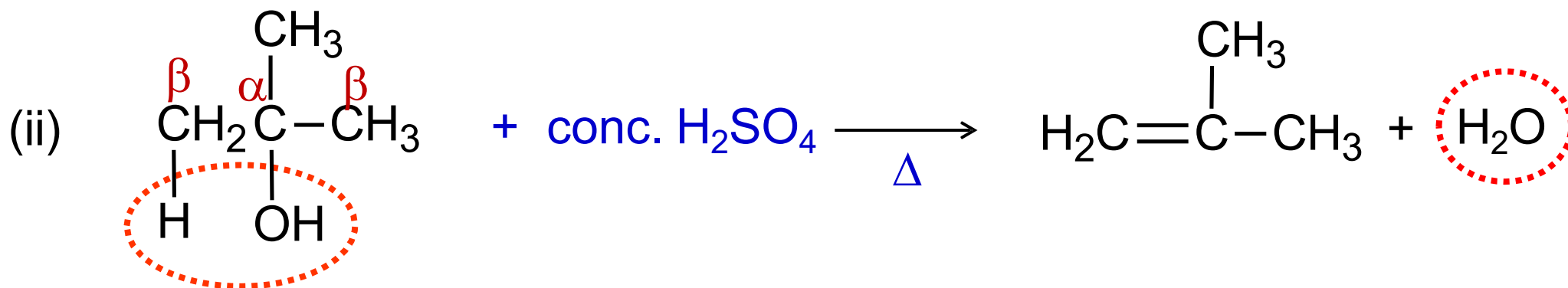
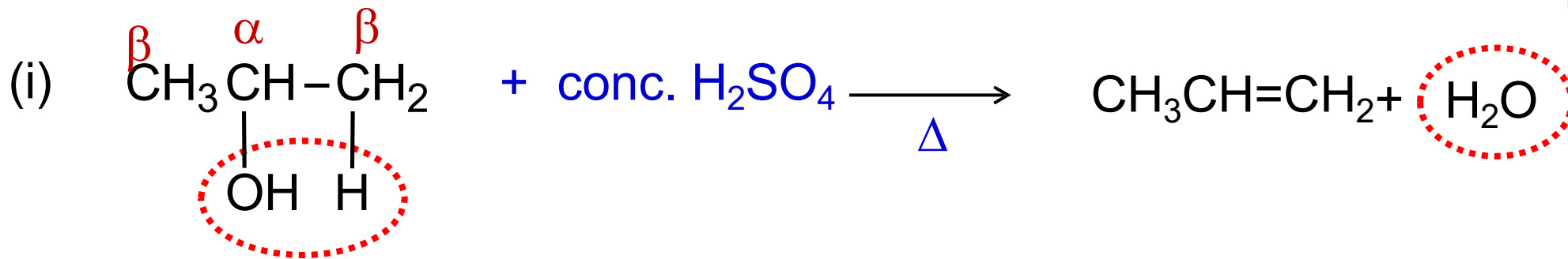
① Dehydration of Alcohol

- Elimination of H_2O from alcohol, an acid catalysed reaction.
- Reagents & conditions: $\text{conc. H}_2\text{SO}_4$, heat or H_3PO_4 , heat.
- General reaction equation:



NOTE

---OH is removed from α -carbon (C bearing the ---OH group) while ---H from the β -carbon (adjacent to α -carbon).



In examples above, both the β -carbons are *identical* thus only *one alkene product* formed

NOTE

For some alcohols with higher number of carbon atoms ($C \geq 4$), the presence of more than one type of β -H enable H_2O molecule to be eliminated in more than one way:

- ☐ Product consists of a mixture of alkenes.
- ☐ The major product can be predicted by using Saytzeff's rule.

Saytzeff's Rule

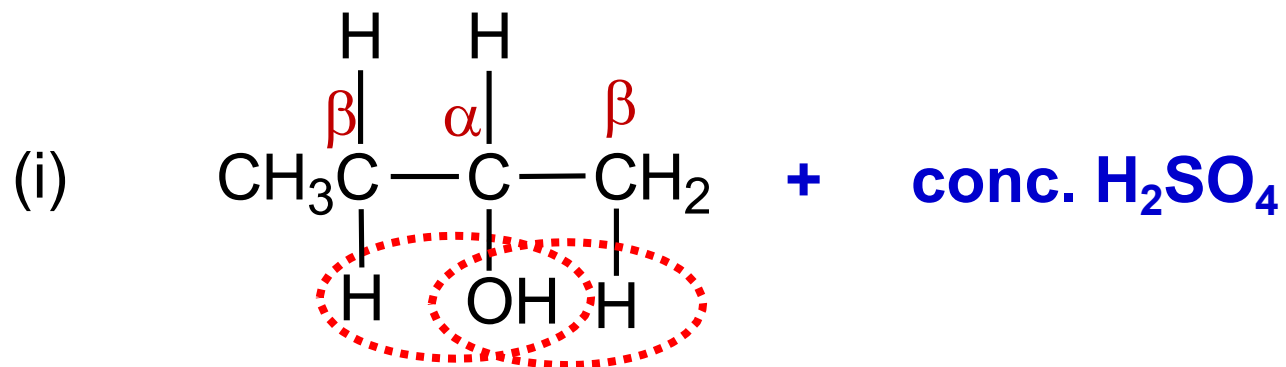
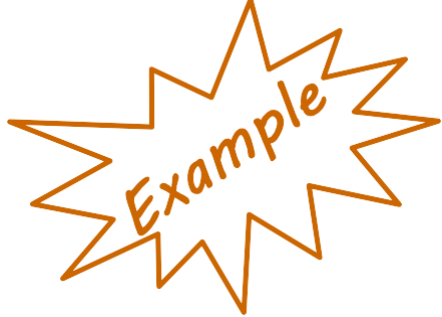


An elimination reaction, will give an alkene with the more highly substituted double bond as the major product.

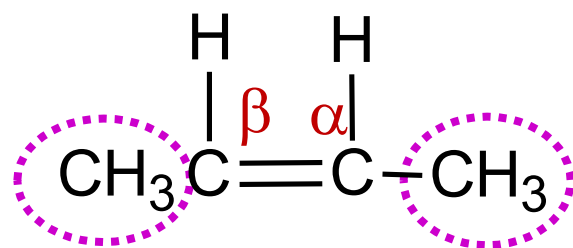
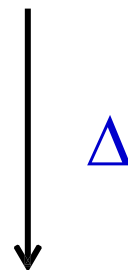


Note:

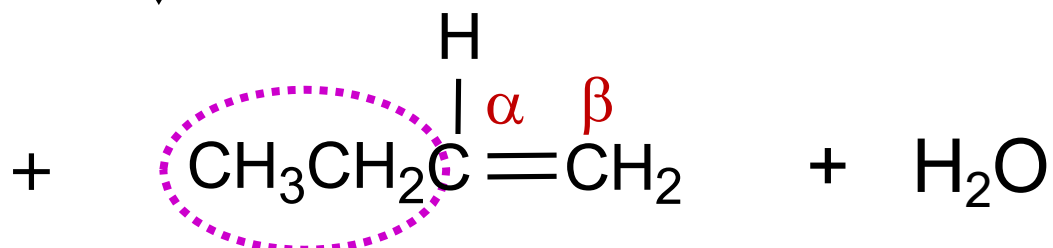
The more highly substituted alkene is one that has greater number of alkyl groups attached to the doubly-bonded carbon.



2-butanol



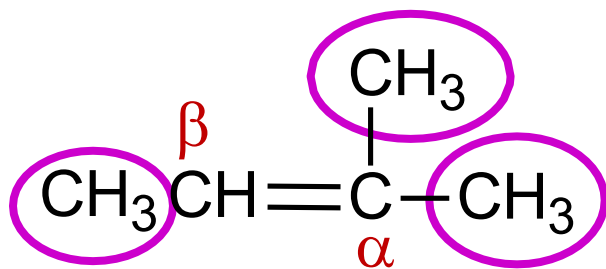
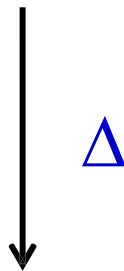
2-butene (2 alkyls)
(*major product*)



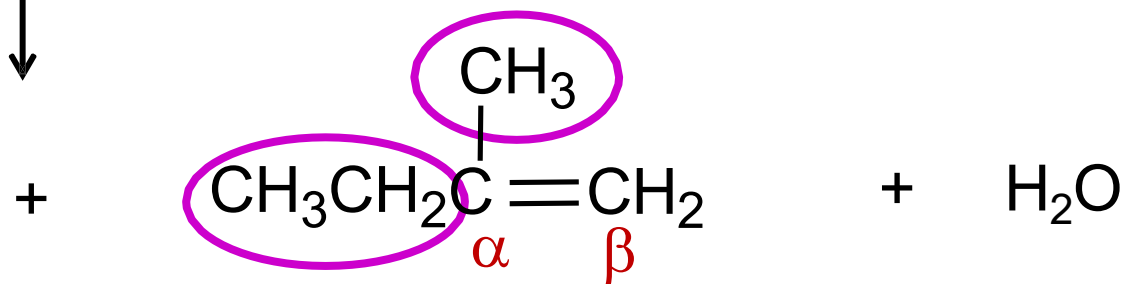
1-butene (1 alkyl)
(*minor product*)

CC(C)(O)CC

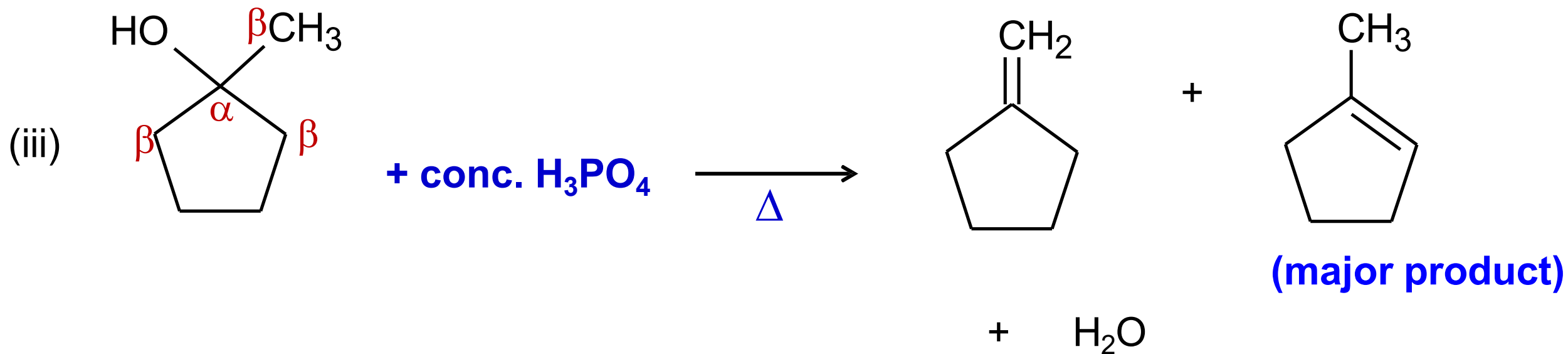
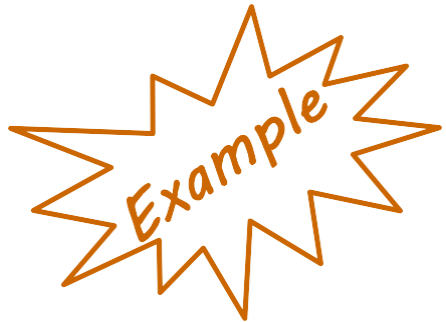
2-methyl-2-butanol



2-methyl-2-butene
(*major product*)

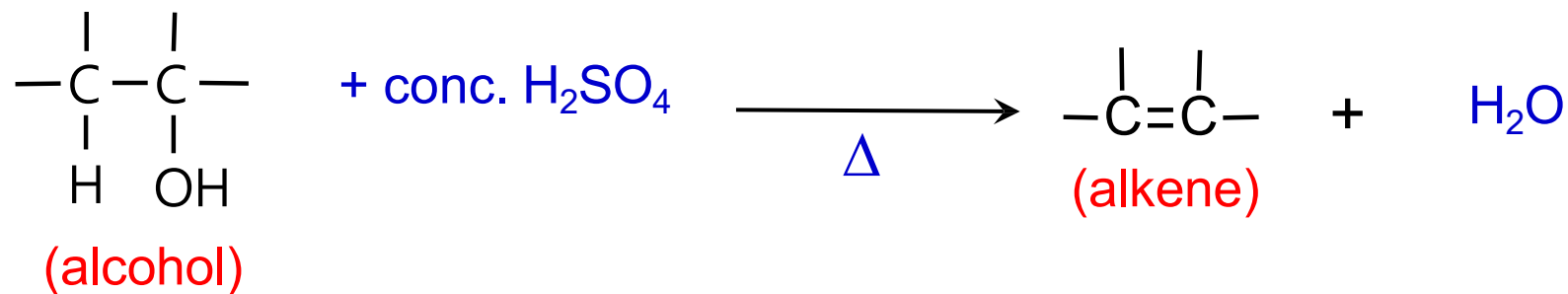


2-methyl-1-butene
(*minor product*)

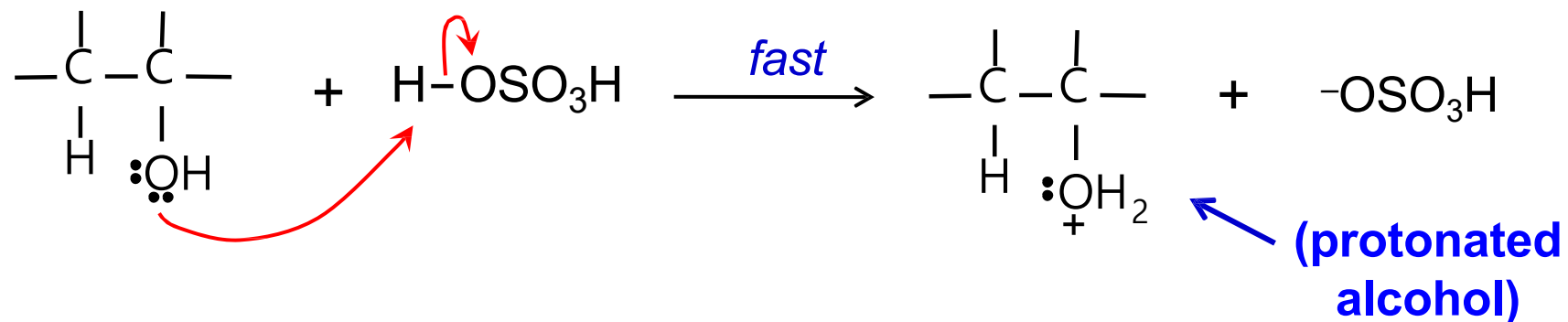


Elimination Mechanism for Dehydration of Alcohol

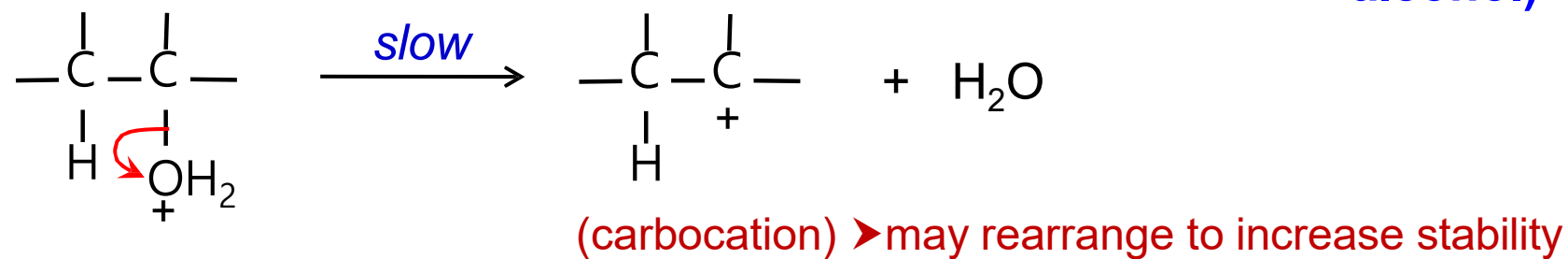
76



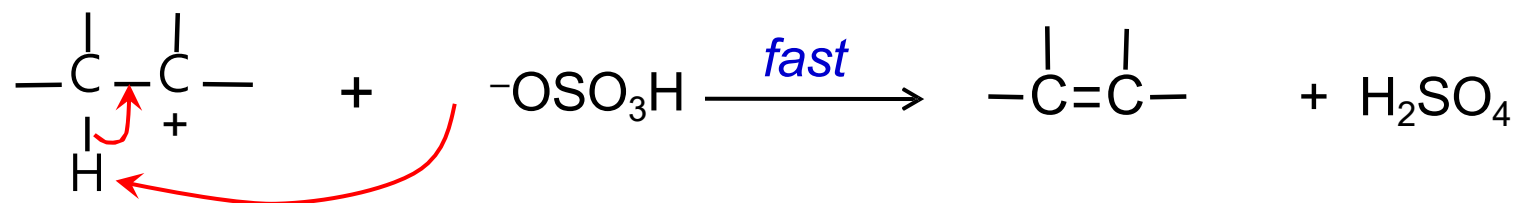
Step 1:
Protonation of alcohol



Step 2:
Formation of carbocation intermediate

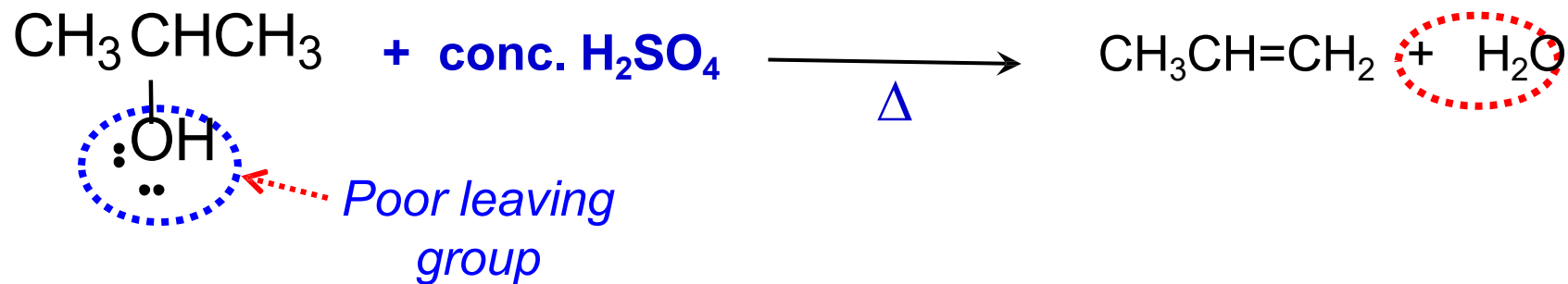


Step 3:
Removal of H⁺ (β-hydrogen) by a base

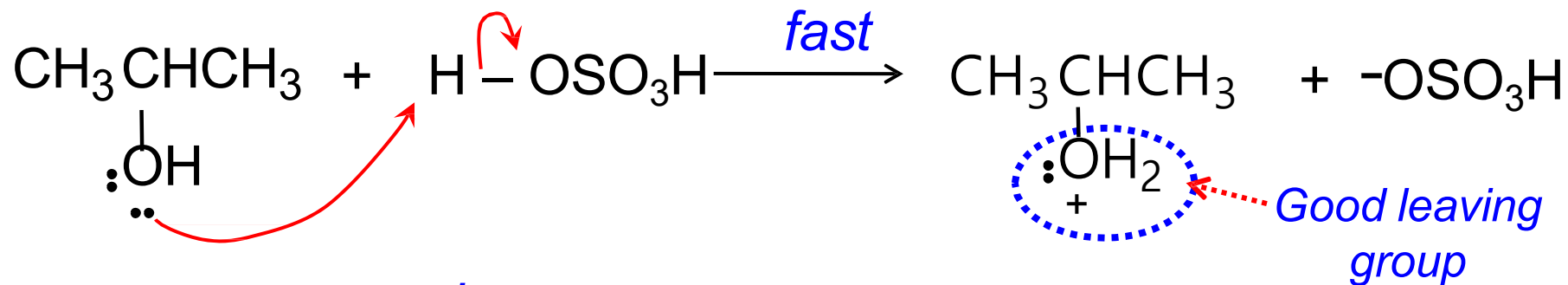


Elimination Mechanism For Dehydration Of Alcohol

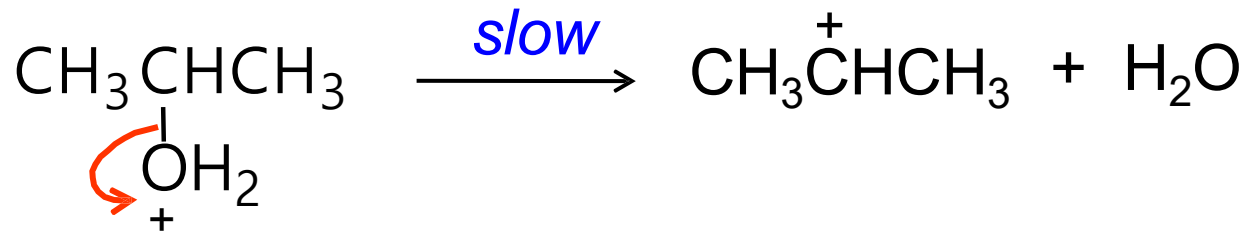
Example 1



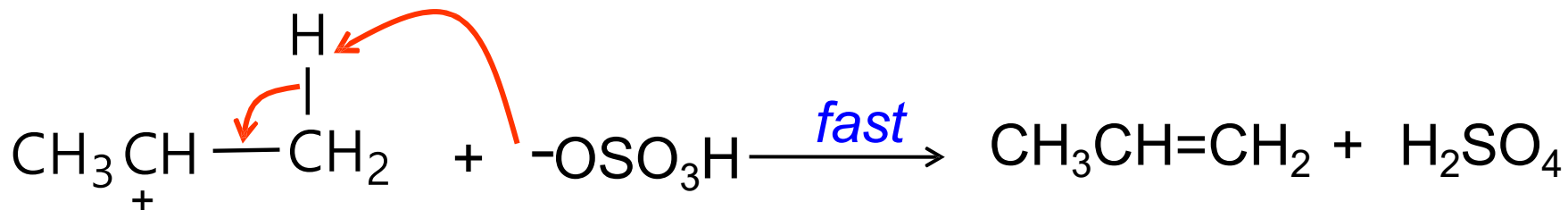
Step 1:
Protonation of alcohol



Step 2:
Formation of carbocation intermediate



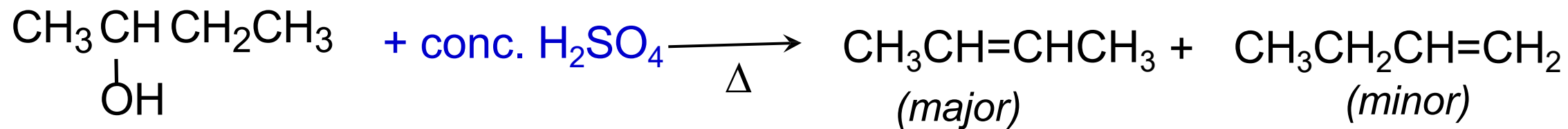
Step 3:
Removal of H^+ (β -hydrogen) by a base



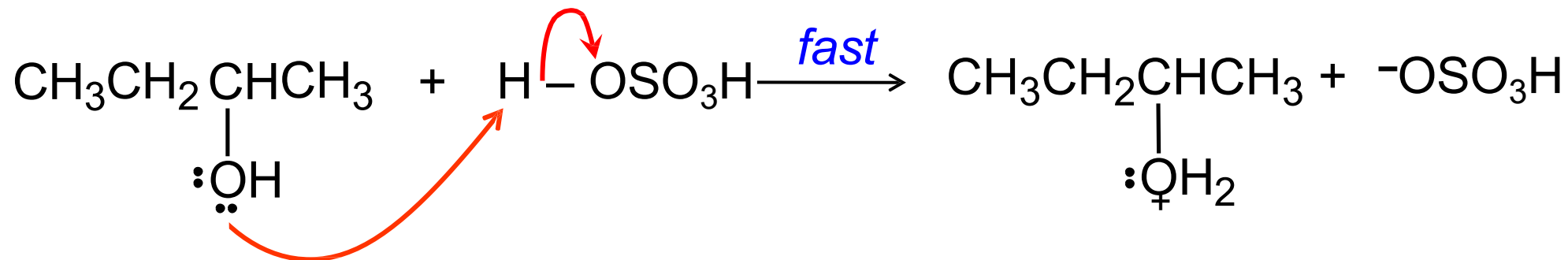
Example 2

Elimination Mechanism For Dehydration Of Alcohol

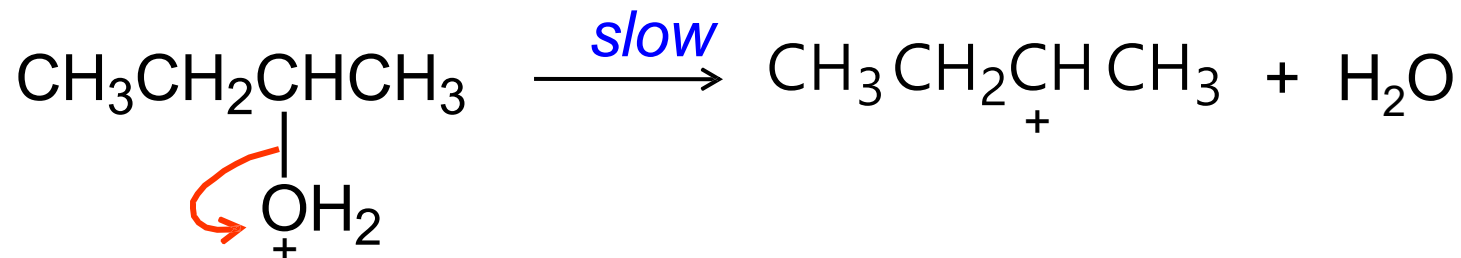
Write the mechanism for the reaction as shown in the equation below:



Step 1:
Protonation of
alcohol

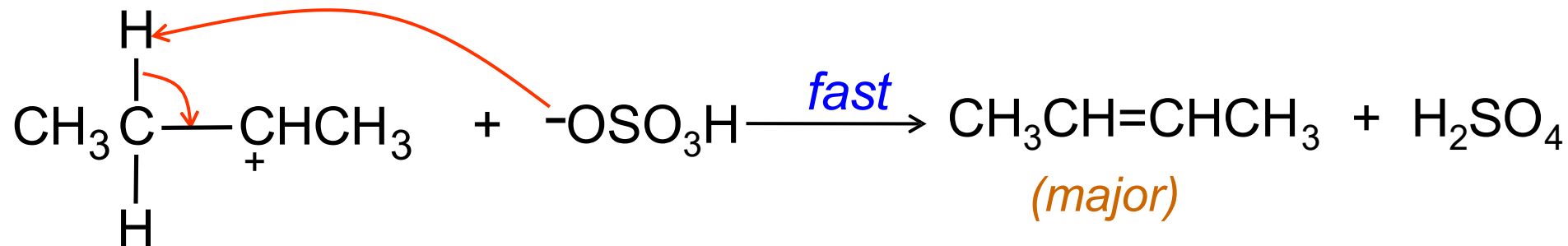


Step 2:
Formation of
carbocation
intermediate



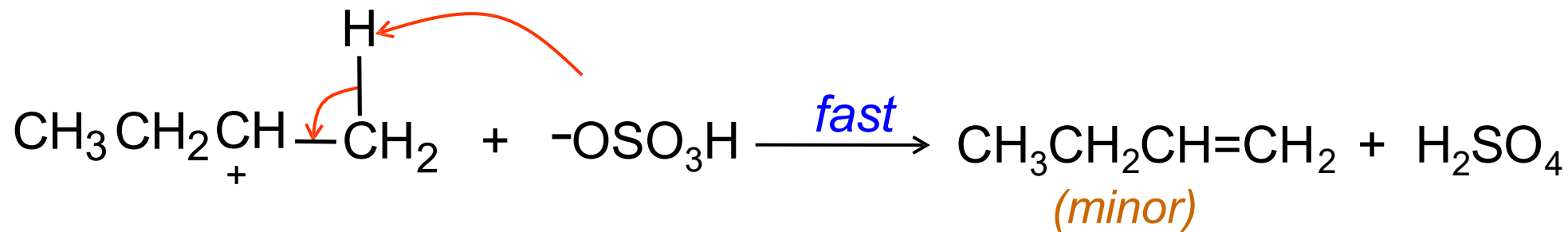
For major product

Step 3:
Removal of H^+
(β -hydrogen) by
a base



For minor product

Step 3:
Removal of H^+
(β -hydrogen) by
a base



NOTE:

- 2-butene is the major product, 1-butene is the minor product.
- Reason: Removal of $2^\circ \beta\text{-H}$ is more favourable than elimination of $1^\circ \beta\text{-H}$ because the alkene produced is more highly substituted, thus more stable.

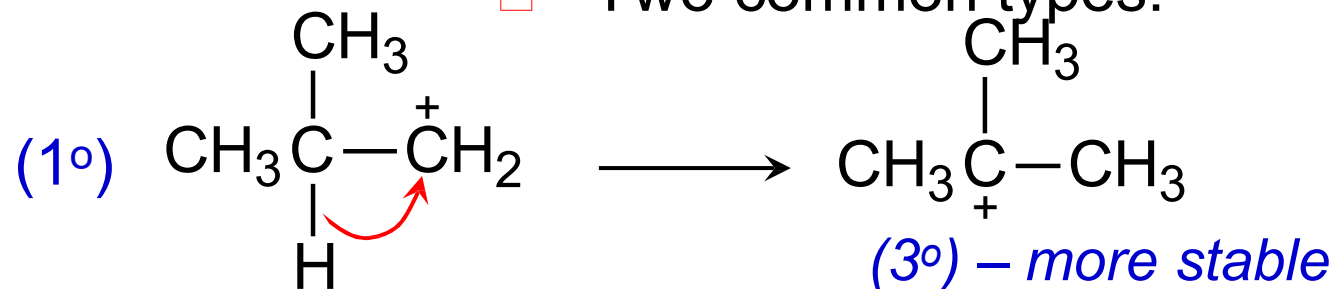
Rearrangement of Carbocation

80

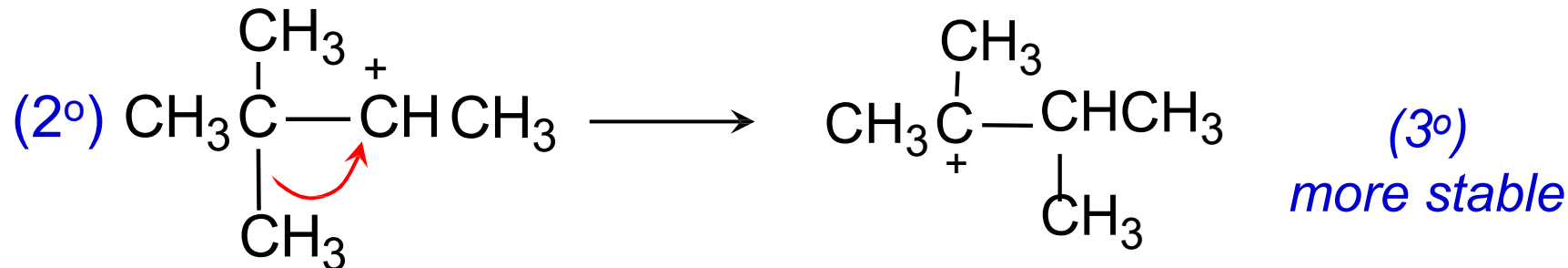
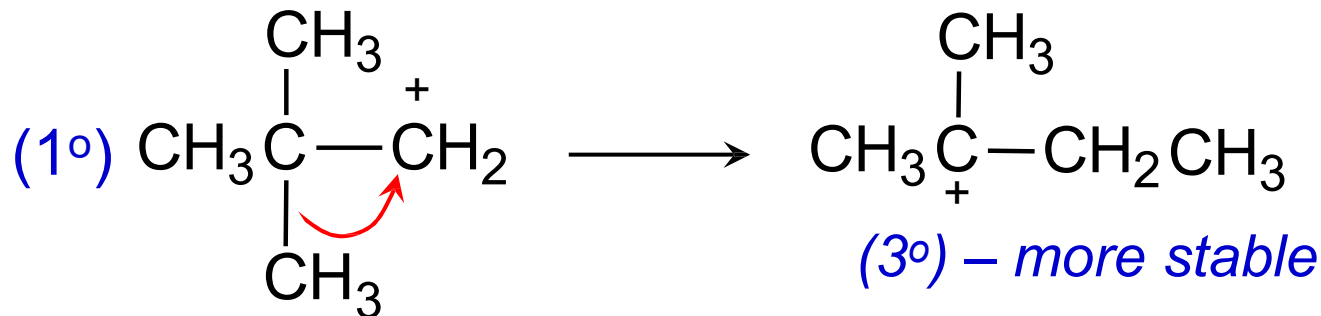
□ Occurs through the **1,2-shifts** i.e the **migration of a group** from the carbon atom adjacent to the positive charge one with

□ Two common types:

(i) 1,2-hydride shift



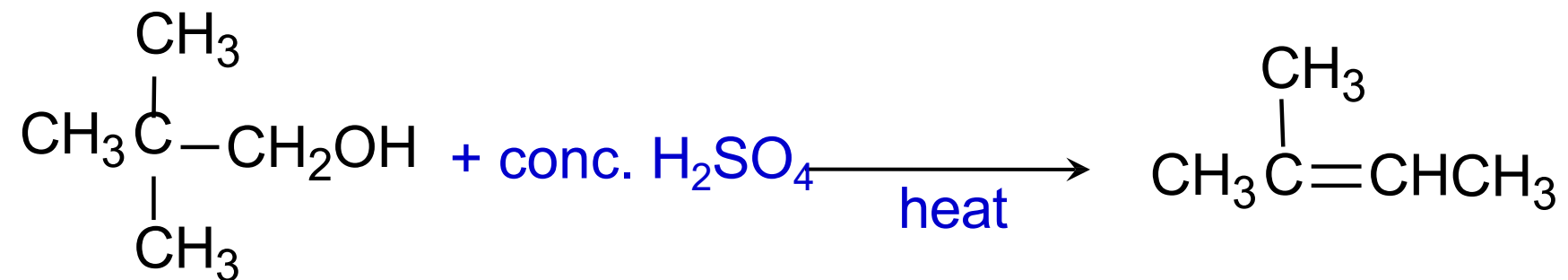
(ii) 1,2-methyl shift



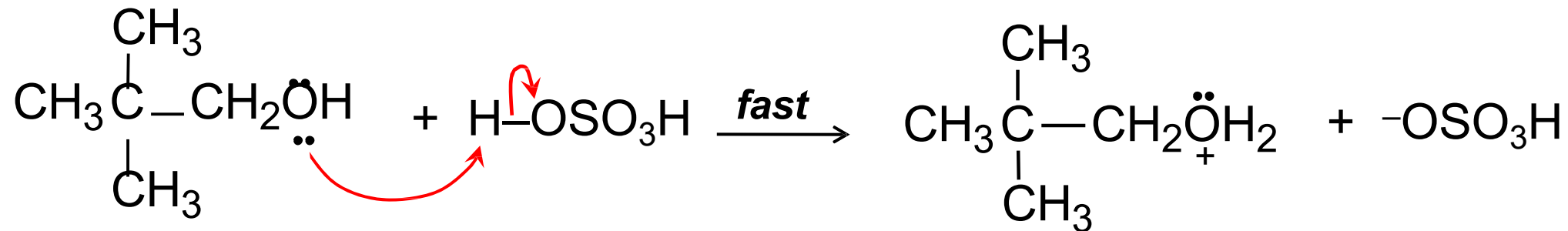
Elimination Mechanism For Dehydration Of Alcohol (with rearrangement of carbocation)



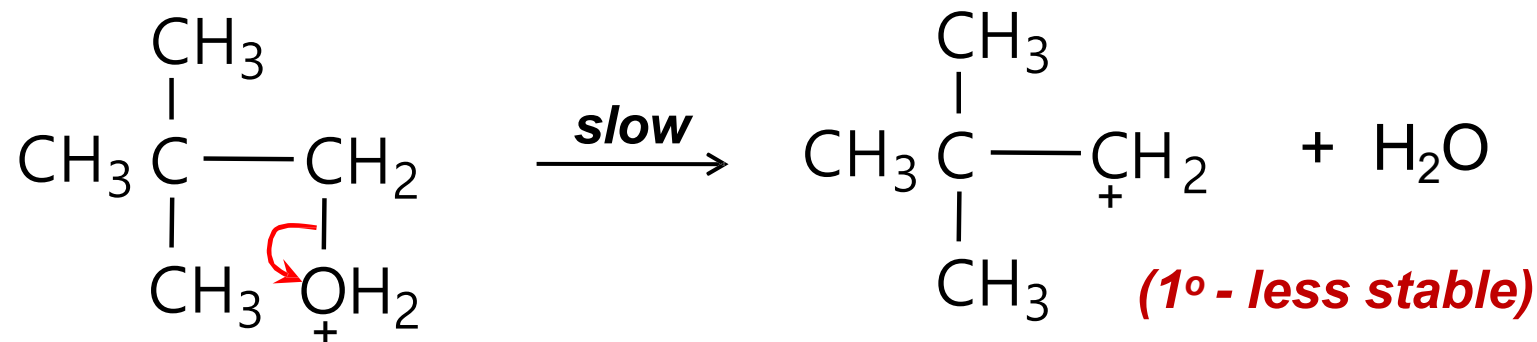
Write the mechanism for the reaction represented by the following equation:



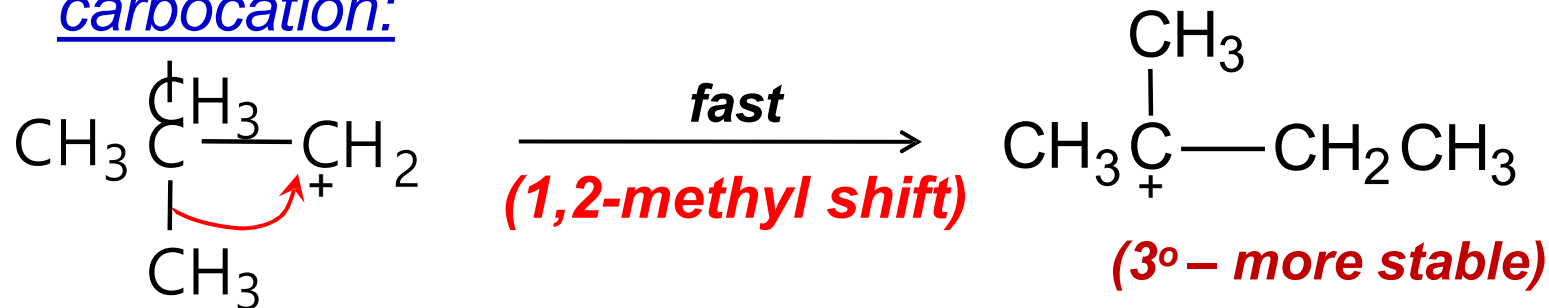
Step 1: Protonation of alcohol



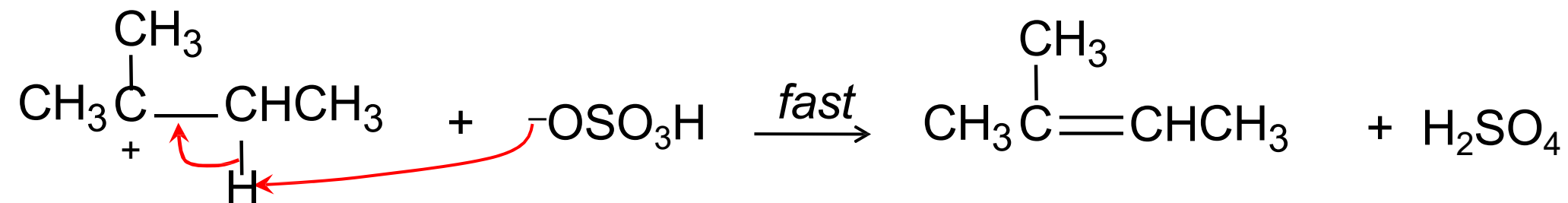
Step 2: Formation of carbocation intermediate



□ Rearrangement of carbocation:



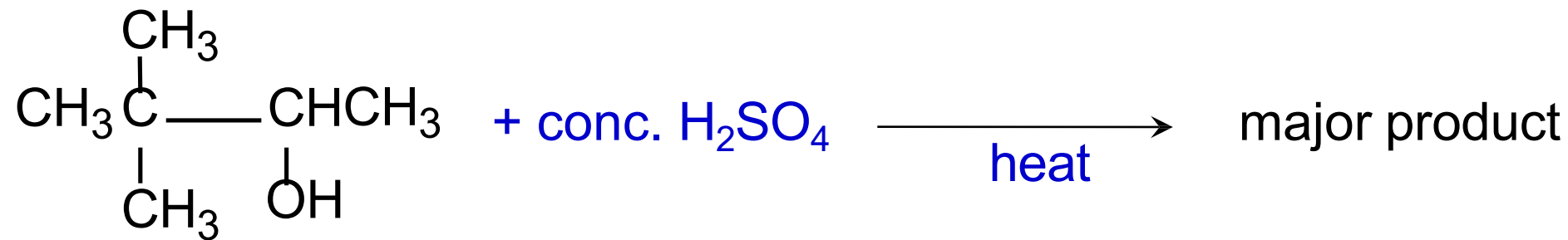
Step 3: Removal of H⁺ (β-hydrogen)



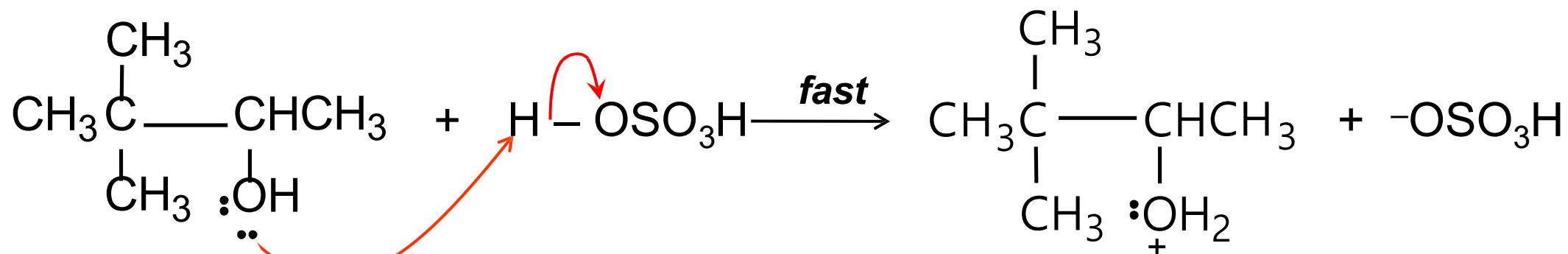
Elimination Mechanism For Dehydration Of Alcohol (with rearrangement of carbocation)



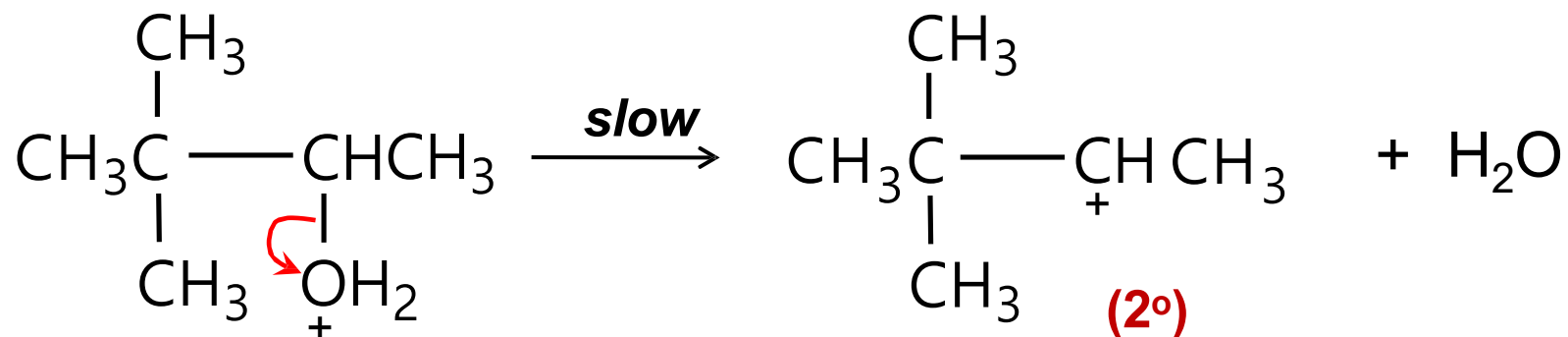
By writing the reaction mechanism, predict the major product of dehydration of the following alcohol:



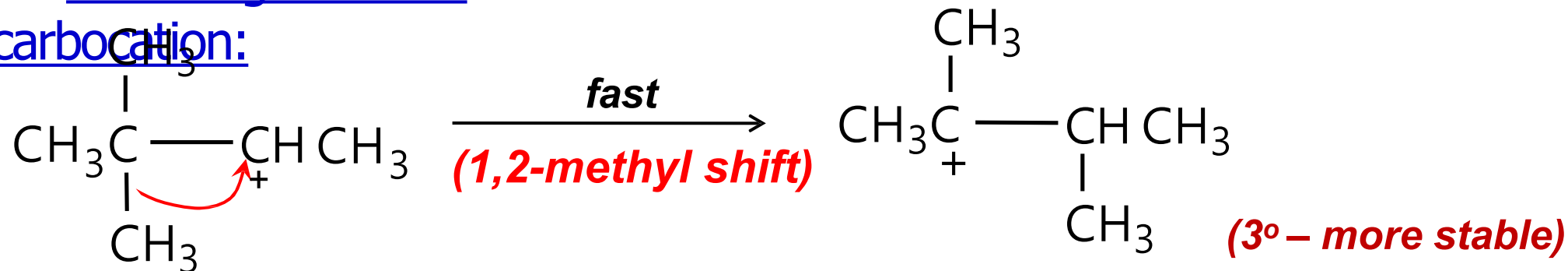
Step 1: Protonation of alcohol



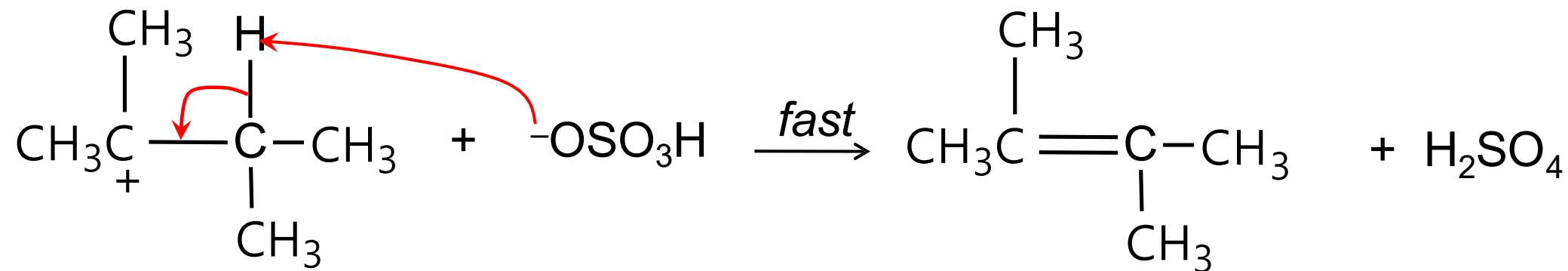
Step 2: Formation of carbocation intermediate



□ Rearrangement of carbocation:

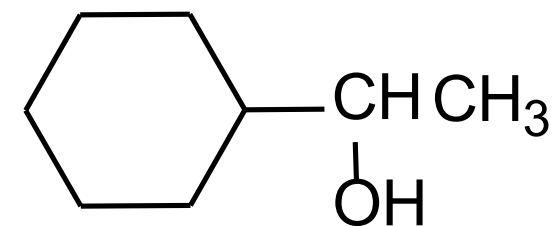


Step 3: Removal of H^+ (β -hydrogen)





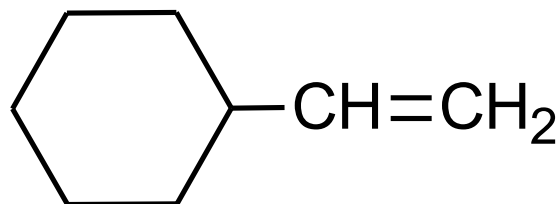
Dehydration of 1-cyclohexylethanol using hot concentrated sulphuric acid yields 3 alkenes. What are they and which is the major product?



(1-cyclohexylethanol)

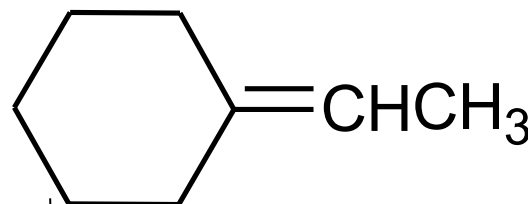
+ conc. H_2SO_4

heat

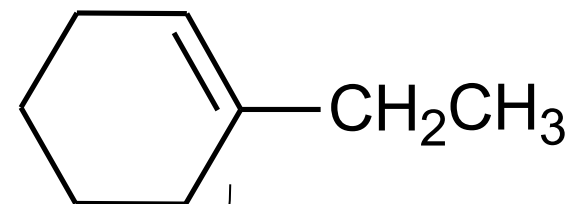


(minor product)

+



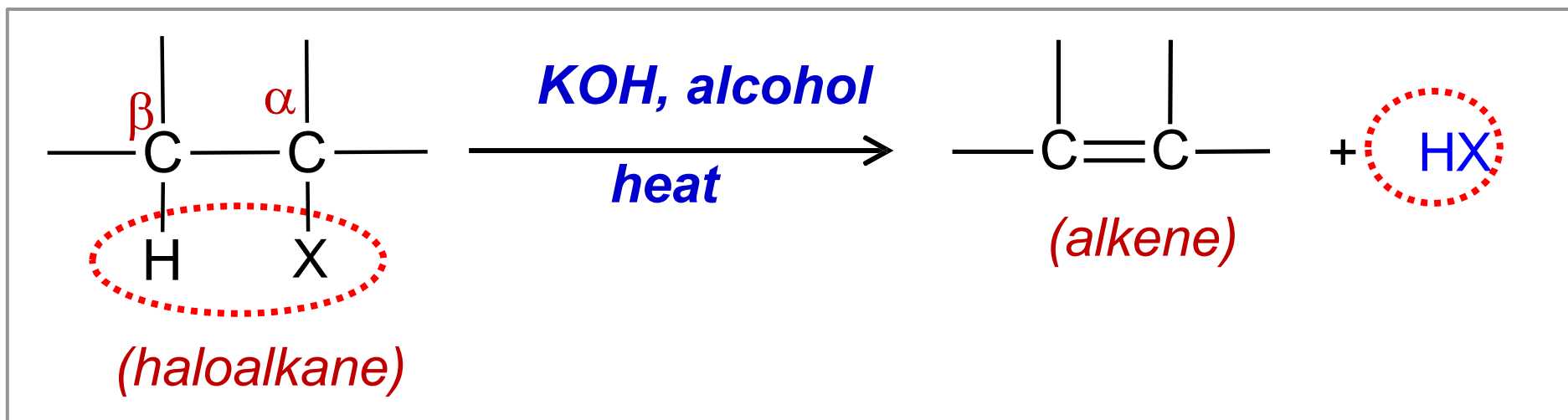
+



(major product)

② Dehydrohalogenation of Haloalkanes

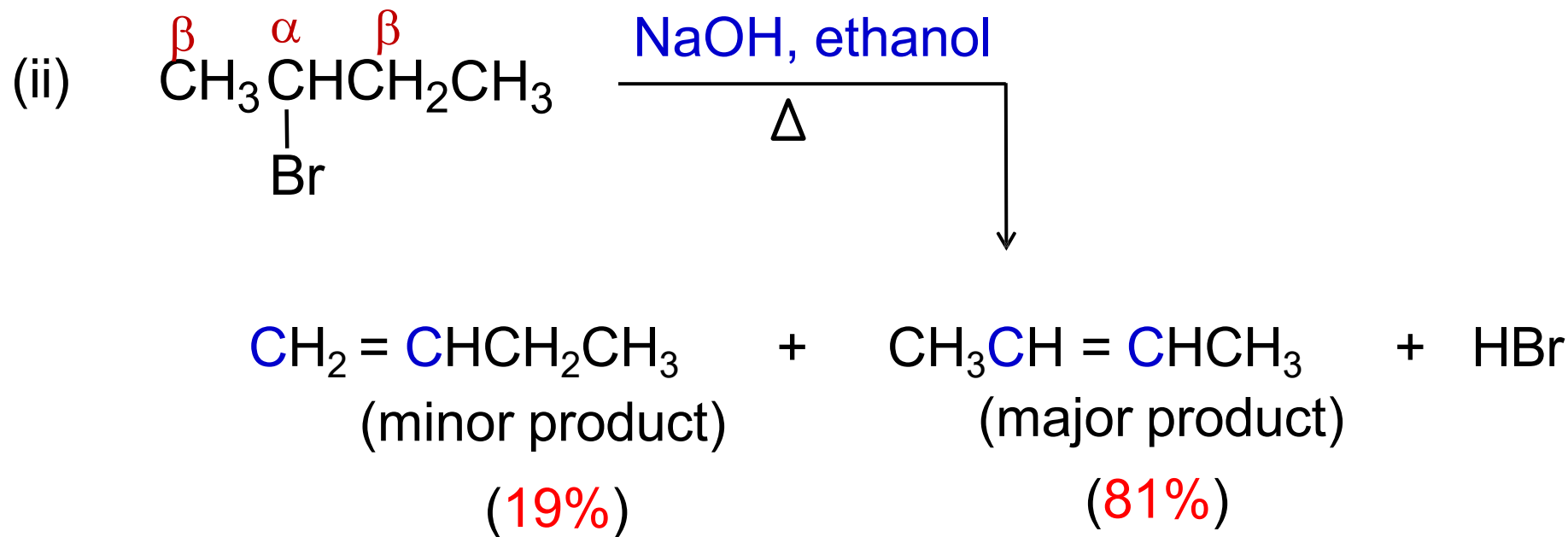
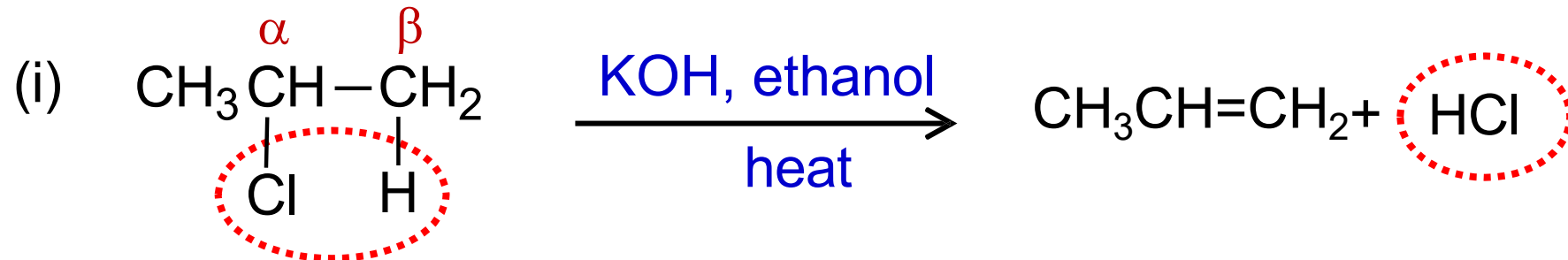
- Elimination of HX from adjacent atoms of a haloalkane (alkyl halides), RX.
- Reagent & condition: Heating RX with a strong base such as KOH or NaOH or sodium alkoxide, RO-Na⁺ in alcohol.
 - General reaction equation:



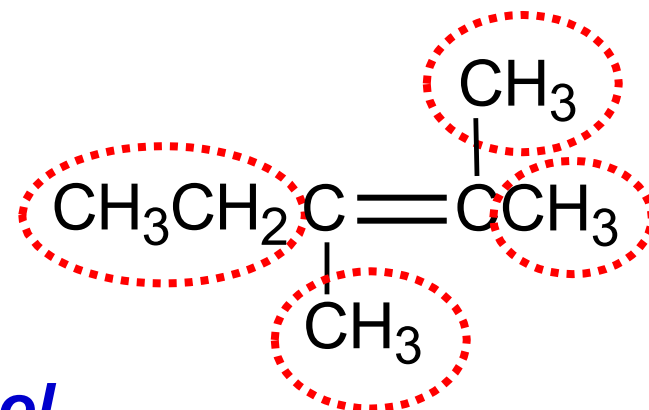
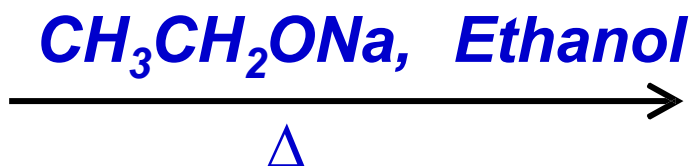
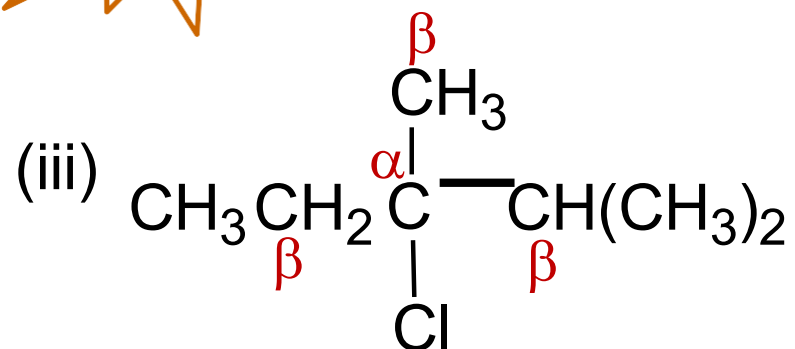
(For unsymmetrical alkene, the major product is determined by Saytzeff's rule)



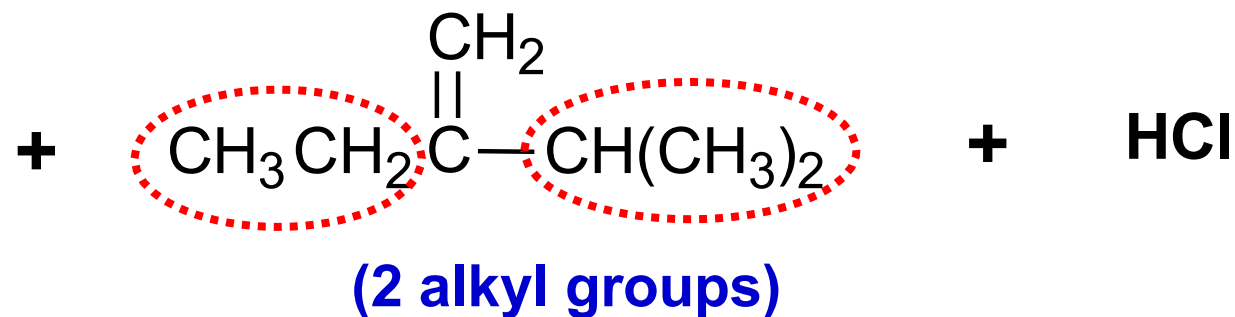
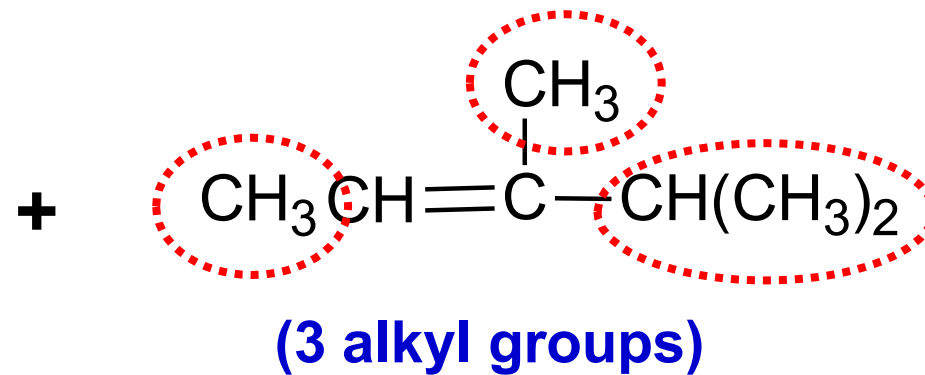
Examples



Example

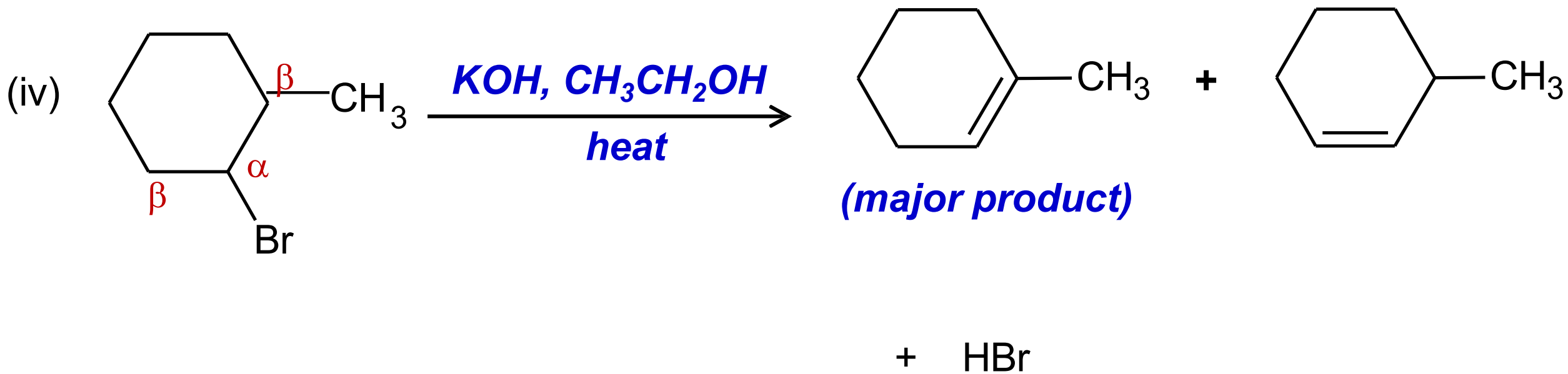


(4 alkyl groups)
(major product)

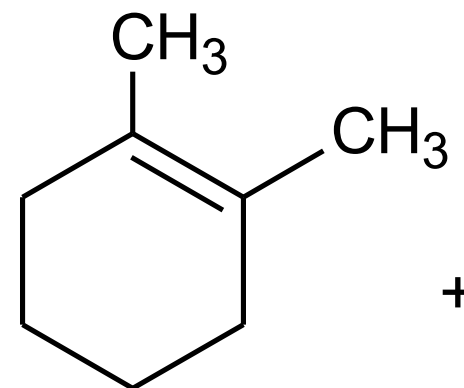
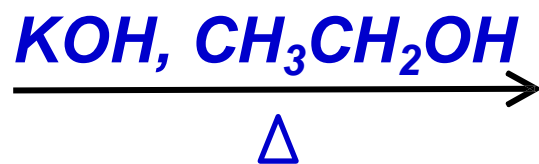
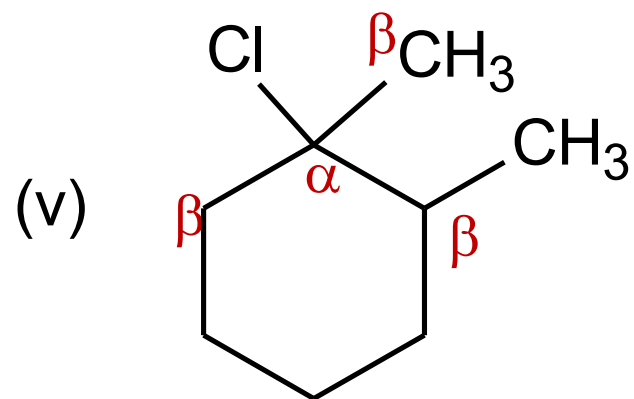




Example

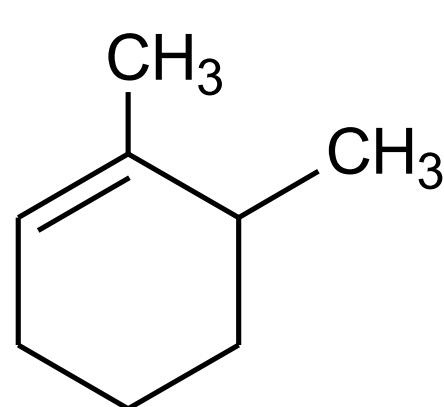


Example

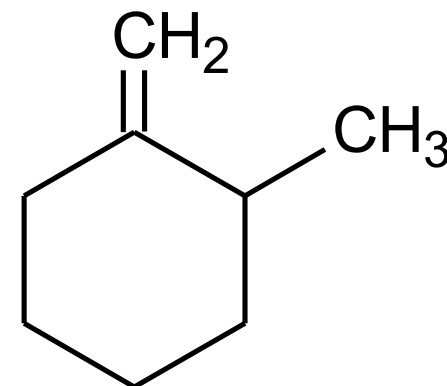


+

(major product)



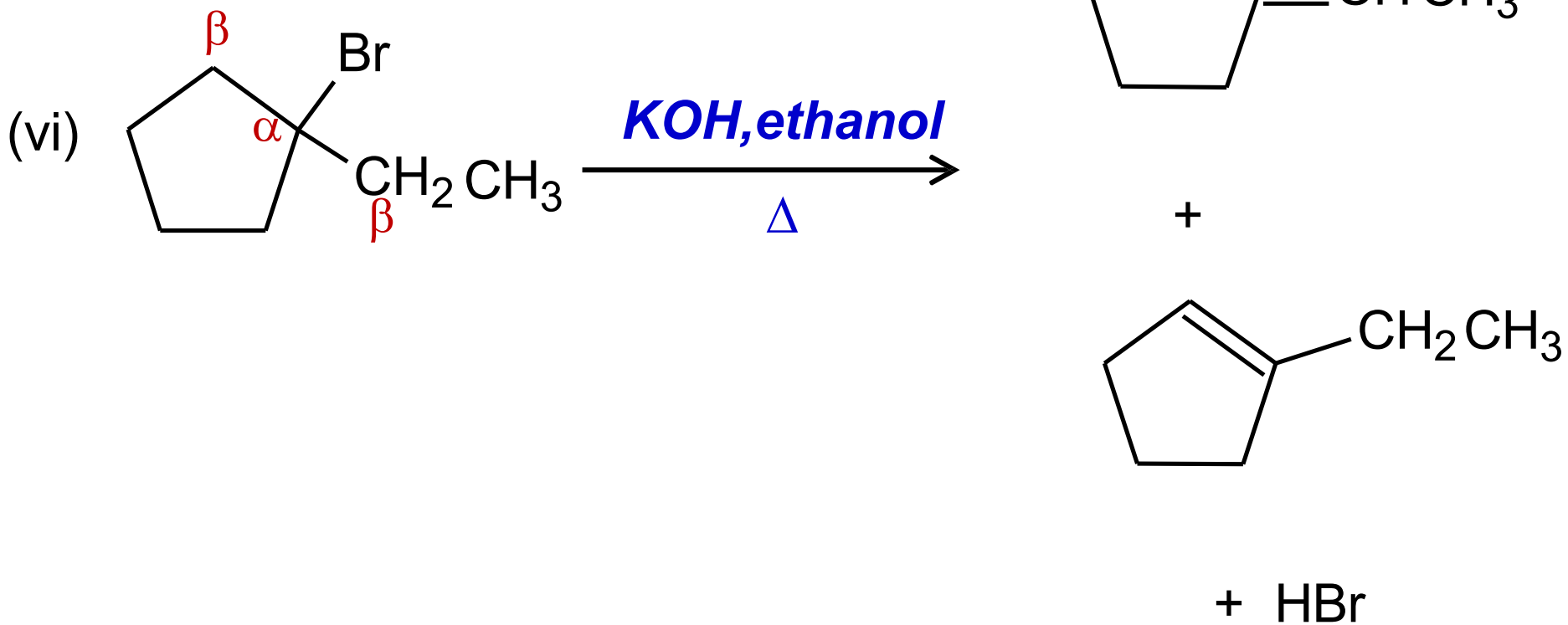
+



+

HCl

Example



Chemical Properties of Alkenes

- Alkenes are **more reactive** than alkanes.

Reason: Alkenes are **unsaturated** compounds
⇒ easily undergo **addition** reactions

- The **C=C** double bond is a site of **high electron density** (nucleophilic site)
⇒ highly susceptible to **electrophilic attack**.

- Type of reaction of alkenes: **Electrophilic Addition Reaction**

REACTIONS OF ALKENES

ELECTROPHILIC ADDITION REACTION

→ (i) Catalytic hydrogenation

→ (ii) Halogenation (Cl_2 or Br_2)

→ In inert solvent, CH_2Cl_2

→ In water, H_2O

→ (iii) Hydrohalogenation (HCl or HBr)*

→ (iv) Hydrobromination in peroxide

→ (v) Acid catalysed hydration *

OXIDATION REACTION

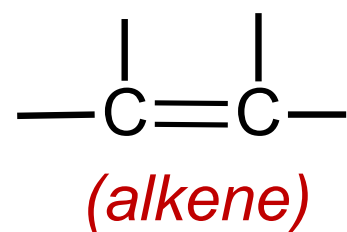
→ (i) With cold alkaline KMnO_4 (Baeyer's test)

→ (ii) With hot acidified KMnO_4

→ (iii) Ozonolysis

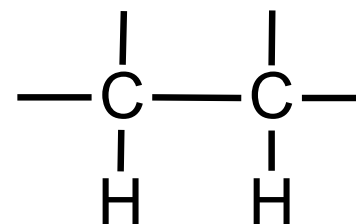
**Comply Markovnikov's rule for unsymmetrical alkenes.*

ADDITION REACTIONS OF ALKENE



H_2 , Pt o @ Ni @ Pd

(hydrogenation)

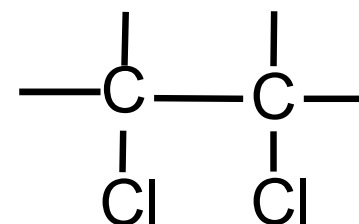


(alkane)

X_2 , CH_2Cl_2

(halogenation)

(X_2 : Cl_2 or Br_2)

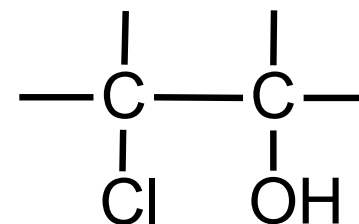


(vicinal dihalide)

□

X_2 (aq)

(halogenation)

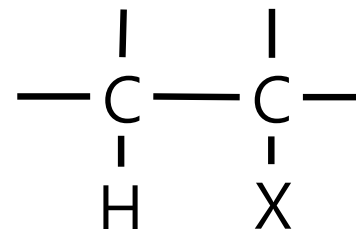


(haloalcohol)

□

HX

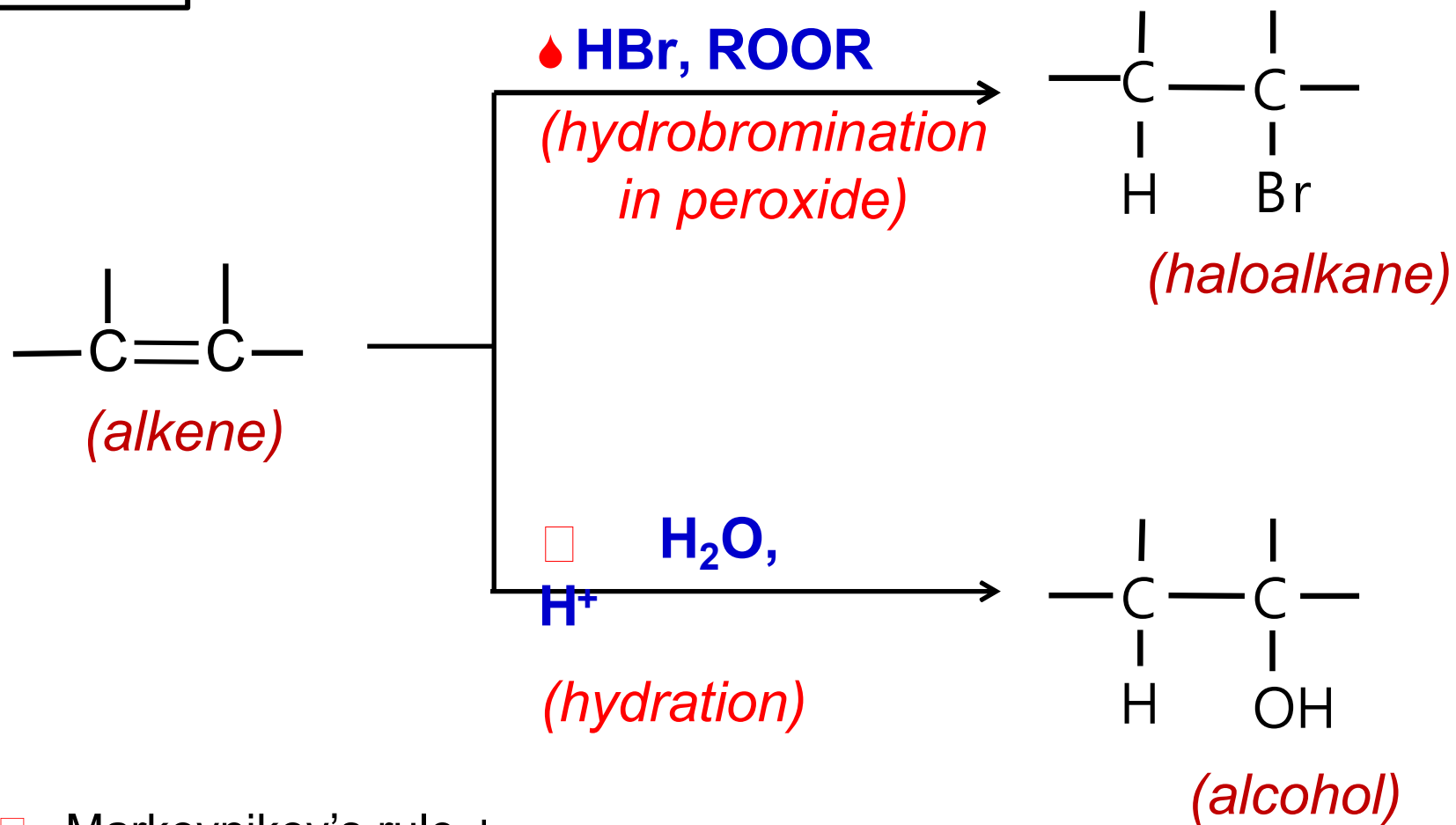
(hydrohalogenation)



(haloalkane)

□ Markovnikov's
rule

ADDITION REACTIONS OF ALKENE



□ Markovnikov's rule + mechanism

🔥 Anti Markovnikov's rule

□ Markovnikov's rule

(Applies to Hydrohalogenation & Hydration reactions)

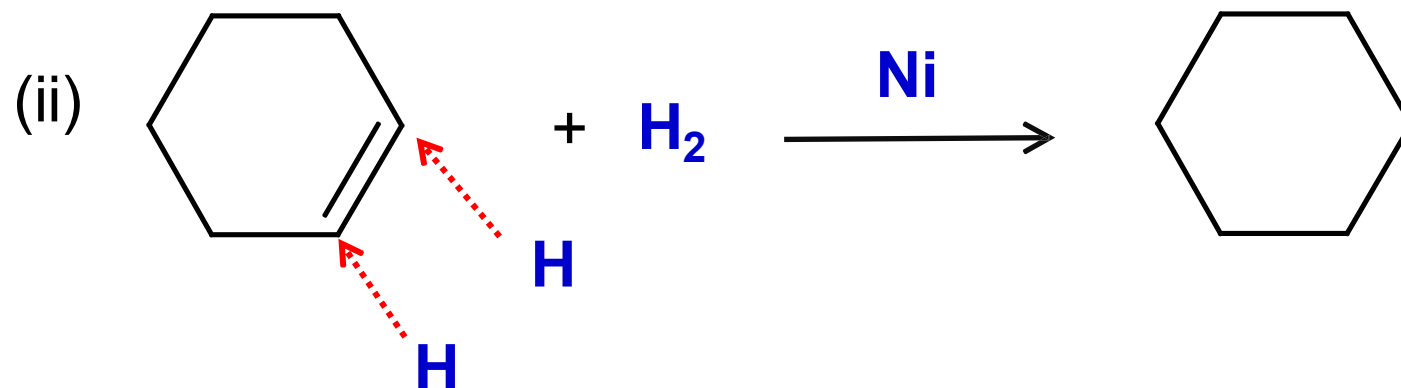
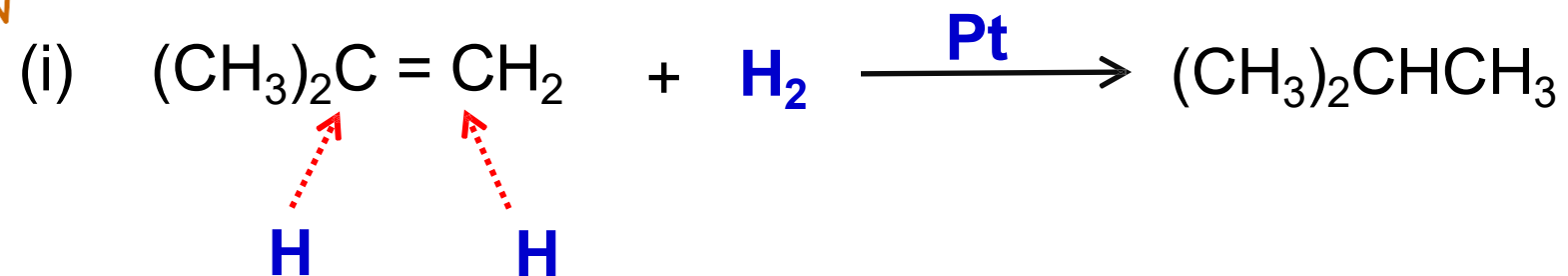
In the addition of a reagent to an unsymmetrical alkene, the more electropositive atom of reagent is added to the doubly-bonded carbon that has greater number of hydrogen atoms.

🔥 Anti-Markovnikov's rule (HBr only)

When an alkene is treated with HBr in the presence of peroxide (ROOR), the hydrogen atom of HBr adds to the doubly-bonded carbon with fewer hydrogen atoms.

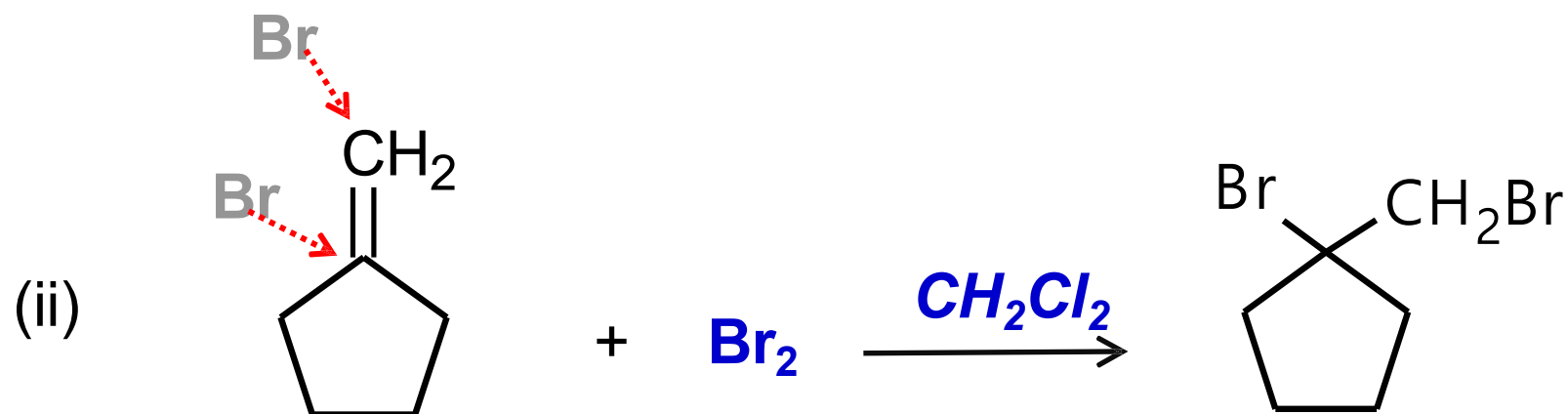
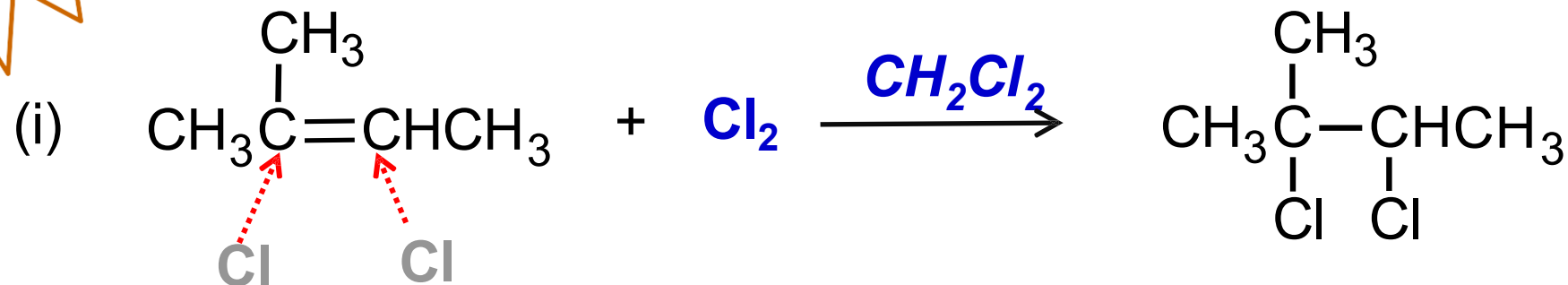
① Catalytic Hydrogenation

Examples



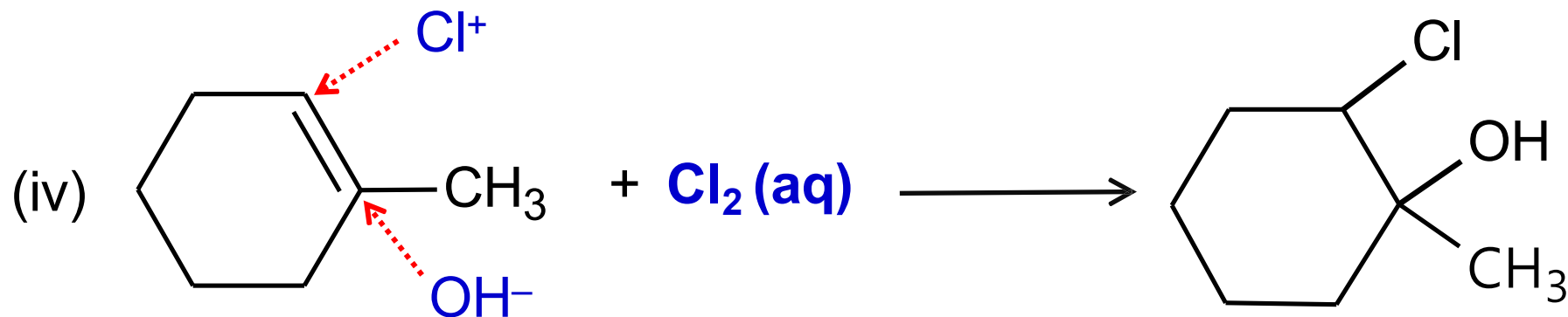
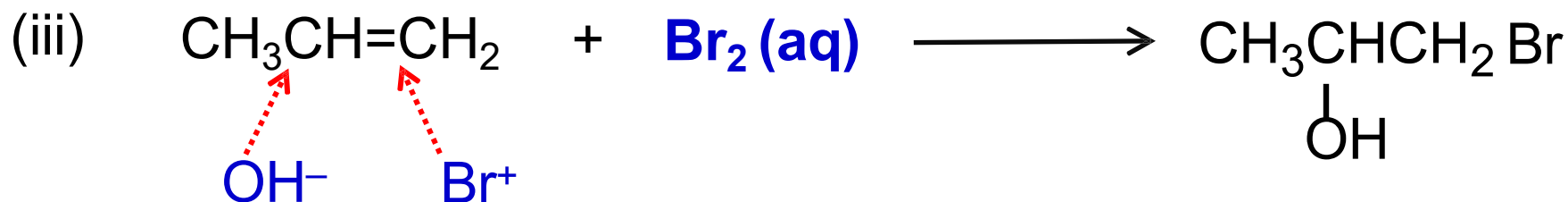
② Halogenation in Inert Solvent

Examples



③ Halogenation in Aqueous Solvent

Examples

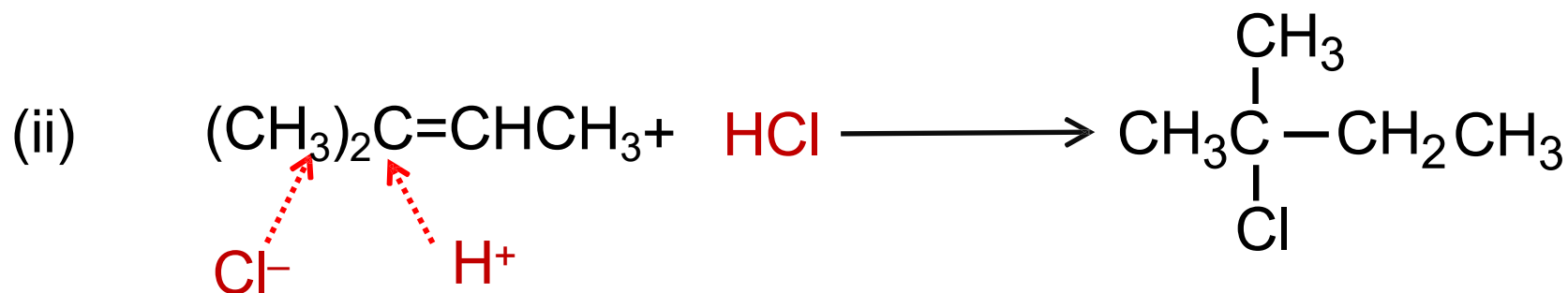
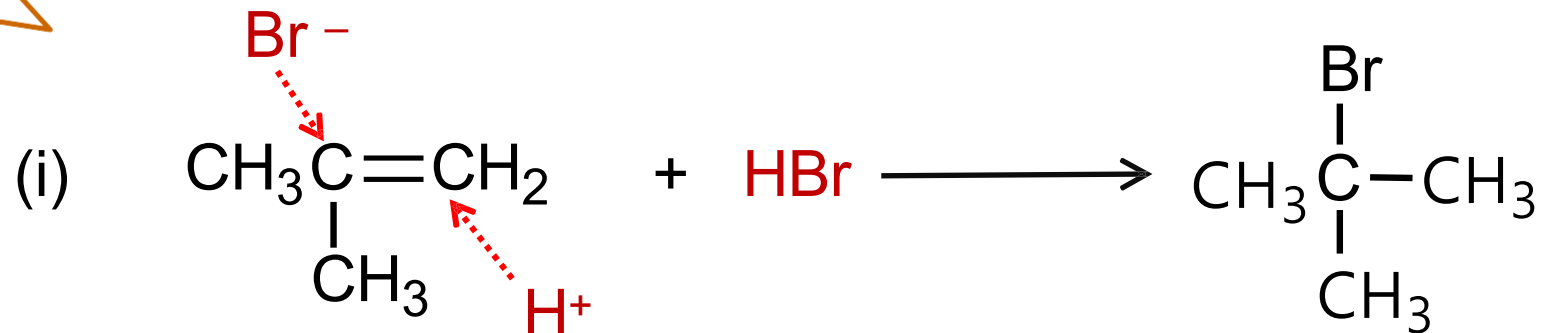


Markovnikov's rule

For unsymmetrical alkenes, the **electrophile**, Cl^+ or Br^+ will attach to the **doubly-bonded carbon** that has **greater number of hydrogen atoms**.

④ Hydrohalogenation

Examples

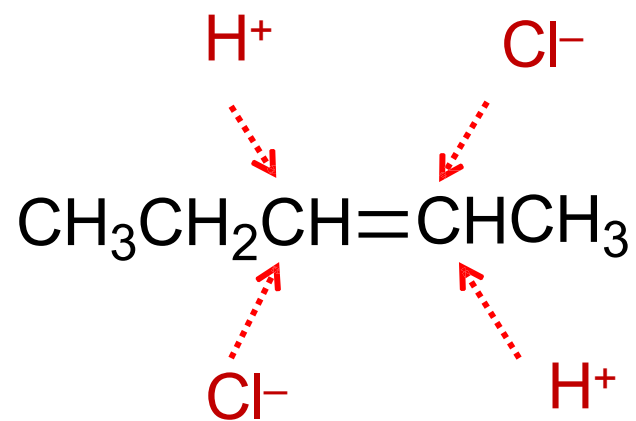


Markovnikov's rule

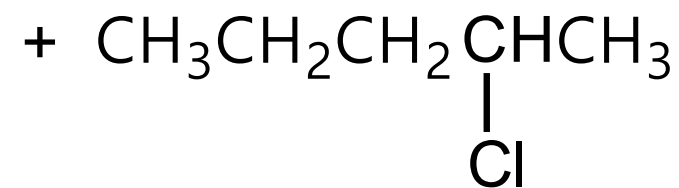
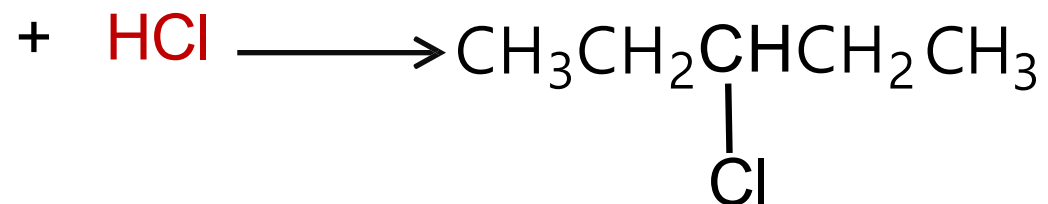
In the addition of electrophilic reagent to an unsymmetrical alkene, the electrophile, H^+ ion will attach to the doubly-bonded carbon that has greater number of hydrogen atoms.

Examples

(iii)

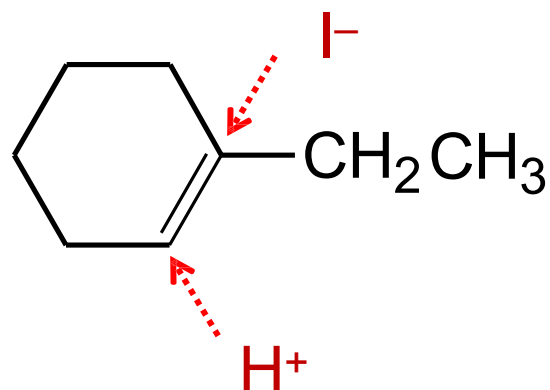


(symmetrical
alkene)

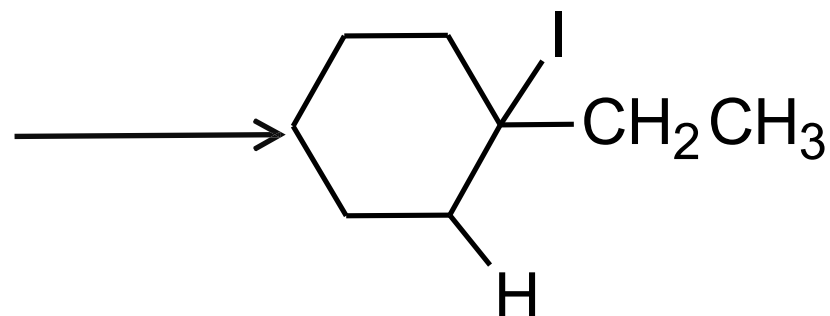


(mixture of 2 products)

(iv)

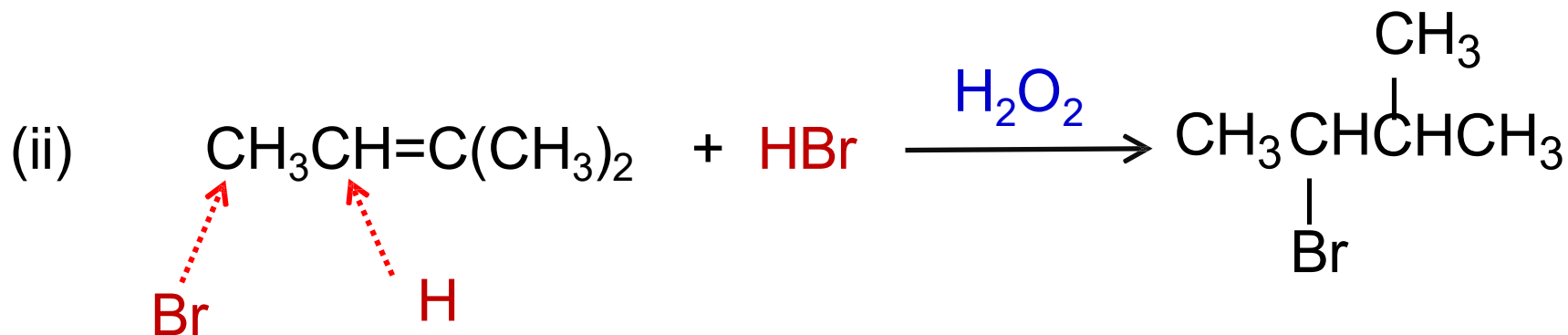
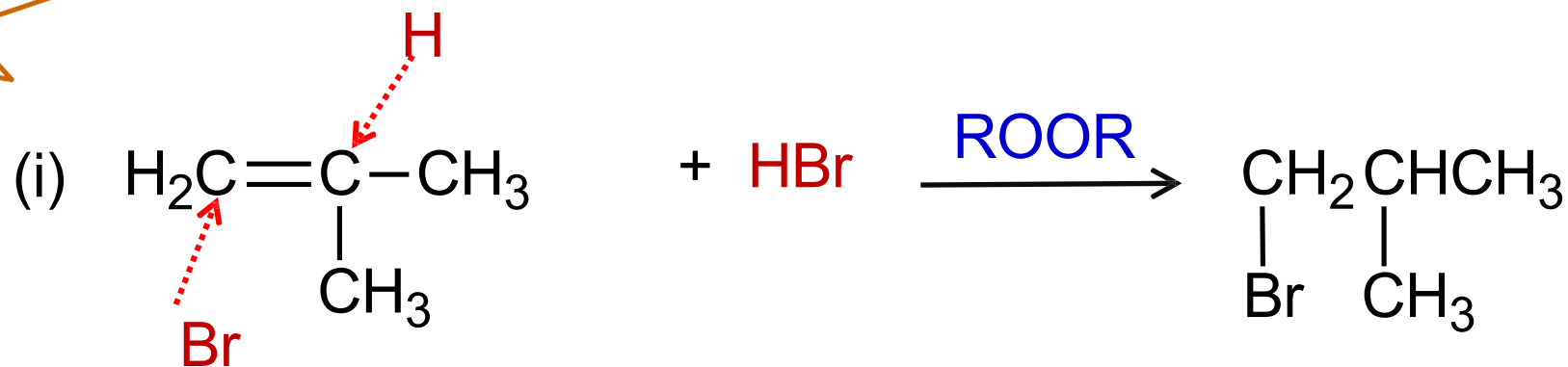


+ HI



⑤ Hydrobromination in the presence of peroxide

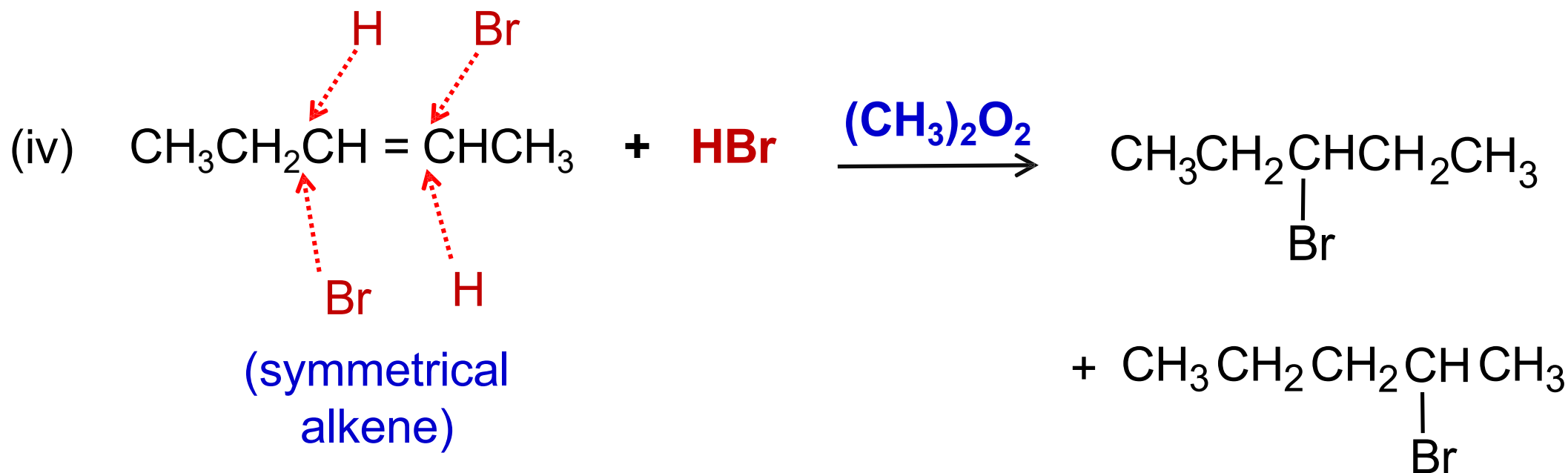
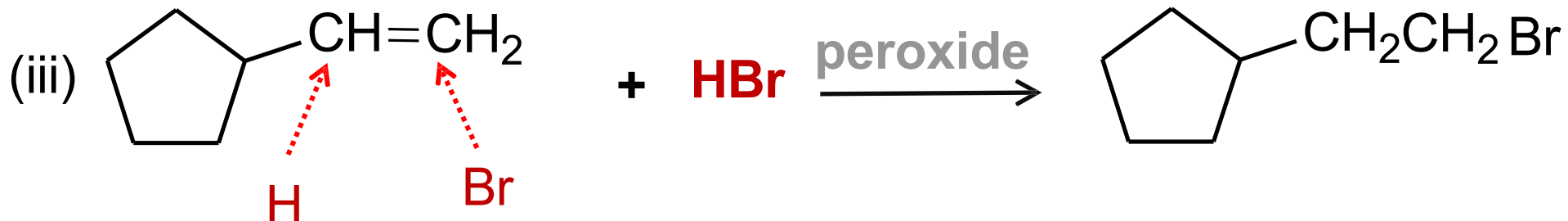
Examples



Anti-Markovnikov's rule

When an unsymmetrical alkene is treated with HBr in the presence of peroxide, ROOR, the hydrogen atom of HBr adds to the doubly-bonded carbon with fewer hydrogen atoms.

Examples

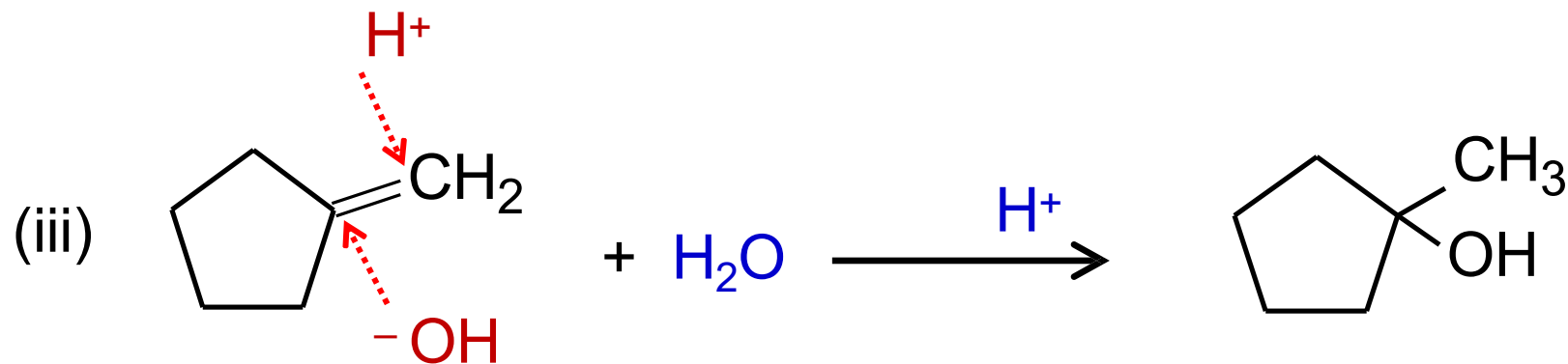
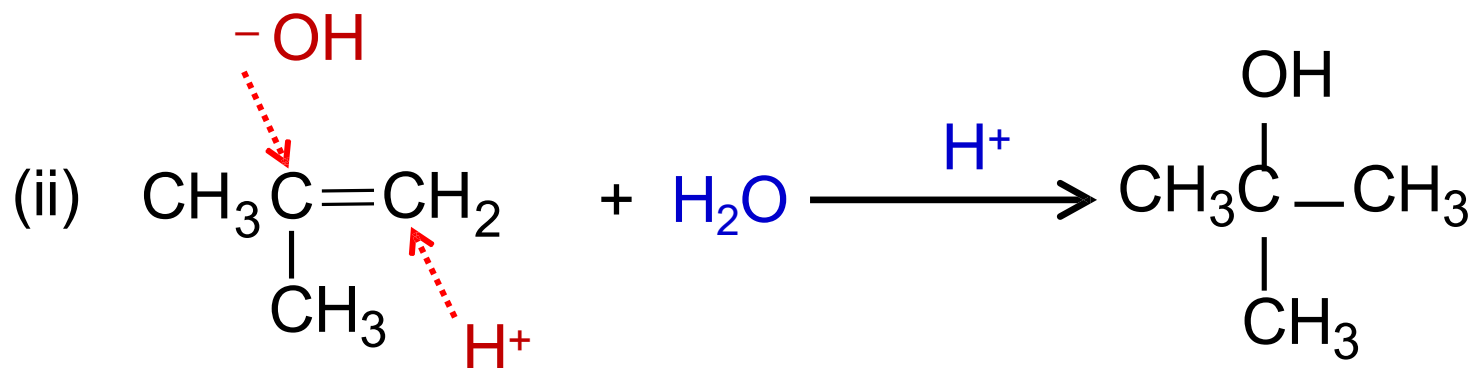
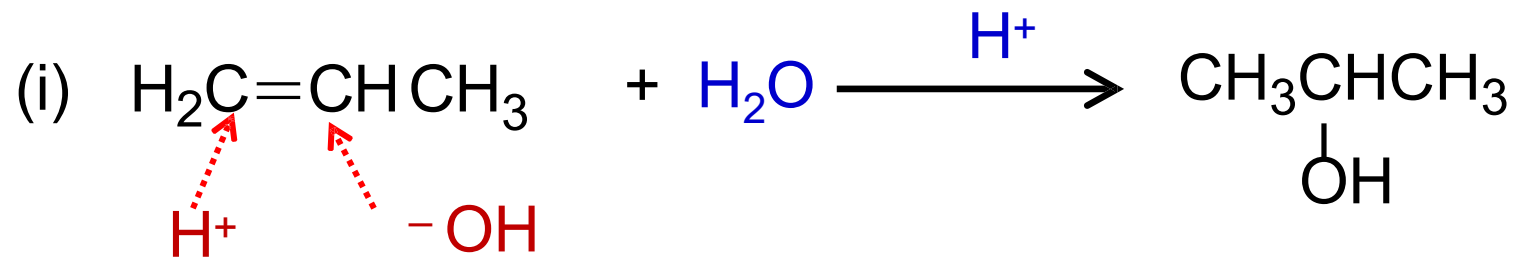


Note : symmetrical alkene forms
a mixture of products.

Unsymmetrical alkenes
comply Markovnikov's rule

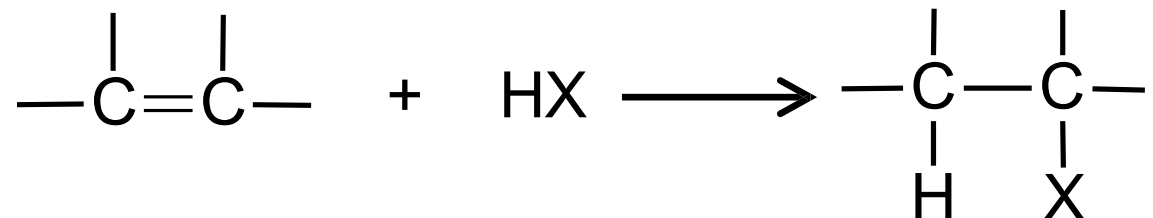
⑥ Acid Catalysed Hydration

Examples

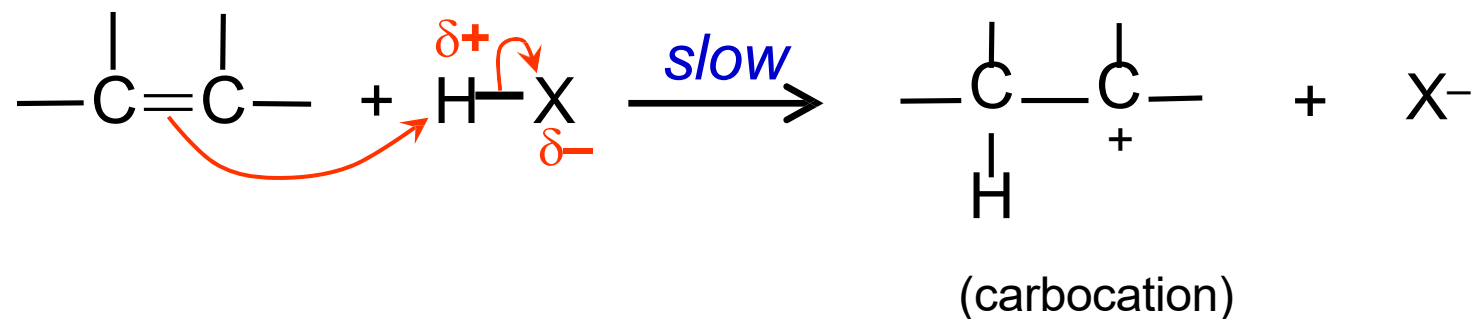


Electrophilic Addition Mechanism for Hydrohalogenation of Alkene

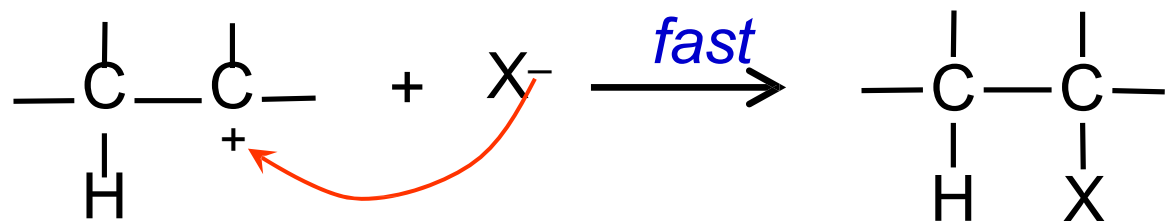
General
Reaction
Equation



Step 1:
Addition of H^+ to
form carbocation



Step 2:
Nucleophile
attack on R^+

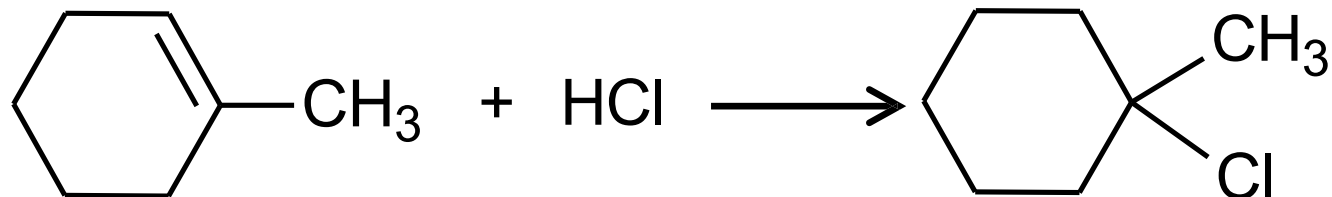


For **unsymmetrical alkene, product is determined by **Markovnikov's rule**.*

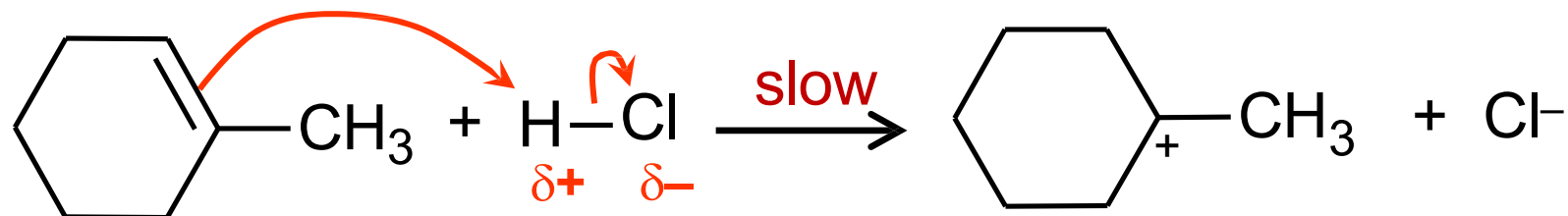
Example

Electrophilic Addition Mechanism for Hydrohalogenation of Alkene

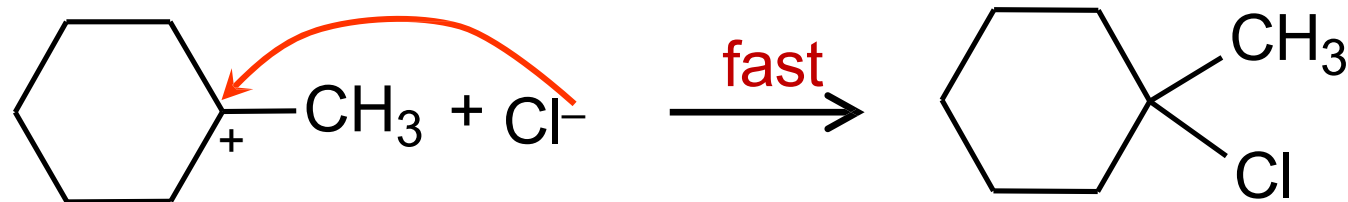
Reaction
Equation



Step 1:
Addition of H^+ to
form carbocation



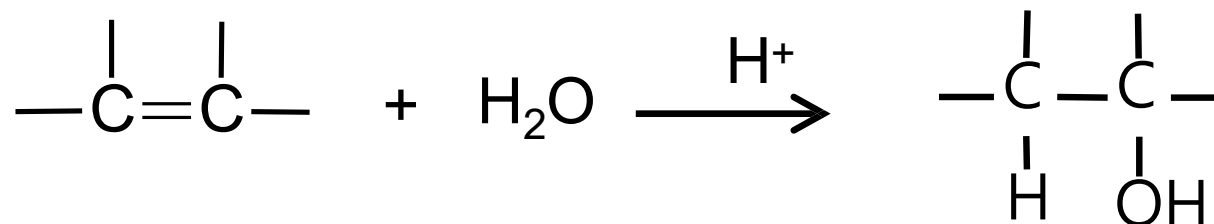
Step 2:
Nucleophile
attack on R^+



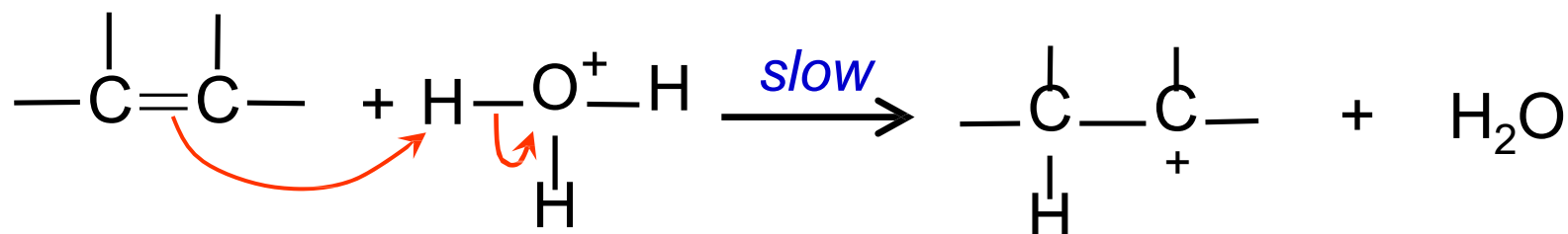
Electrophilic Addition Mechanism for Hydration of Alkene

10
9

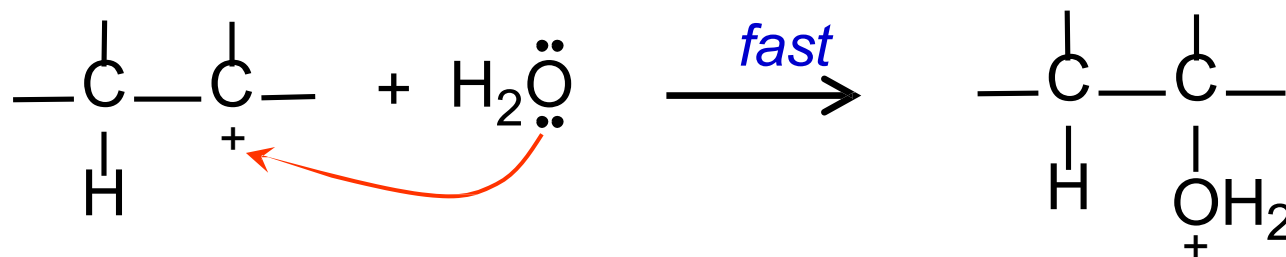
General
Reaction
Equation



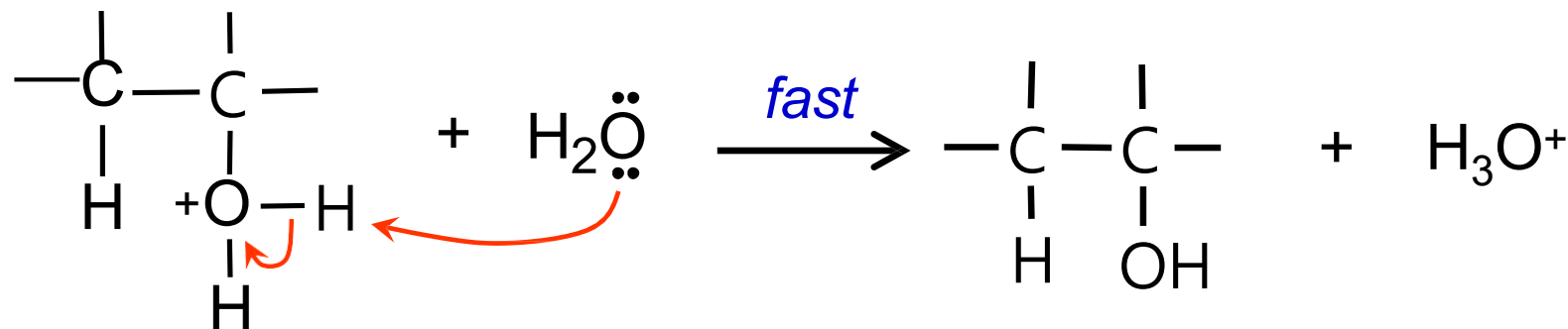
Step 1:
Addition of H^+ to
form carbocation



Step 2:
Nucleophile
attack on R^+

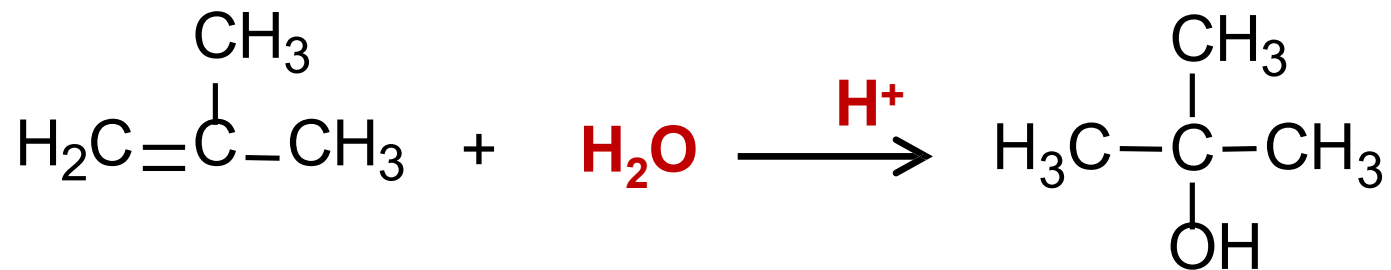


Step 3:
Loss of H^+

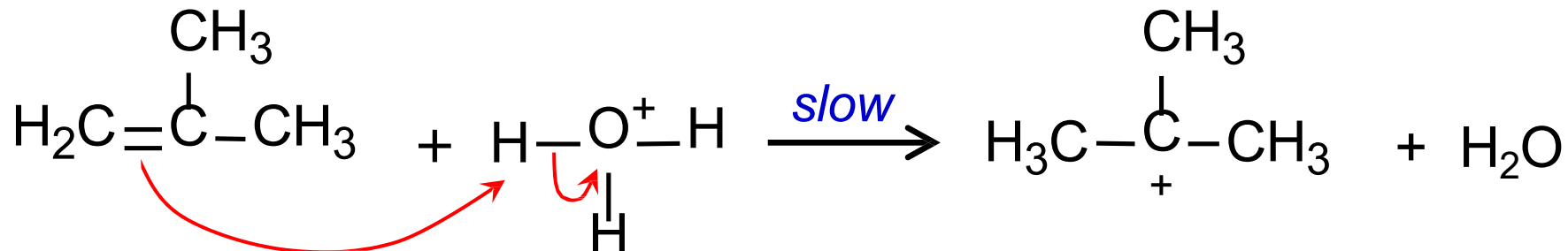


Example

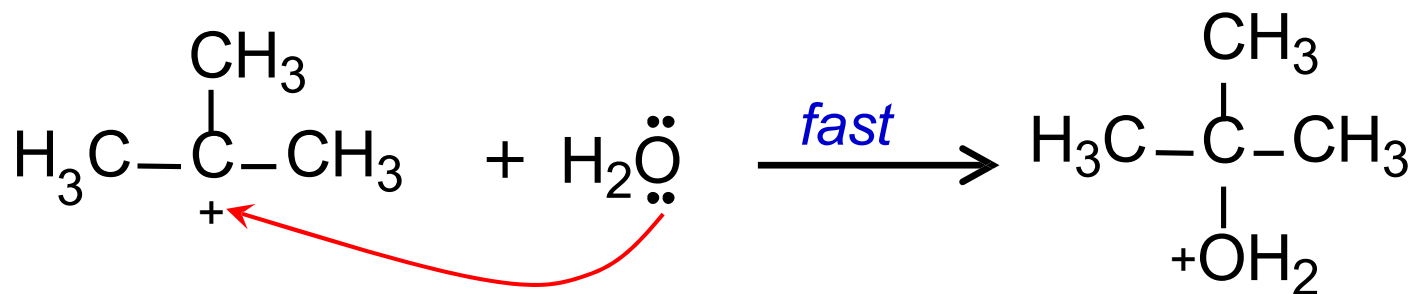
Reaction
Equation



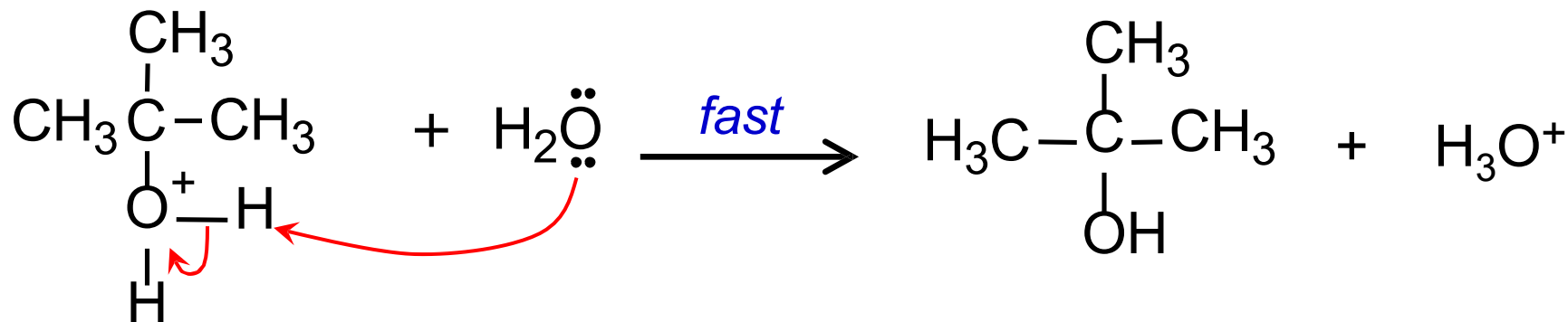
Step 1:
Addition of H^+ to
form carbocation



Step 2:
Nucleophile
attack on R^+



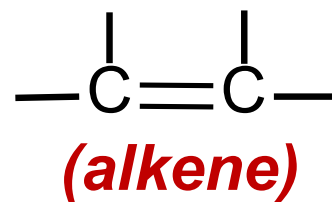
Step 3:
Loss of H^+



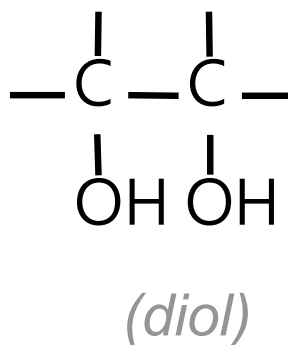
OXIDATION OF ALKENES

- The C = C in alkenes can be oxidised when treated with oxidising reagents.
- The products of oxidation depend markedly on the type of oxidizing reagent and the conditions of the reactions.

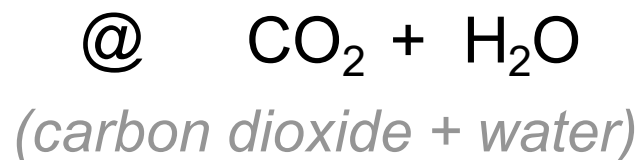
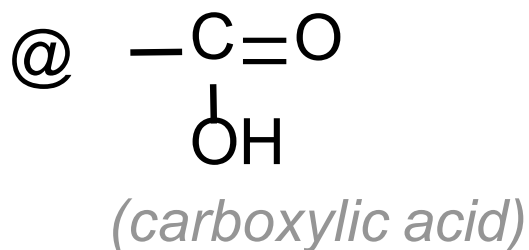
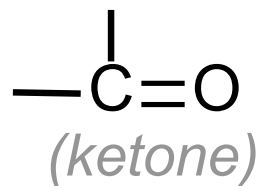
Oxidation of Alkenes



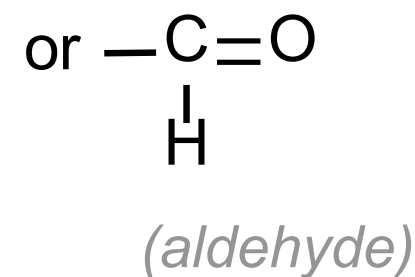
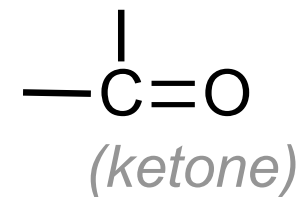
KMnO₄, OH⁻(cold)
Baeyer's Test



KMnO₄, H⁺
(heat)

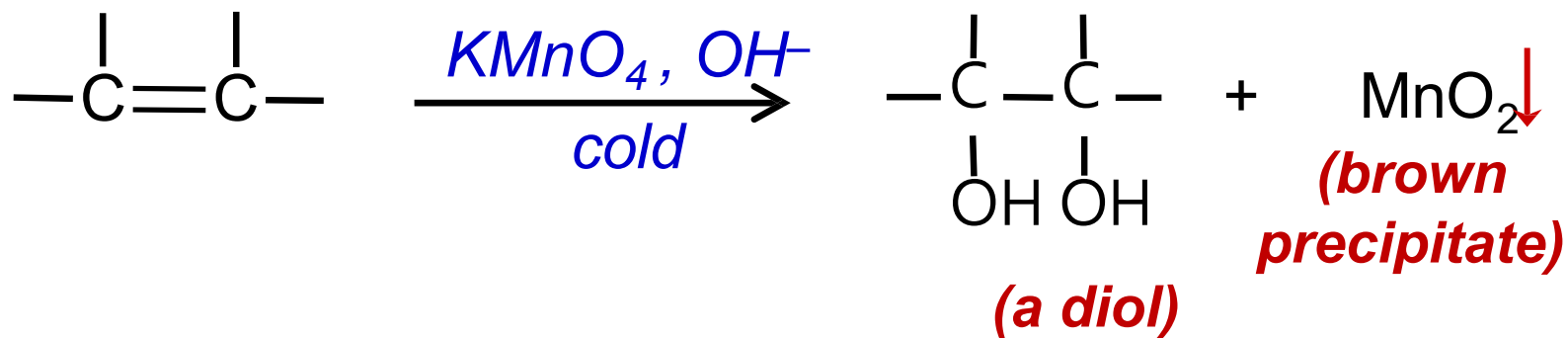


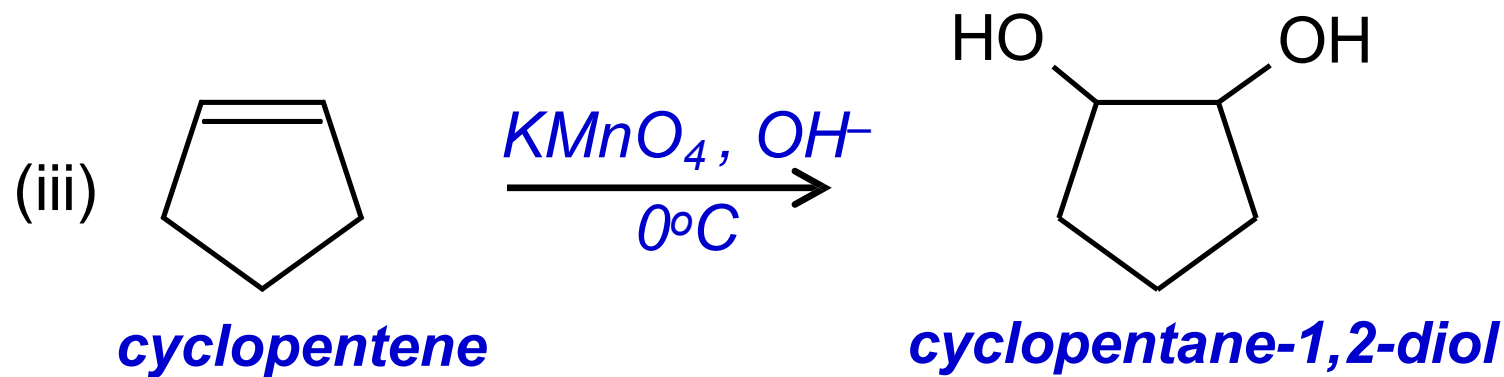
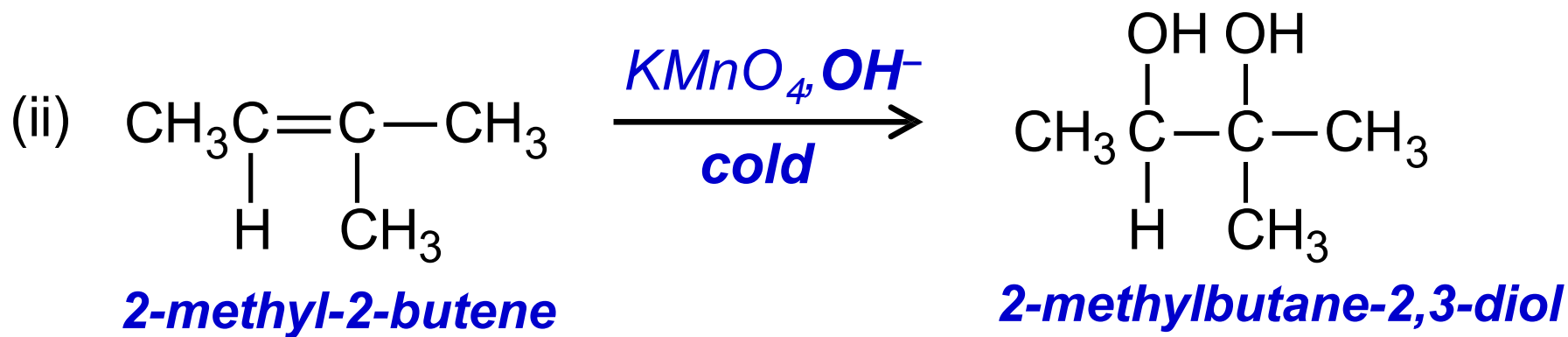
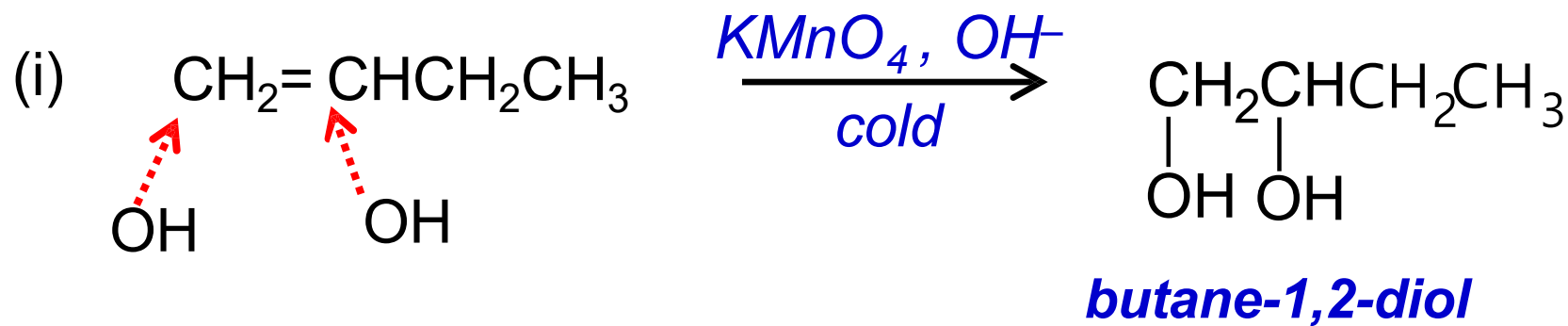
i) O₃
ii) H₂O, Zn @ (CH₃)₂S



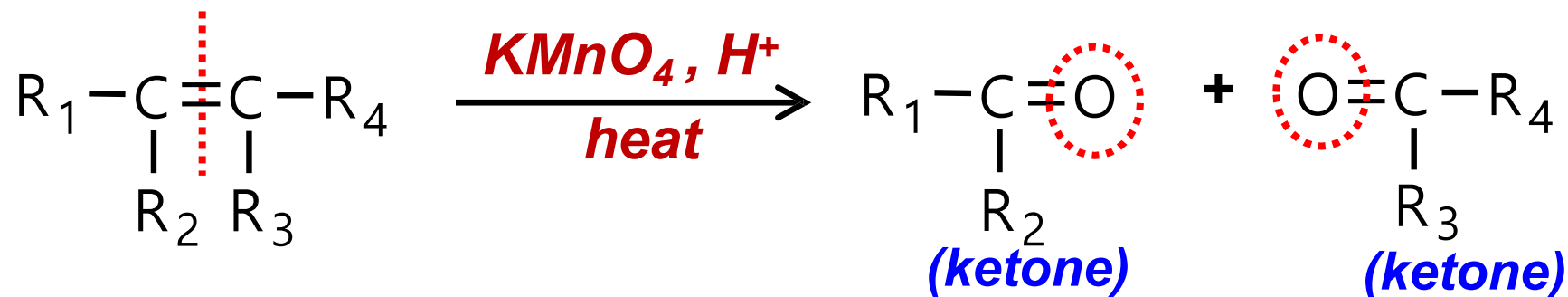
1 With cold dilute KMnO_4

- Alkenes are oxidised to 1,2-diol (glycol).
- This reaction is called the **Baeyer's test**.
- It can serve as a **test for the presence of $\text{C}=\text{C}$** , and is commonly used to **differentiate alkenes from alkanes**.
- Positive test observations: Purple solution of KMnO_4 is **decolourised** and **a brown precipitate** (MnO_2) formed.

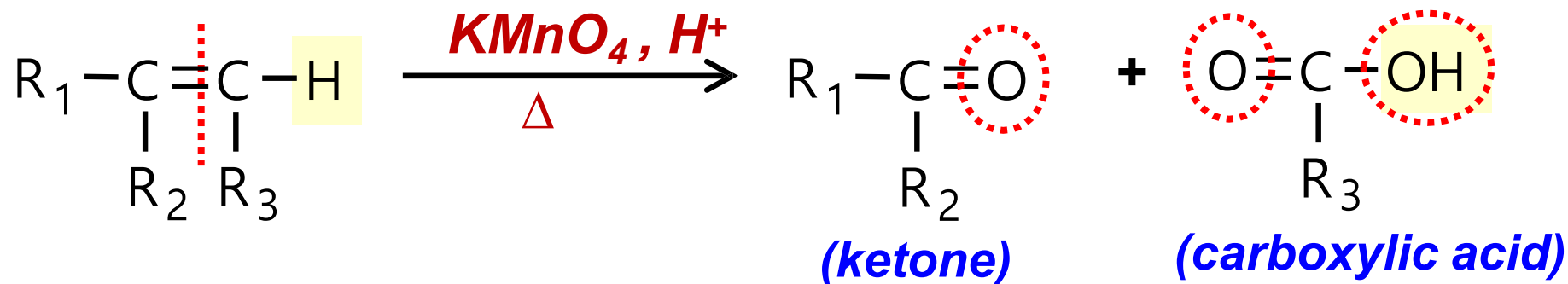




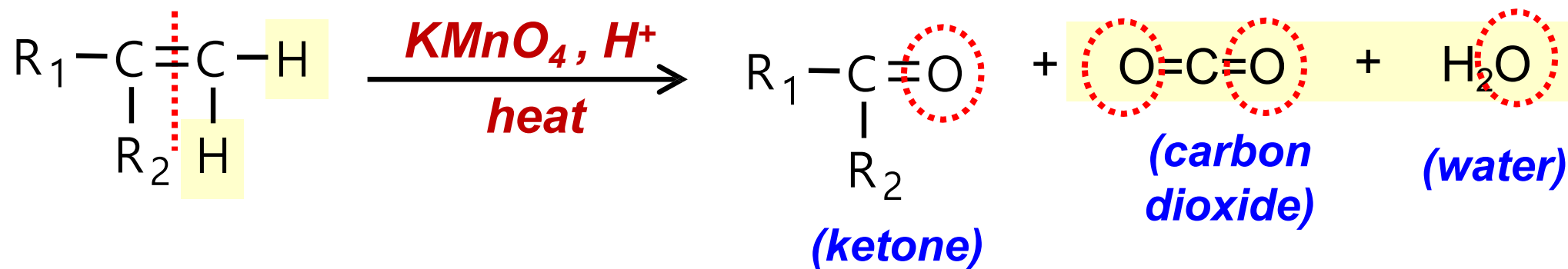
PRODUCTS OF OXIDATIVE CLEAVAGE (WITH HOT ACIDIFIED KMnO_4)



Note - If the doubly bonded C attached **two alkyl groups**, it will be oxidised to **ketones**.

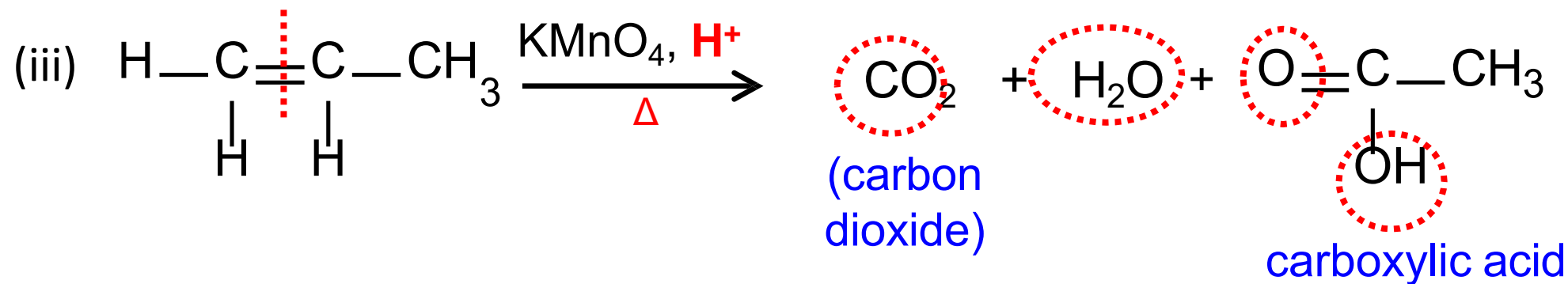
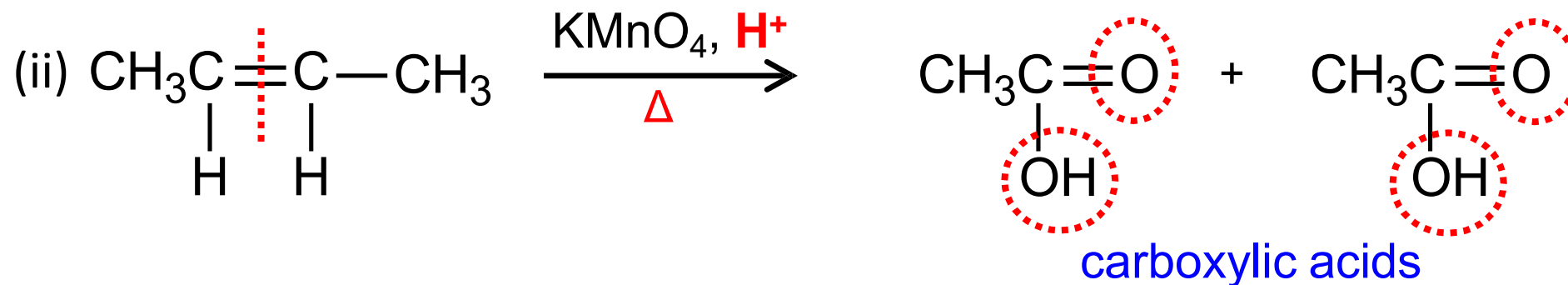
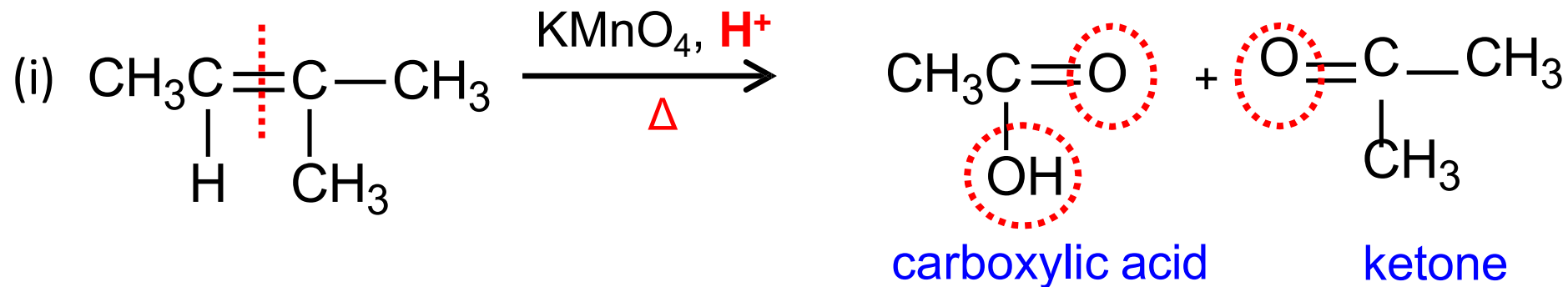


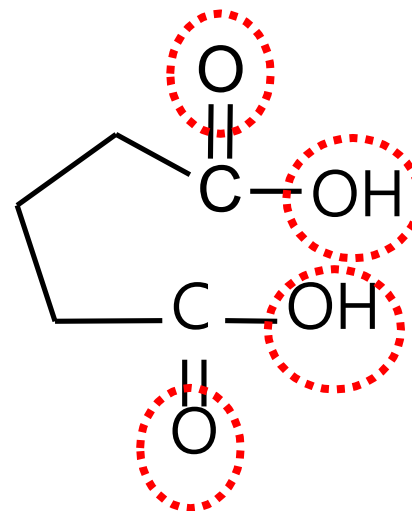
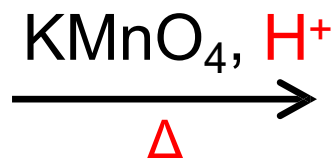
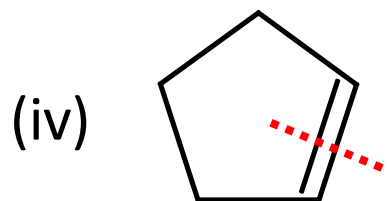
Note - If the doubly bonded C attached to **a H atom**, it will be oxidised to **carboxylic acid**.



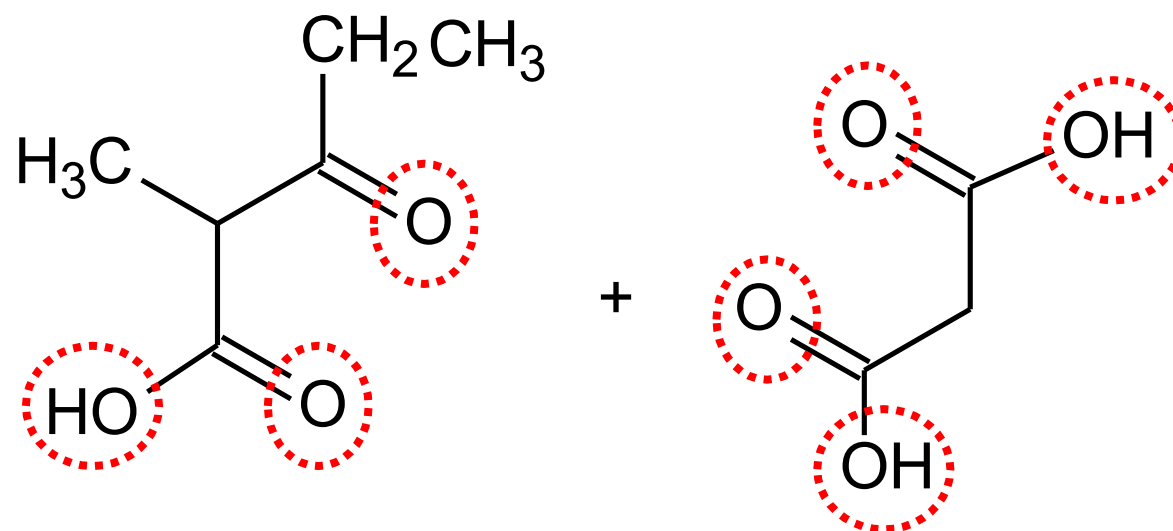
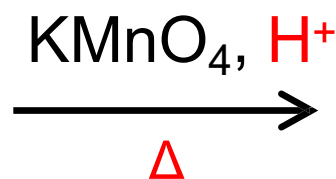
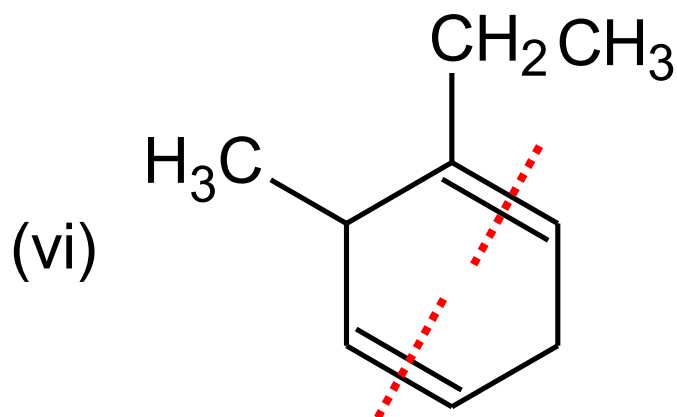
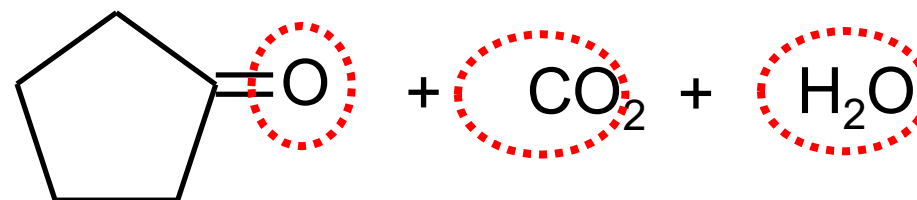
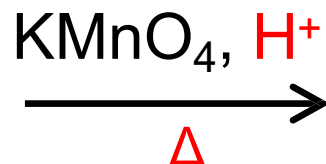
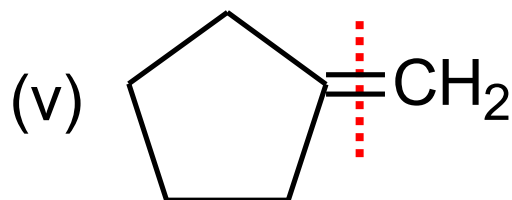
Note - If the doubly bonded C attached two H atoms, it will be oxidised to carbon dioxide and water (cleavage on the C=C and two C–H bonds).

Examples

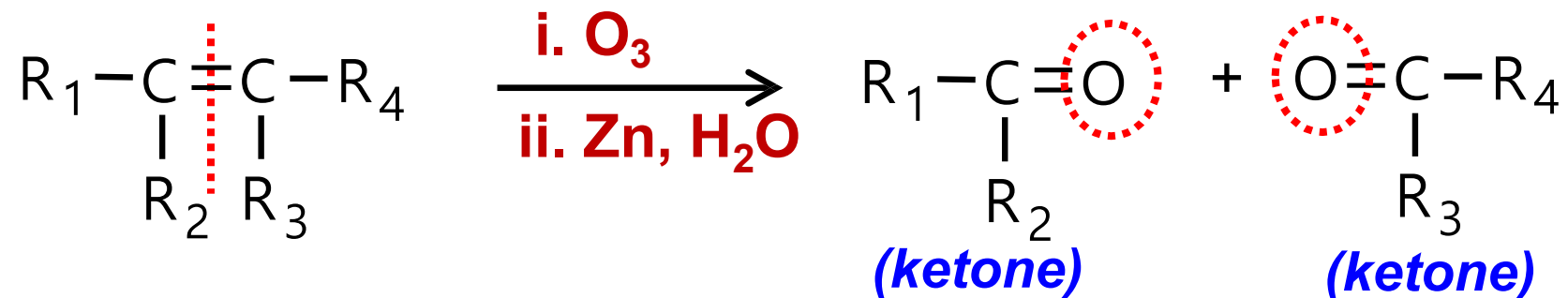




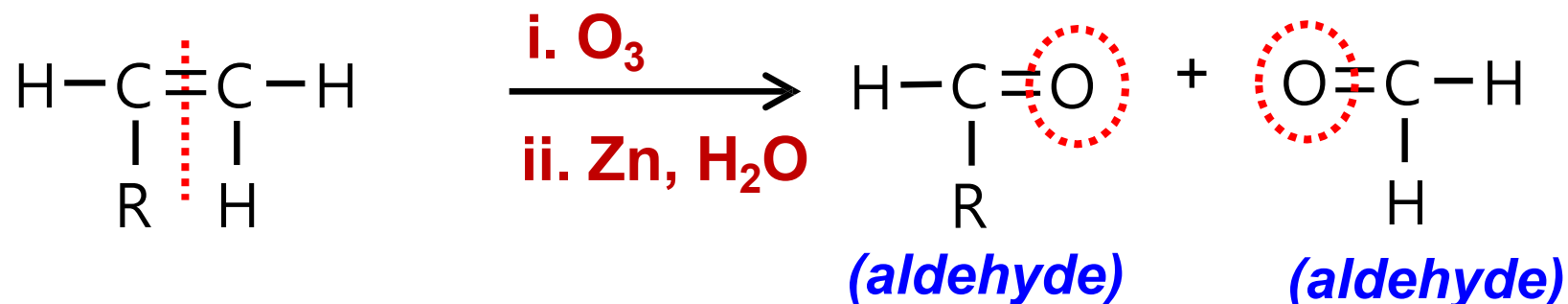
dicarboxylic acid



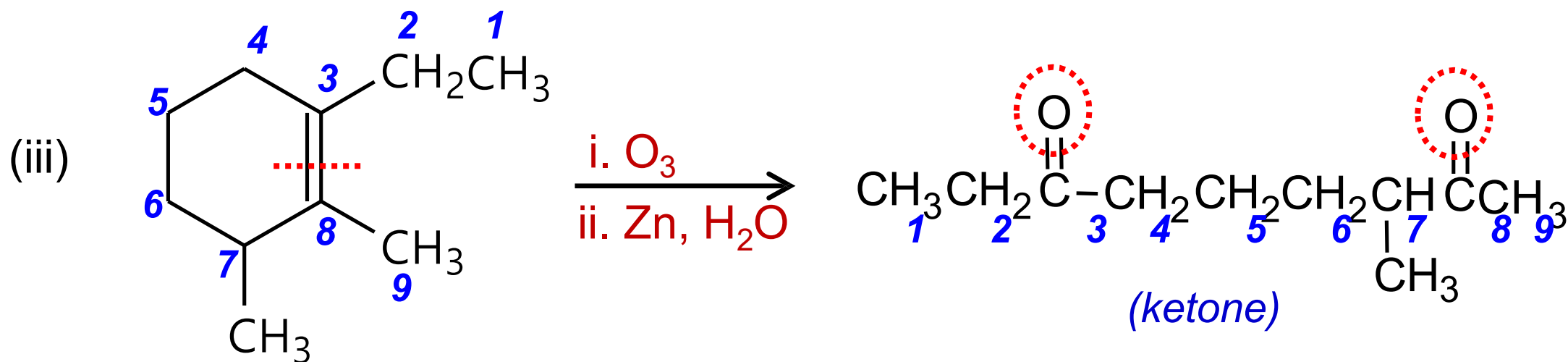
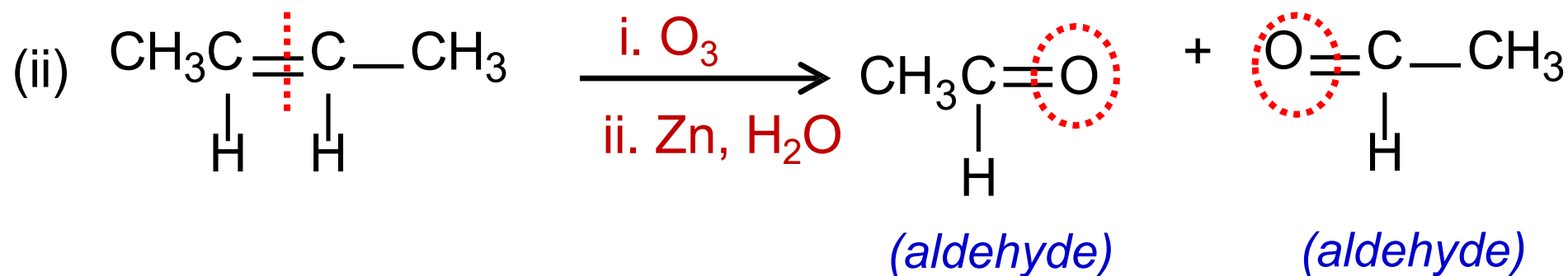
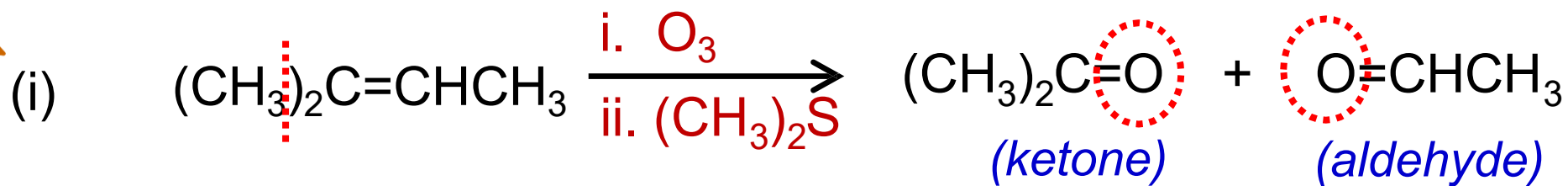
PRODUCTS OF OXIDATIVE CLEAVAGE (OZONOLYSIS)



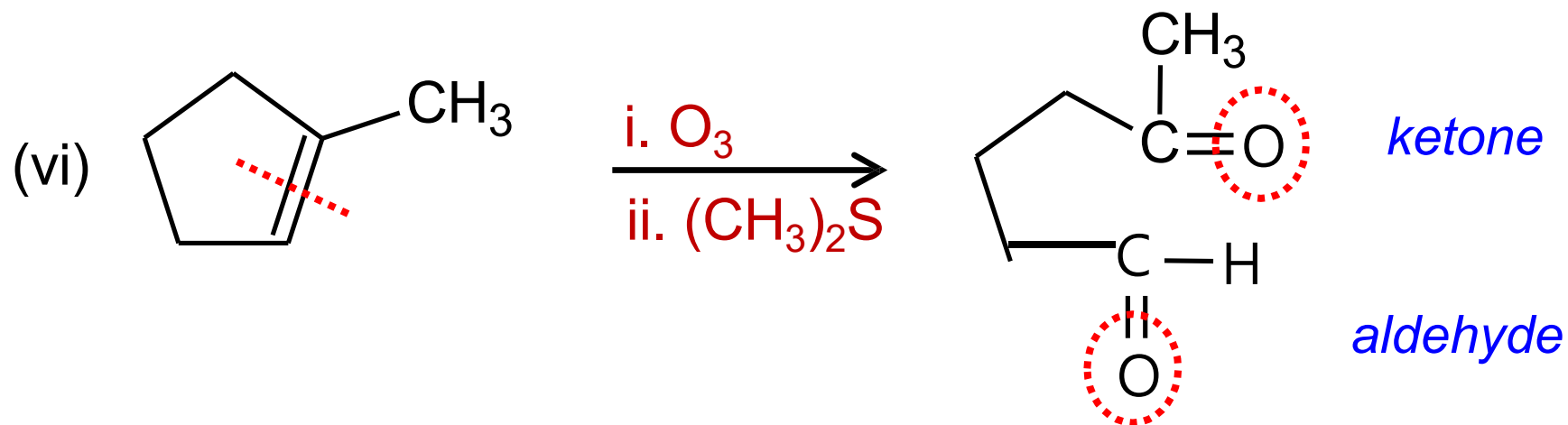
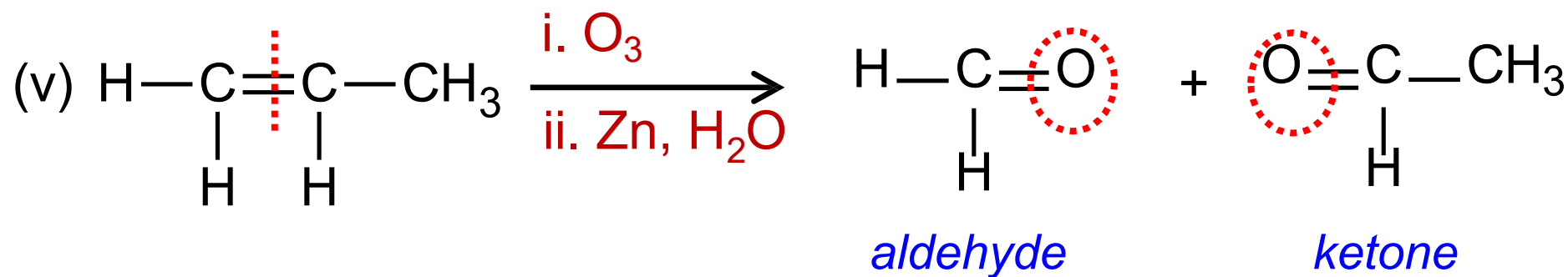
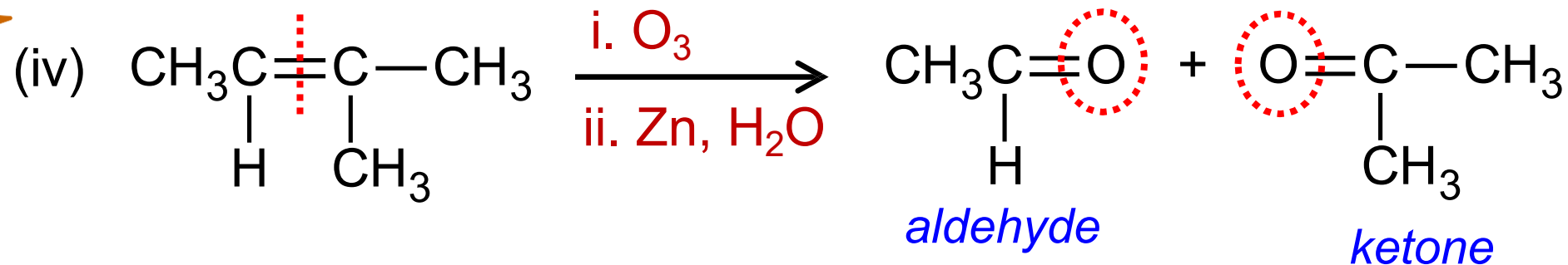
Note - If the doubly bonded C attached **two alkyl groups**, it will be oxidised to **ketones**.



Note - If the doubly bonded C attached to **one or two H atoms**, it will also be oxidised to **aldehyde**.



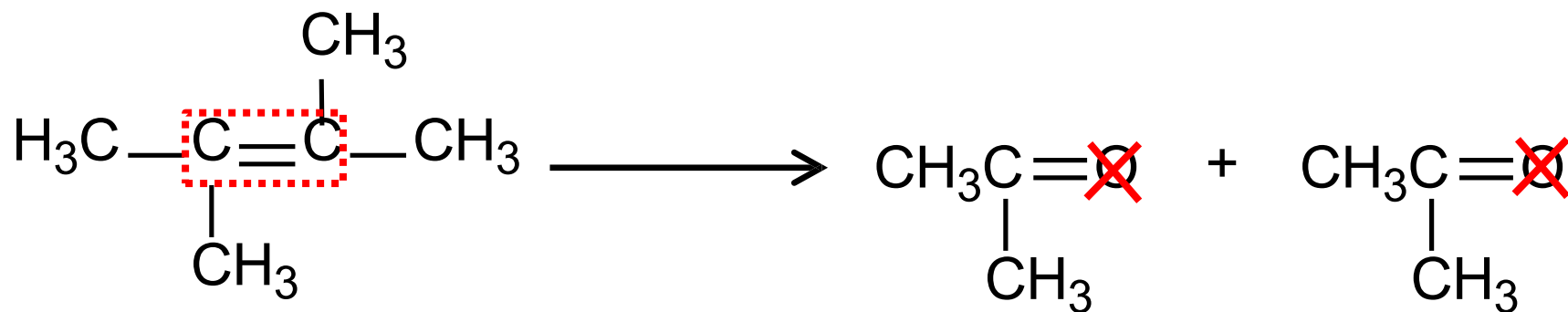
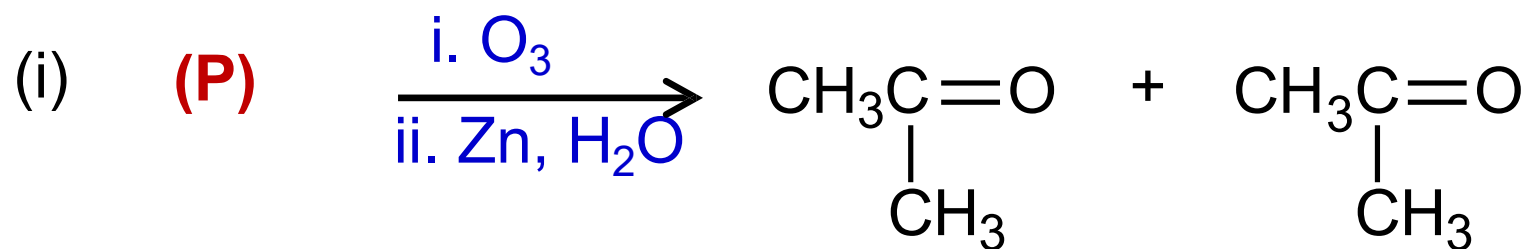
Examples



The oxidative cleavage of alkenes via ozonolysis can be used to establish the **location of the double** bond in an unknown alkene.

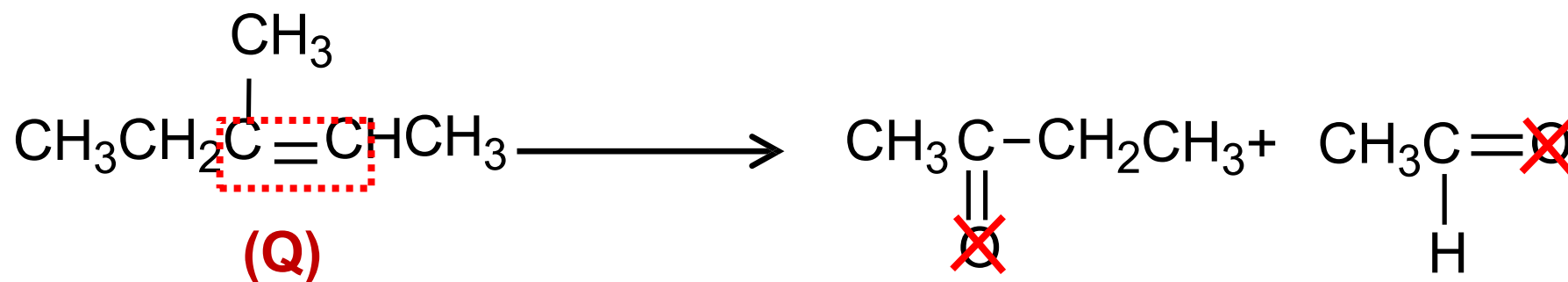
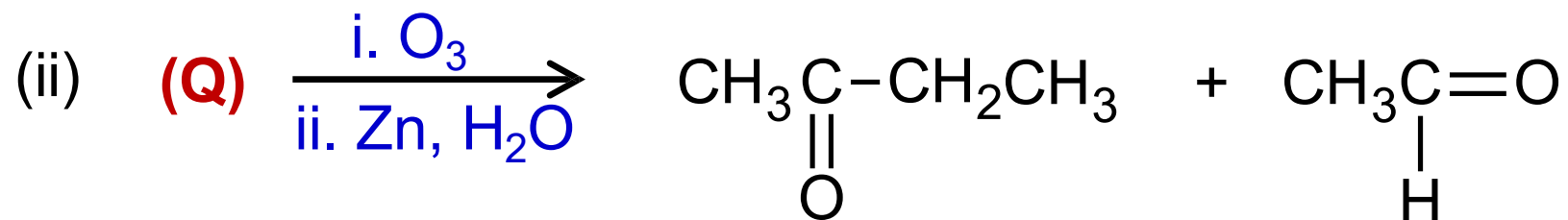
Example

Write and name the structures of the alkenes which give the following products through ozonolysis:



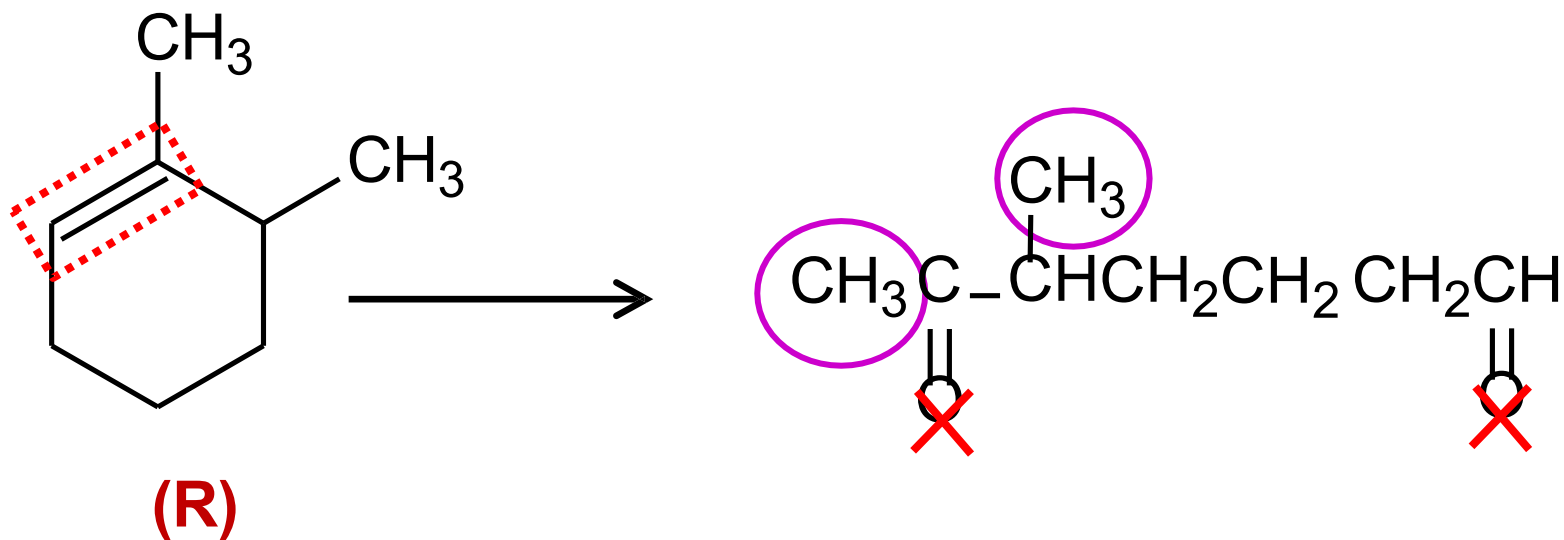
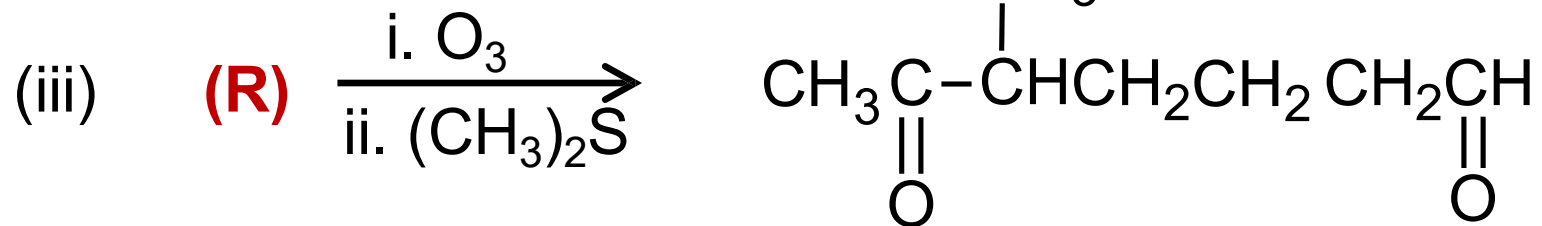
(P)

2,3-dimethyl-2-butene


 Example


3-methyl-2-pentene

Example

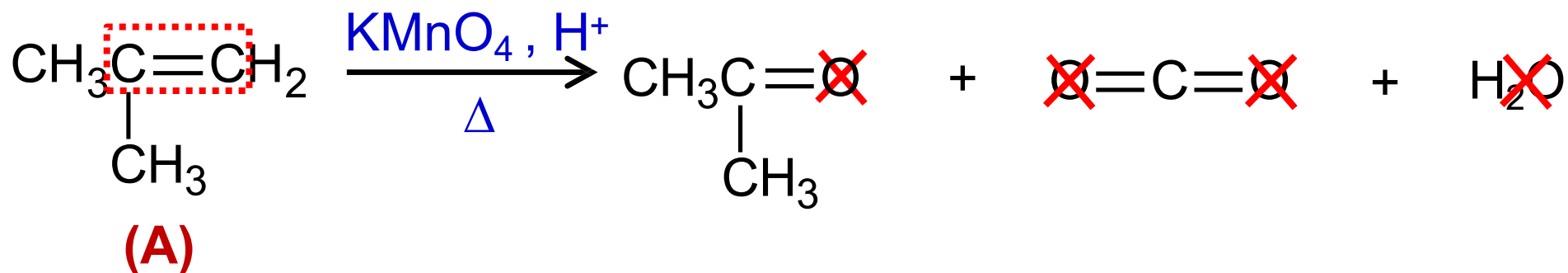
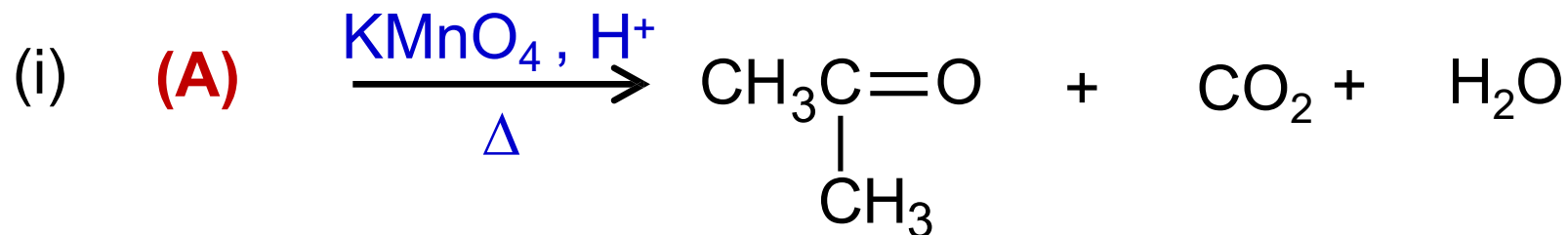


1,6-dimethylcyclohexene

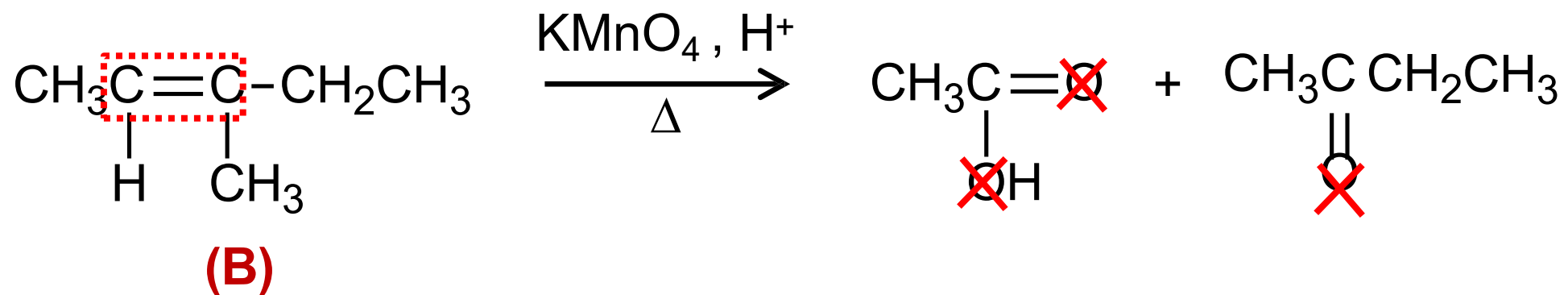
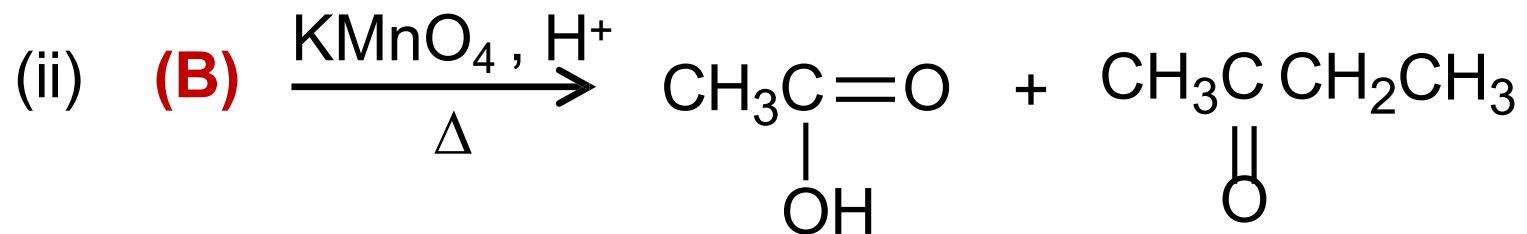
The oxidative cleavage of alkenes using hot acidified KMnO_4 can also be used to establish the **location of the double** bond in an unknown alkene.

Example

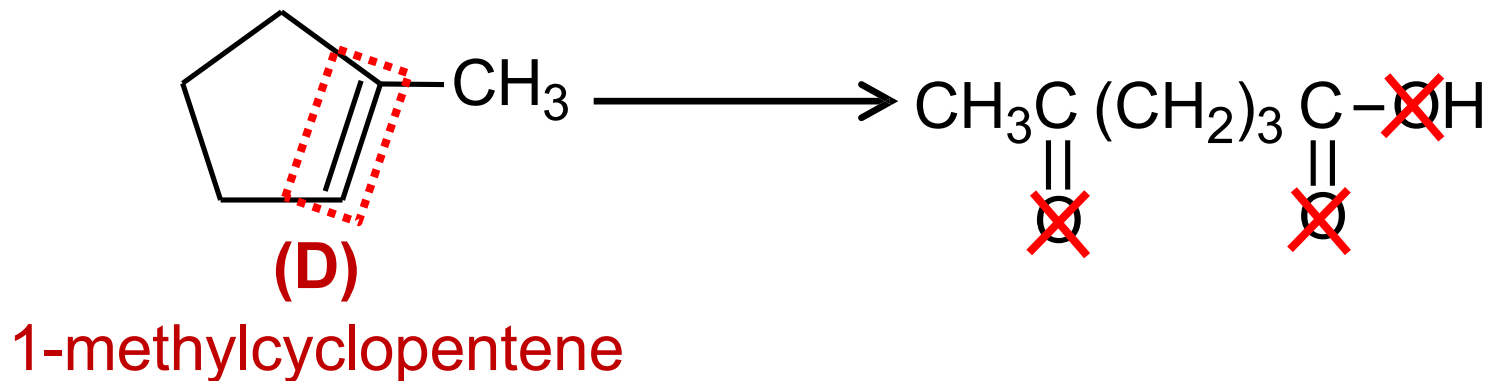
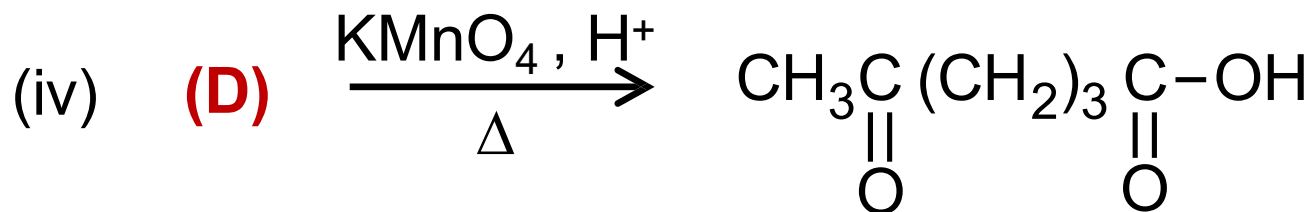
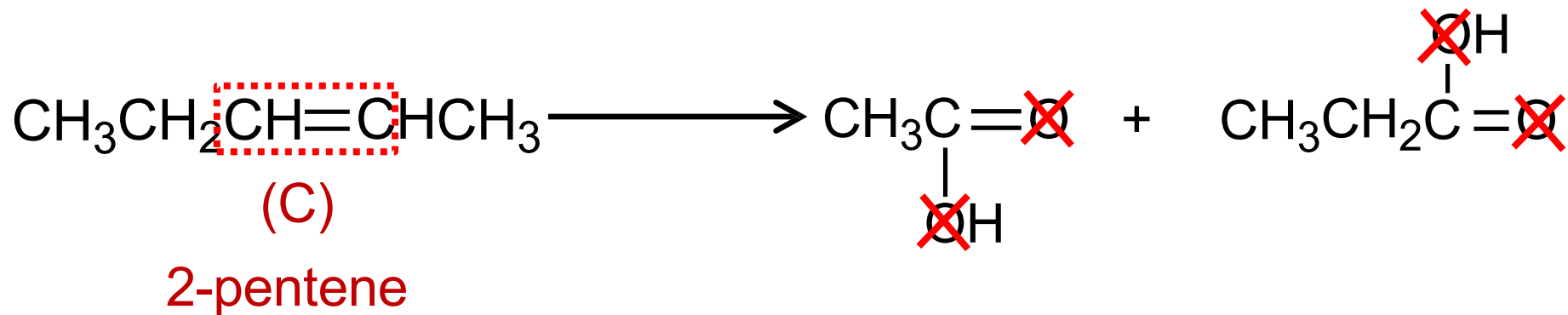
Give the **structures and IUPAC names** of the **alkenes** that give the following products when treated with hot acidified KMnO_4 :



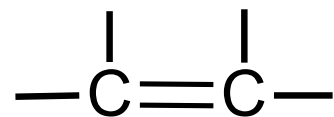
2-methylpropene


 Example


3-methyl-2-pentene

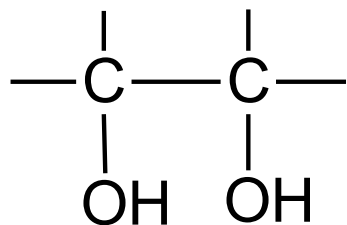

 Example


UNSATURATION TEST FOR ALKENES



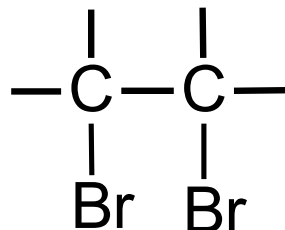
(alkene)

**KMnO₄, OH⁻
(cold)**



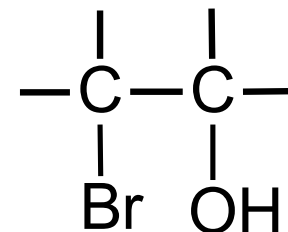
*Purple solution
decolourised and a
brown precipitate
(MnO₂) formed*

**Br₂, CH₂Cl₂
(in the dark)**



*Reddish-brown
solution
decolourised*

Br₂ (aq)

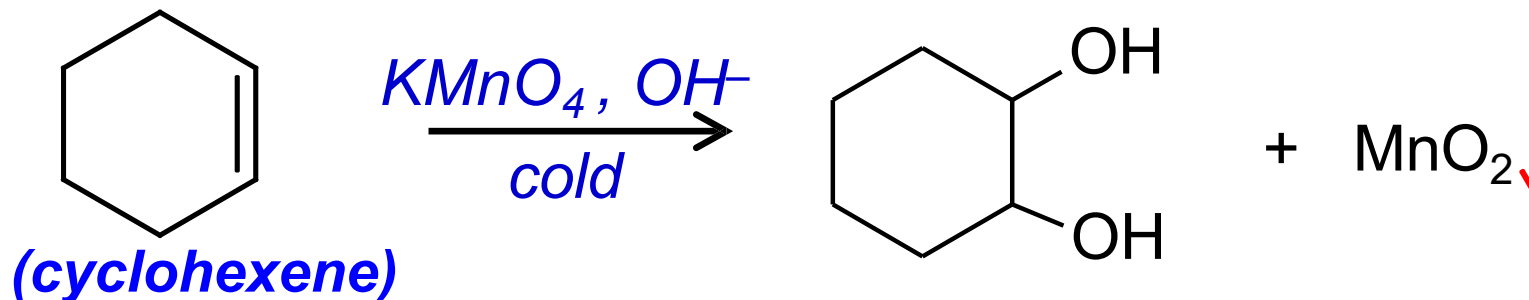


*Reddish-brown
solution
decolourised*

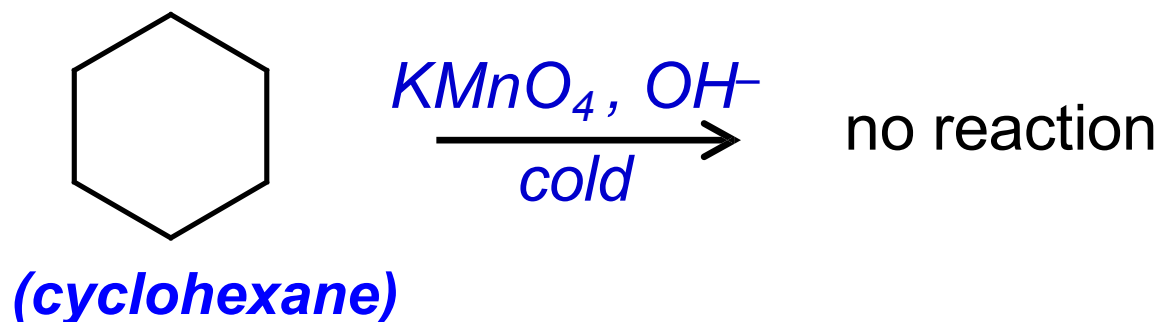
Example

(ii) Suggest a suitable chemical test to distinguish cyclohexene from cyclohexane. Give the equations and observations.

Baeyer's test



Observation: Purple solution of KMnO_4 is decolourised and a brown precipitate formed.

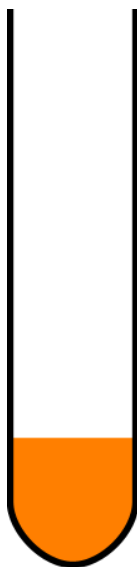


Observation: Purple solution of KMnO_4 remained unchanged.

Bromine in CH_2Cl_2

Procedure:

(i) Place a solution of bromine in a test tube

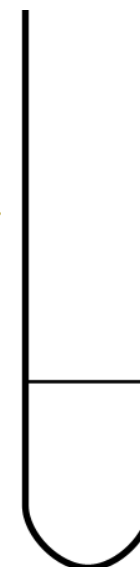


reddish
brown

(ii) Add the hydrocarbon to be tested and shake the tube



(iii) If the reddish brown colour disappears then the hydrocarbon is an **alkene**

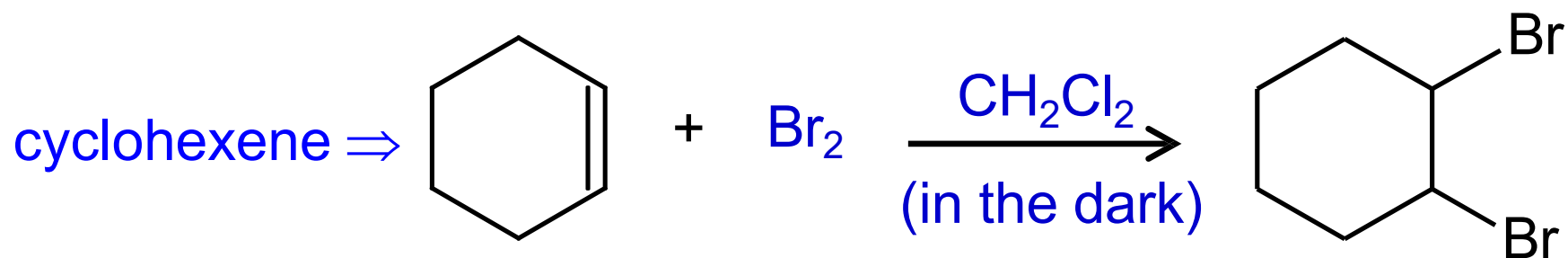


colourless

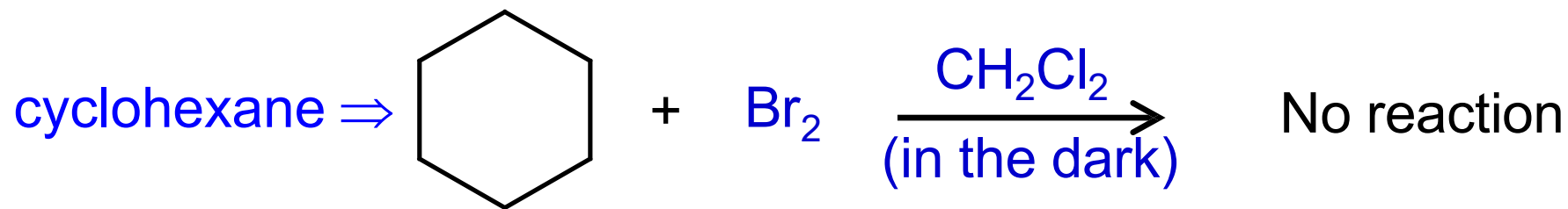
Example

- (i) Describe a suitable chemical test which can be used to distinguish cyclohexene from cyclohexane.

Bromine in CH_2Cl_2 (*in the dark condition)



Observation: Reddish-brown colour of bromine is decolourised.

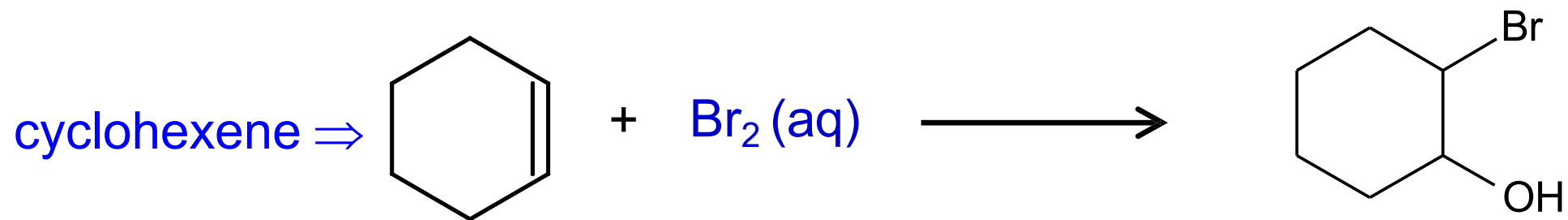


Observation: Reddish-brown colour of bromine remained unchanged.

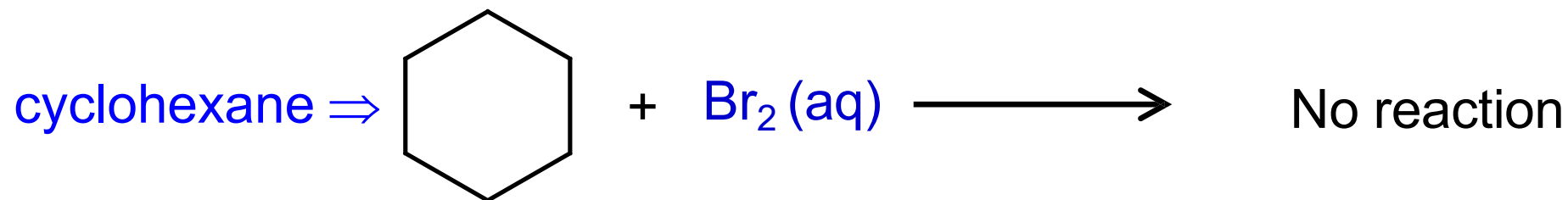
Example

- (i) Describe a suitable chemical test which can be used to distinguish cyclohexene from cyclohexane.

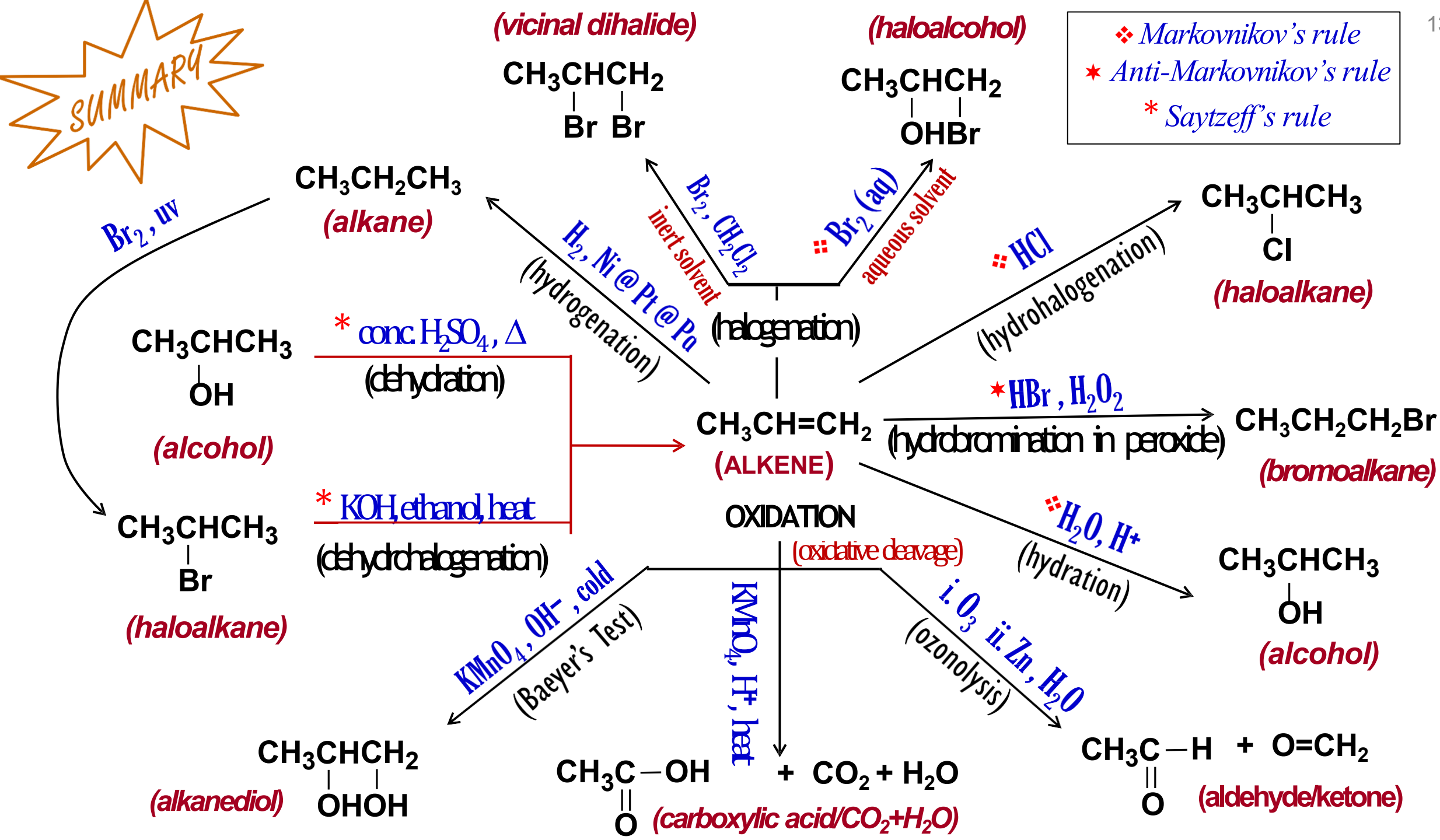
Bromine water, Br₂ (aq)



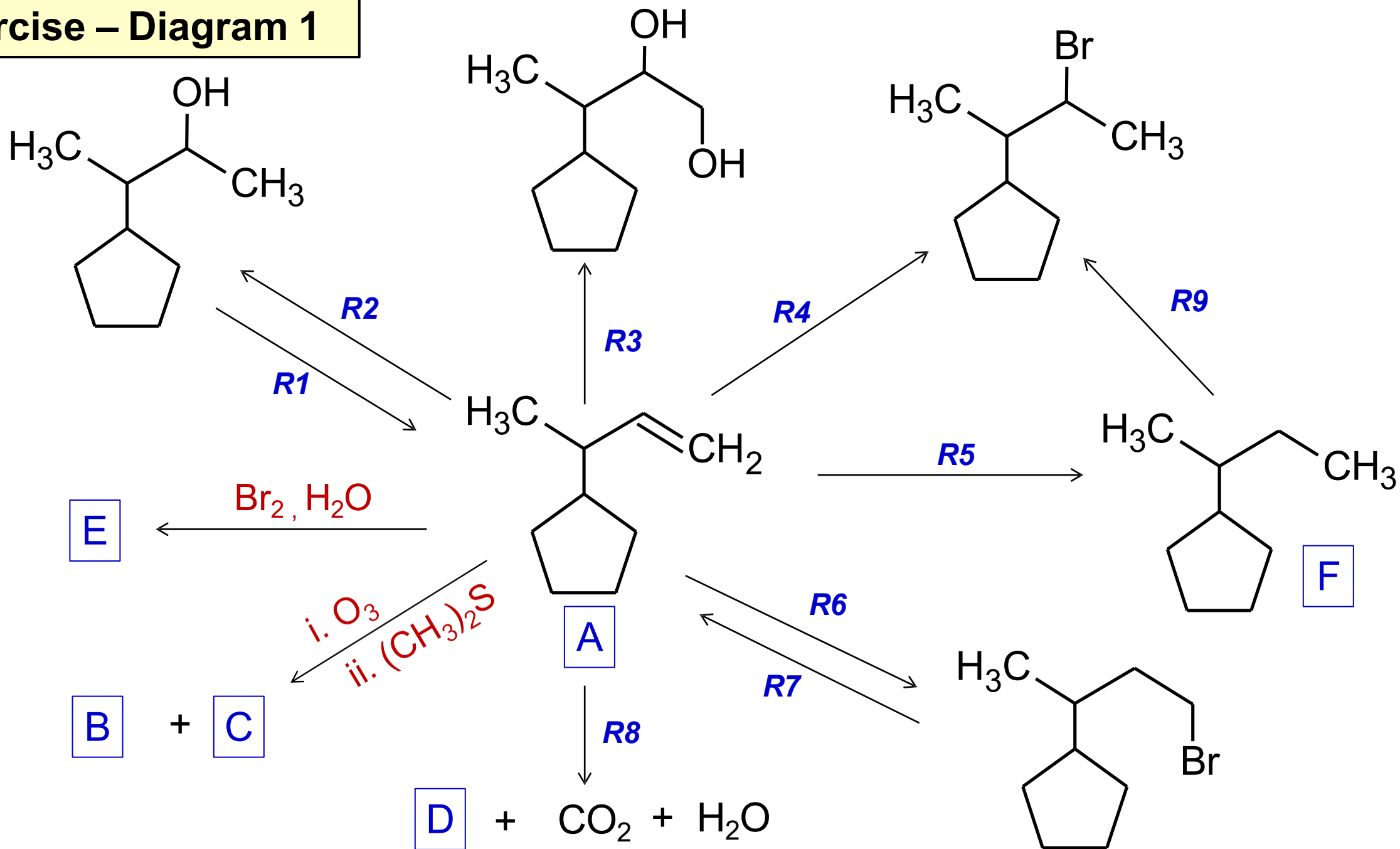
Observation: Reddish-brown colour of bromine is decolourised.



Observation: Reddish-brown colour of bromine remained unchanged.



Exercise – Diagram 1



With reference to the reaction scheme in Diagram 1:

- i) Name the alkene A.
- ii) Give the reagent and condition labelled from *R1* through *R9*.
- iii) Give the structures of compound B, C, D and E.
- iv) Write the mechanism for reactions *R1*, *R2* and *R4*.
- v) Describe a suitable chemical test to distinguish compound A from F.

Monomer and Polymer

Monomer

* The molecule from which a polymer is synthesised.

Polymerisation

* The process that combine monomers to form a polymer.

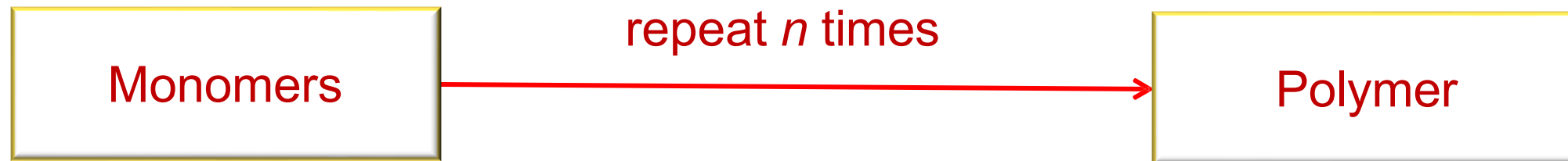
Polymer

* A macromolecule (large) that is made up of small repeating units.

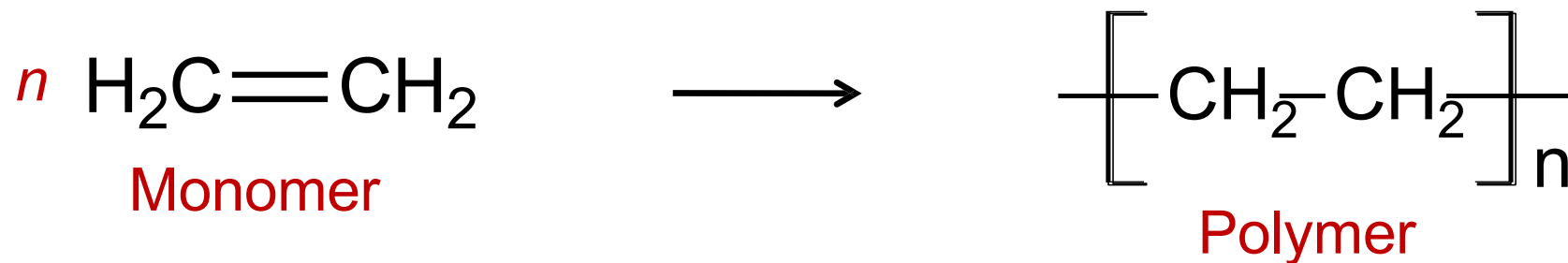
Repeating Units

* The actual structure that is being repeated in the polymer chain.

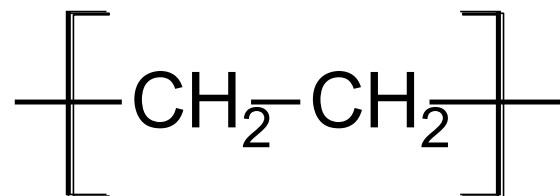
Monomer and Polymer



Example:



Repeating Units



Synthetic Polymers

Addition Polymerisation

- Produce **polyalkene**
 - (i) Polyethylene
 - (ii) Polyvinyl chloride (PVC)
 - (iii) Polystyrene
 - (iv) Teflon

Condensation Polymerisation

- Produce **polyamide**
 - (i) Kevlar
 - (ii) Nylon 6
 - (iii) Nylon 6,6
- Produce **polyester**
 - (i) Dacron
 - (ii) Terylene

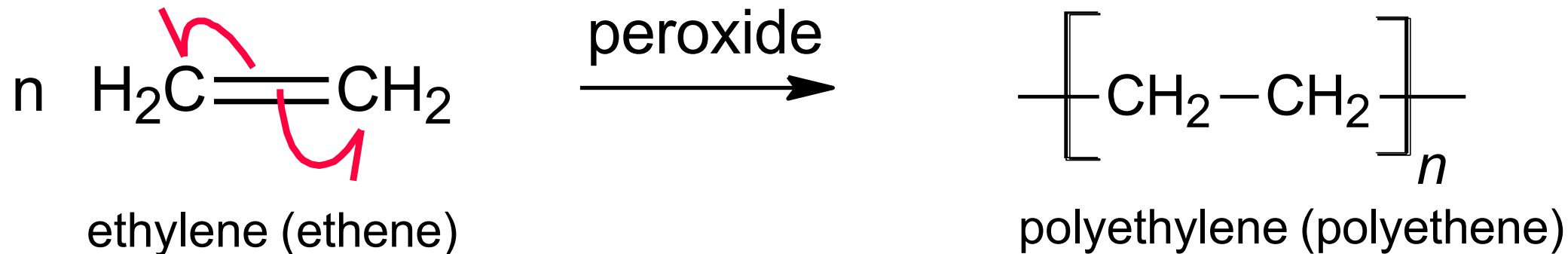
It will be discussed in Chapter 10
(**CARBOXYLIC ACID & ITS
DERIVATIVES**)

Addition Polymerisation

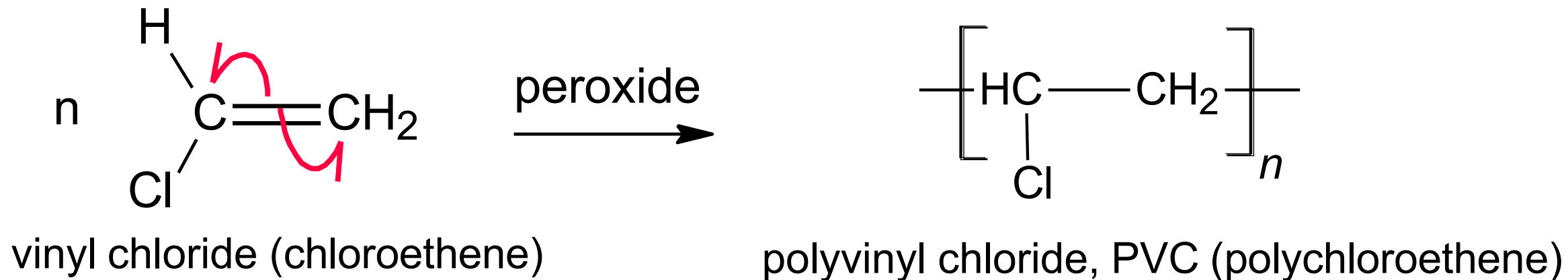
- The addition reaction in which monomers with double bonds are joined together by covalent bonds to form a large molecule of polymer without a loss of smaller molecule.
- $C=C$ are broken and replaced by $C-C$ single bonds.
- Called addition polymers.
- Free radical such as peroxide ($ROOR$) as initiator in addition polymerisation.

Example

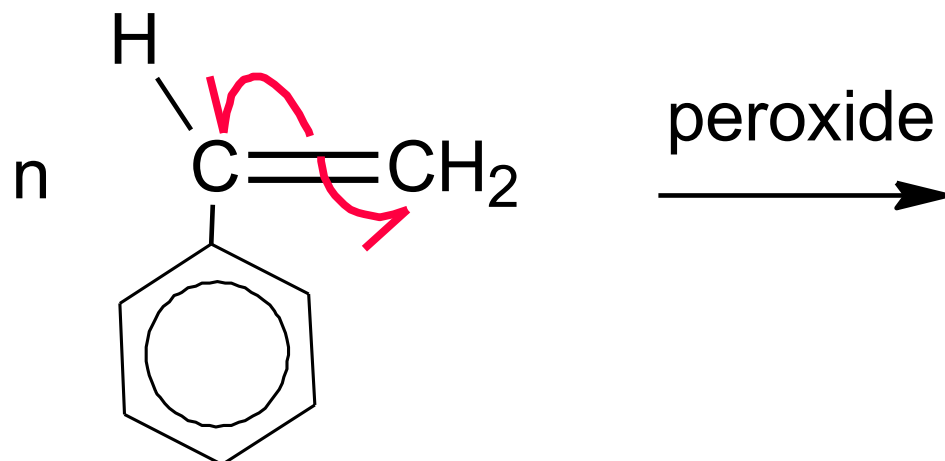
i) Formation of polyethylene



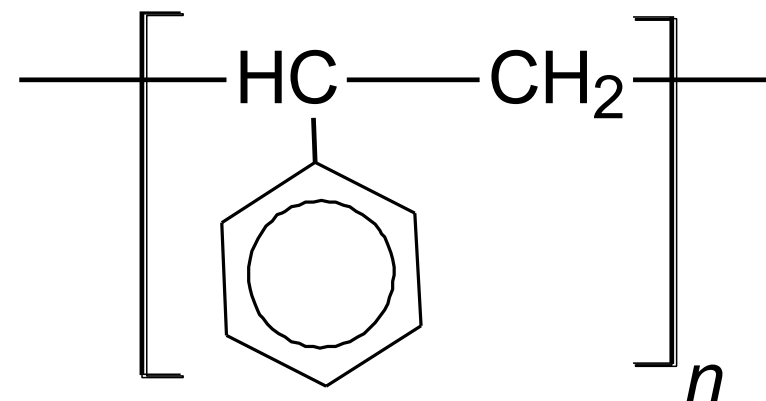
ii) Formation of polyvinyl chloride



iii) Formation of polystyrene

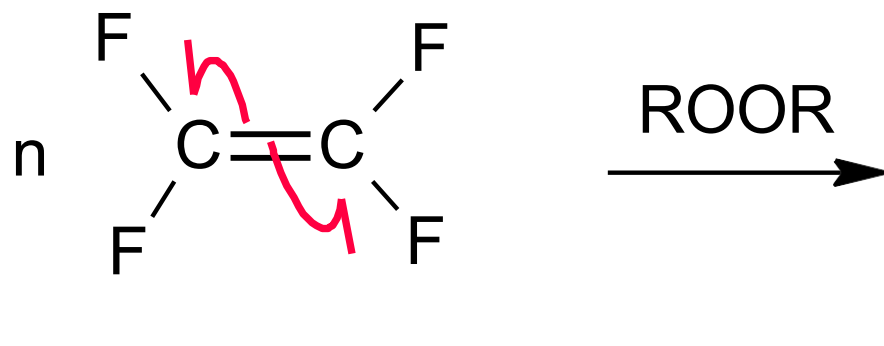


styrene (phenylethene)

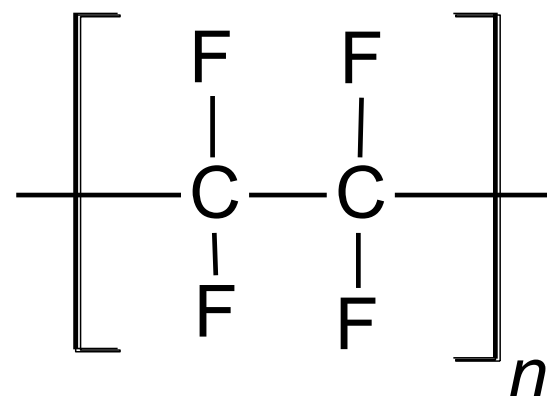


polystyrene (polyphenylethene)

iv) Formation of Teflon



tetrafluoroethene



Teflon (polytetrafluoroethene)

The slide features abstract geometric shapes in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains overlapping triangles in various shades of gray.

THE END